



Spectrum PowerCC
ICCP Protocol

05.00

Reference Guide

History

Preface

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Note

Please observe safety notes and warnings for your own safety.

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History

Version	Date	Modification
05-0-01	03.2009	Spectrum PowerCC 5.00
04-2-03	03.2007	Adding description of remaining ICCP data types (new Chapter 3-2)
04-2-02	02.2007	New standardized layout
04-2-01	09.2006	Implementing review comments.
04-2-00	09.2006	Update to SPC 4.2 version. Security functions added
04-1-01	04.2005	Corrected acc. to Variance SPC400007520
04-1-00	10.2004	Formal update to SPC 4.1
04-0-00	05.2004	Update to SPC 4.0

Preface

Safety Notes

This manual does not constitute a complete catalog of all safety measures required for operating the product in question because special operating conditions might require additional measures. For application specific safety measures refer to the respective project documentation. However, it does contain notes that must be adhered to for your own personal safety and to avoid damage to property. These notes are highlighted with a warning triangle and different keywords indicating different degrees of danger as stated in the Warning Conventions.



Warning

WARNING indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.

- ✧ Please follow all advice instructions to prevent death or serious injury.
-



Caution

CAUTION used with the safety alert symbol indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury.

- ✧ Please follow all advice instructions to prevent minor or moderate injury.
-



Note

NOTE used without the safety alert symbol indicates a potential situation which, if not avoided, may result in an undesirable result or state.

Qualified Personnel

Only competent and authorized personnel should work with this product after becoming thoroughly familiar with all warnings, safety notices, operating instructions and maintenance procedures.

Use As Described

The product must not be used for any other purposes than that described in the technical documentation. If it is used together with third-party devices and components, these must be recommended or approved by Siemens AG. The successful and safe operation of this product is dependent on adequate transportation and proper handling, storage, installation, operation and maintenance.

Scope

This Reference Guide provides specific and detailed information on how to use a particular product or product component:

- Typical workflows and operations
- Description of all user interfaces and displays



Note

Please note that the screenshots used in this document contain sample data which may not be available in some systems.

Typical Users

This Reference Guide is designed for users that are already familiar with operational and technical aspects of power generation and power transmission and distribution as well as Spectrum PowerCC product concepts:

- **System Engineers**

The System Engineer is able to install and to customize the system. He needs deep knowledge about the internal structures and processes of the network control system.

- **Operators**

The Operator is controlling the network or the plants with the system therefore he needs to know how the network control system has to be operated.

- **Data Engineers**

The Data Engineer translates the real network of a utility or the generation plants to the data model of the network control system. He also maintains the data model.

- **Technical Specialists**

The Technical Specialist is responsible for a special technical area which is related to the network control system. This can be the planning department, which plans changes in the network and uses network analyses applications of the network control system or the RTU department, which has to connect the RTU to the network control system. The Technical Specialist has to know how specific functions of the network control system work in detail.

Structure of this Manual

- **Introduction:**

Basic information about the Spectrum PowerCC component

- **Workflow / Operations:**

Main use cases, workflows and operational steps for optimal use of the component

- **References:**

Reference and description of user interfaces and dialogs

Documentation Conventions

Example	Meaning
File > Open	Selecting the menu item Open on the File menu
F1	Pressing the F1 function key on the keyboard
Control+o	Holding down the Ctrl key and pressing the letter o on the keyboard

Example	Meaning
Control+Shift+o	Holding down the Ctrl key and the Shift key and pressing the letter o on the keyboard
Left-click	Clicking the left mouse button
Right-click	Clicking the right mouse button
Shift-click	Holding down the Shift key on the keyboard and clicking the left mouse button

Questions?

This document is part of a set of manuals that describes the complete Spectrum PowerCC product.

Various other product modules may be mentioned or discussed in this document. For more detailed information or if you have any questions about these Spectrum PowerCC products, please contact your Siemens representative.

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1 Introduction

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1.1 Inter-Control Center Communication

The ICCP is designed to allow **data exchange** over WANs between **control centers** (ICCP nodes). Examples of other control centers include neighboring control centers, power pools, regional control centers, and generators.

Data exchanged includes real-time information such as measured analog, digital, and accumulator values, along with supervisory control commands. Data exchange occurs between a host of one control center and the host of another control center. The data exchange path (ICCP link) may traverse one or more intervening communications processors.

ICCP has been designed to provide a general purpose communication protocol for use within the power industry. ICCP is an international standard officially known as IEC 60870-6 TASE.2. The ICCP standards are "companion standards" of the Manufacturing Message Specification (MMS) and ICCP is a user of MMS services. The intent of ICCP is to provide a standard protocol to augment the existing (several) data exchange protocols, thereby simplifying the problems of data exchange between control centers.

An individual control center, in its interactions with other control centers, may act as client, server, or both. Regardless of the architecture of the system in use at a control center, when acting as a server, the control center appears as a single entity to clients. A server control center may interact with multiple clients.

1.2 Communicating with Remote Control Center

1.2.1 Establishing Associations

Checklist

Here is a check list of things that need to be done in order to get the ICCP association established between the PowerCC ICCP system and the Remote Control Center.

- Gather the information listed in the table in Section [6.1 Mandatory Configuration Parameters](#).
This table lists the parameters that you need to exchange with your remote partner in order to bring up the ICCP association.
Use this information for configuring the ICCP Remote, Link, Local AR, and Remote AR in the Information Model Managerment (IMM). Also see Section [5.3.2 Accessing the Communication Editor for ICCP](#).
For additional information, refer to [Reference Guide Information Model Management \(IMM\)](#).
- Define the Software Package for ICCP. See Section [5.4.1 ART Configuration](#) for details.
- Stop and Restart Spectrum PowerCC.
- Update the osill2.cfg file using the OSIFileCreator. See Section [5.5.2 Maintaining osill2.cfg File](#) for details.
- Bring up the ICCP UI and enable the ICCP subsystem and ICCP remote. See Section [4.2.1 Accessing ICCP Functionality](#) for more information on ICCP UI.

1.2.2 Exchanging Data

Once the association with the remote has been established, it is time to configure the data to be exchanged.

- Determine SCADA measurements to be exchanged. ICCP is used to send and receive SCADA measurements defined in the Network Model. If these measurements are not already present, they need to be created.
- For SCADA measurements being provided to the remote, ICCP Block 1&2 Server objects need to be defined on the remote.
See Section [5.2.3 ICCP Configuration Data Types and Attributes](#) for more information on the ICCP Block 1&2 Server type. The ICCP Communication Editor in IMM can help the user configure Server objects.
- For SCADA measurements coming from the remote, ICCP Block 1&2 Client objects need to be defined on the remote.

If the Remote Control Center has configured their Server objects, the ICCP Remote fetch feature can be used to fetch this information from the remote (See Section [2.1 Remote Fetch](#) for more details).

The remote fetch process will create two .xml files, one containing a simple Network Model that contains SCADA measurement place holders. The second is the ICCP Block 1&2 Client definitions that go along with the simple Network Model. Import the SCADA .xml file followed by the ICCP .xml file. At a later time, the ICCP Block 1&2 Client object can be linked to the appropriate measurement in the Network.

The ICCP Communication Editor in IMM can also be used to help with the configuration of the Client objects.



2 Workflow

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2.1 Remote Fetch

Workflow for Remote Fetch

The ICCP protocol standard defines how the local system can retrieve the names and types of the Block 1&2 server objects that the remote control center is providing. PowerCC provides a way for a data engineer to use this capability to configure the corresponding client information on the local system.

Because the Remote Fetch impacts the realtime ICCP association with the remote control center, the data engineer and operator need to work together to complete this workflow. In most cases, the data engineer determines when the Remote Fetch is needed and the workflow should be executed.

Workflow

- Data engineer sets Remote Fetch Trigger.
- Operator activates Remote Fetch. Produces .xml files.
- Data engineer applies resulting .xml files into IMM and system.
- Operator/data engineer uses Runtime Explorer to view the data from a remote machine.
- Data engineer creates IMM job to modify associations of IcClientBlock1And2Object instances so they link to the final location in the Network Model.



3 Operations

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3.1 Connecting and Disconnecting Remote Systems

Connect and Disconnect Remote Systems

- Select ICCP Main Display tab.
- Select ICCP Node that you desire to Connect or Disconnect.
- Right-click to open the Connection Control context menu.
- Choose "Connect" to change the administrative status from **Off** to **On**.
- Choose "Disconnect" to change the administrative status from **On** to **Off**.

3.2 Enabling and Disabling the ICCP Subsystem

Select ICCP Subsystem in ICCP Main Display

- Click the **On** button in the ICCP Subsystem area to enable the selected ICCP Subsystem.

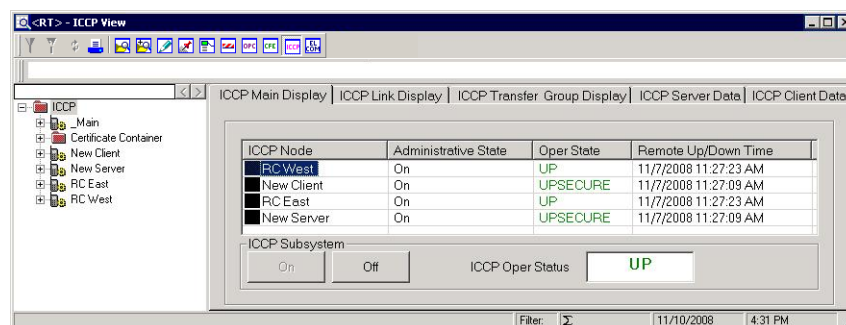


Figure 3-1 ICCP Subsystem On Button

- Click the **Off** button in the ICCP Subsystem area to disable the selected ICCP Subsystem.

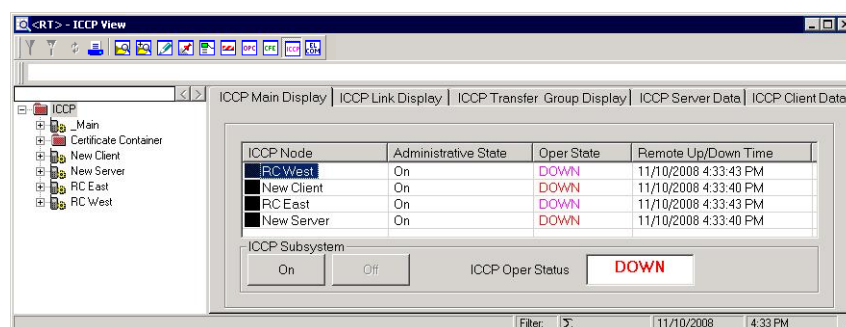


Figure 3-2 ICCP Subsystem Off Button



Note

The **On** button is disabled when the ICCP subsystem is enabled.

The **Off** button is disabled when the ICCP subsystem is disabled.

A read-only field that shows the current status of the ICCP subsystem.

3.3 Remote Fetch

Prerequisite

For remote fetch, the local and remote systems must have their ICCP subsystems configured and the ICCP communications established between the two systems. The remote system has configured the ICCP Server Objects that it will be providing to the local system.

Setting Remote Fetch Trigger

- The data engineer determines that it is necessary to fetch the object names from a specific remote system. The data engineer calls up Runtime Explorer, and then goes to the ICCP view.

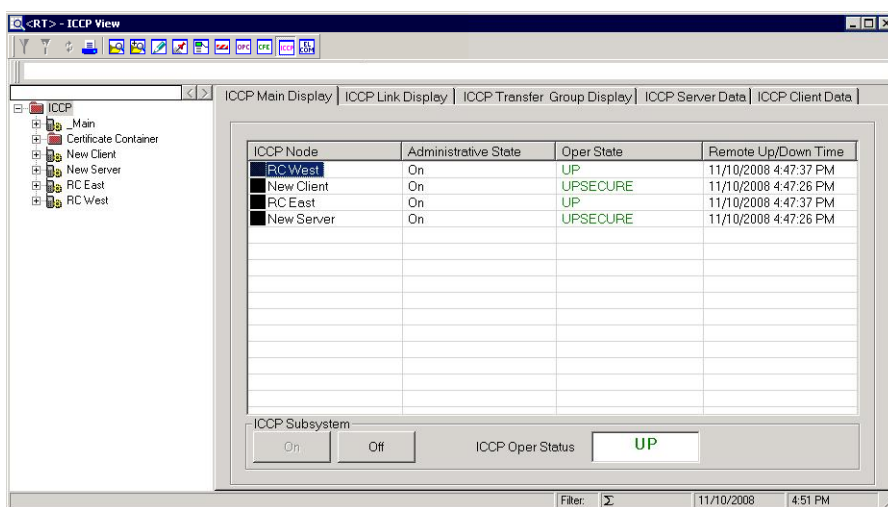


Figure 3-3 Runtime Explorer: ICCP View

- The data engineer right clicks on the remote where he wants to perform the fetch. In addition to the Connect/Disconnect that currently appears, a Remote Fetch selection will also appear. This is only selectable if the user has IMM Instance Data Access Rights. The user selects this option.

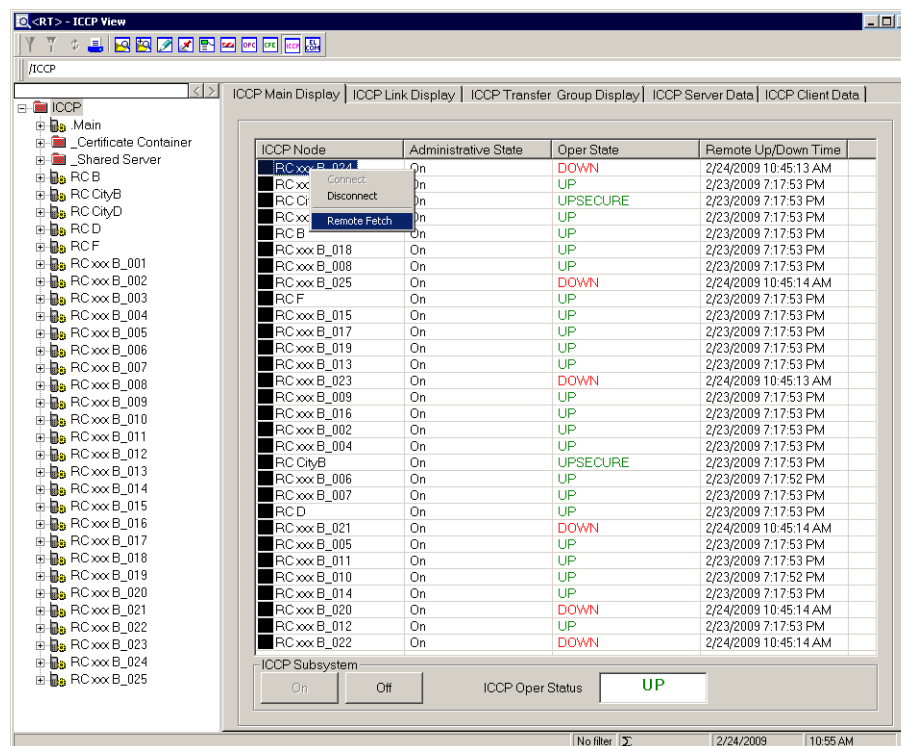


Figure 3-4 Remote Fetch Display

**Note**

A Warning Dialog Box will appear telling the user that they are about to initiate the remote fetch.

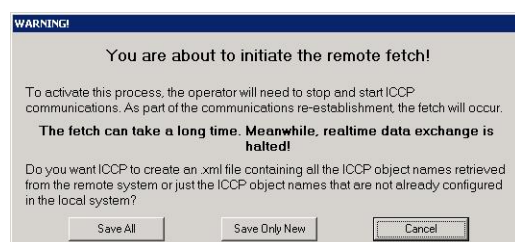


Figure 3-5 Remote Fetch Warning Dialog Box

The user can choose one of three choices:

- Save All
complete .xml file created of all objects
- Save Only New
.xml file containing just the objects that ICCP does not already know about (objects not already present in the local database)
- Cancel

This trigger only notifies ICCP that it needs to perform the fetch the next time the link comes up

The following figure shows where the .xml files are saved on the ICCP Master server.

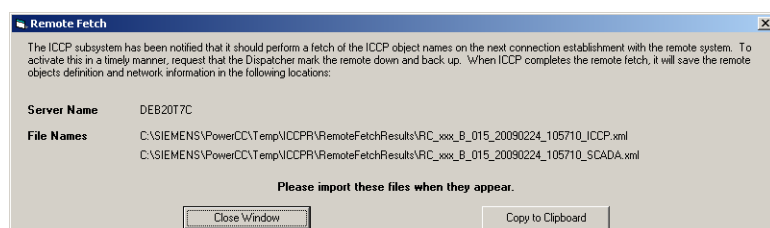


Figure 3-6 Remote Fetch XML Files Display

Activating Remote Fetch

The data engineer requests the operator/dispatcher to turn communications with the specific remote system off and then back on. This is the real trigger for ICCP to retrieve the information.

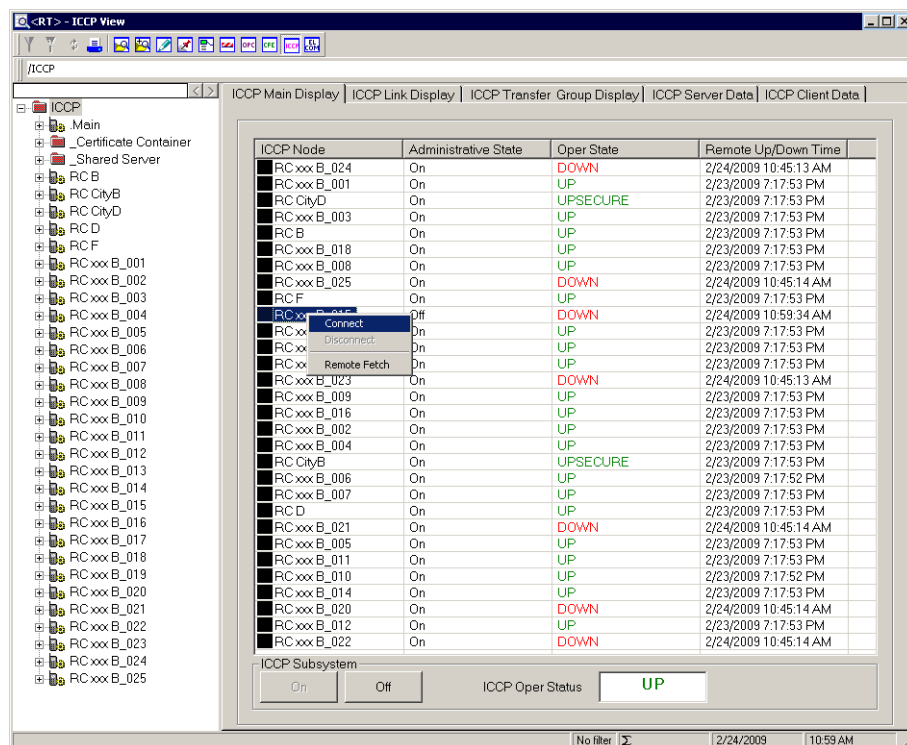


Figure 3-7 Disconnect from Remote System

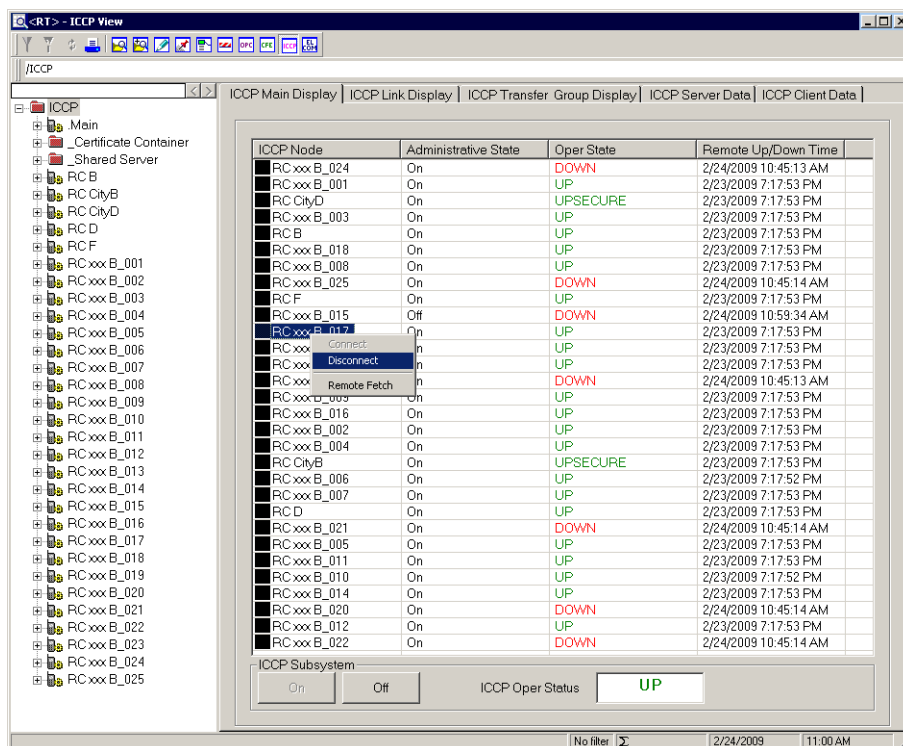


Figure 3-8 Reconnect to Remote System

- When communication with the remote system is re-established, one of the first things ICCP will do is fetch the object names and types from the remote server. Depending on the selection made by the data engineer, ICCP will either create two .xml files containing all objects, or two .xml files of just the new objects. One .xml file contains a simple 'pseudo' network model which will link with the ICCP objects. The second file contains the ICCP object information
- ICCP writes the .xml files to the following directory on the 'Master' ICCP server where the ICCPR process for the remote is running: %SR5_TEMP%\ICCP\RemoteFetchResults\[xml file name]. The file names are 'remote host name'_'date'_ICCP.xml and remote host name'_'date'_SCADA.xml. The 'date' is the timestamp of when the remote fetch that created the files was initiated. The _SCADA indicates the file contains the 'pseudo' network model and _ICCP indicates the file contains the ICCP Client information.

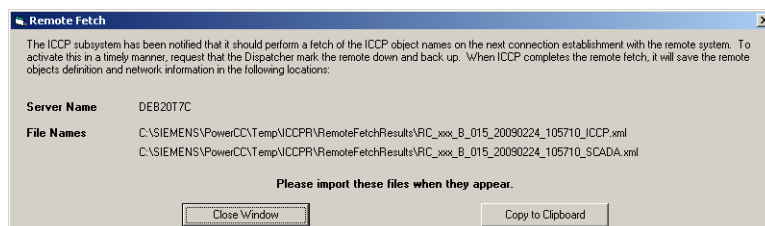


Figure 3-9 Remote Fetch XML Files Dialog

**Note**

The .xml files should be written to a directory on the Master ICCP server.

Applying Resulting XML files

- The data engineer will go to IMM and create a job. Within the job, the data engineer will first import, from the indicated directory, the SCADA.xml followed by the ICCP.xml. After the import is complete, the data engineer can prepare and activate the job.

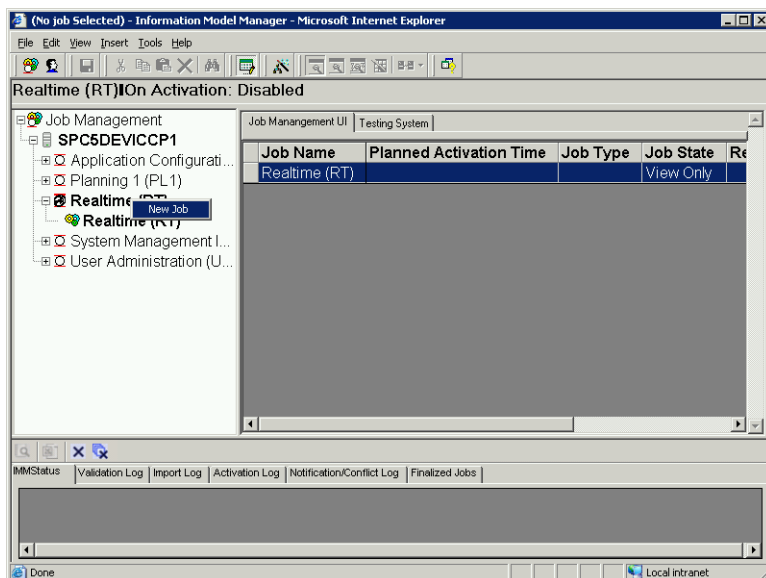


Figure 3-10 Create New IMM Job

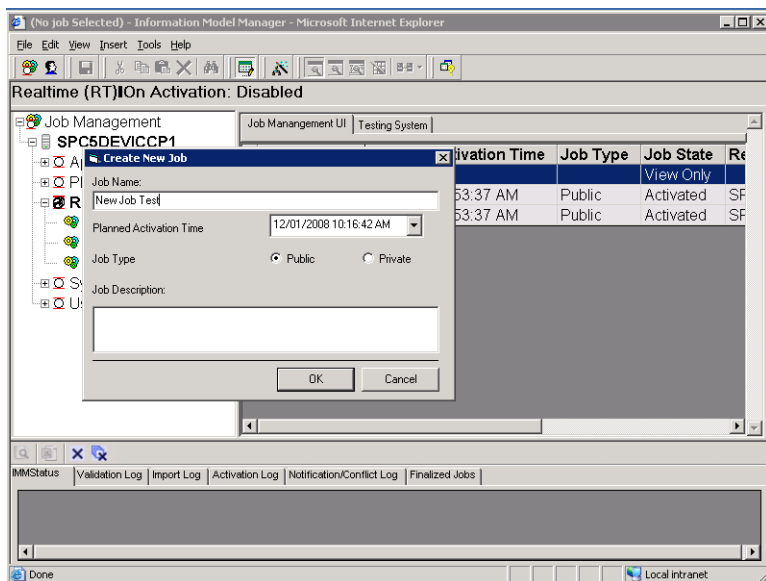


Figure 3-11 IMM New Job Creation

Once the job has been activated, it is up to the data engineer to move and backup the .xml file.

Viewing Data Using Runtime Explorer

- The imported xml files include a 'pseudo' substation inserted in the network model for the ICCP points coming from the specific remote. This substation was derived from the ICCP point name and ICCP point

type. An easy way to check out the values coming from the remote is to view these values in the Runtime Explorer using the Substation View.

The location of the Substation is a fixed location: **Net-ICCP > Substations ICCP > [remote host name]**.

Creating IMM Job

- The data engineer creates IMM job to modify associations of IcClientBlock1And2Object instances so they link to final location in Network Model.

When you are ready to have ICCP update the value of the true measurement within the Network Model, the user will create another job in IMM.

In this job, they need to modify the associations of the measurement from IcClientBlock1And2Object instance. Instead of having it associated with the measurement in the simple station, they modify the association so it is linked to the appropriate measurement in the Network Model.

3.4 ICCP Tag Creation for Operator

Creating a tag to send across ICCP is no different than creating a tag on other systems. It is important to note that only the 'Title' text gets propagated to the remote system.

For additional information, refer to [6.5.1 Tag Modification](#).

Here is an example of how to set a tag:

Setting the Tag

- From the spectrum PowerCC display,
- Open **Runtime Explorer > Substation View – RO > Net-E**
- Expand **Substations Generation > Sub Gen A**
- Select **Bay Gen 11**
- Select measurement and Right Click, choose **Apply Marker**

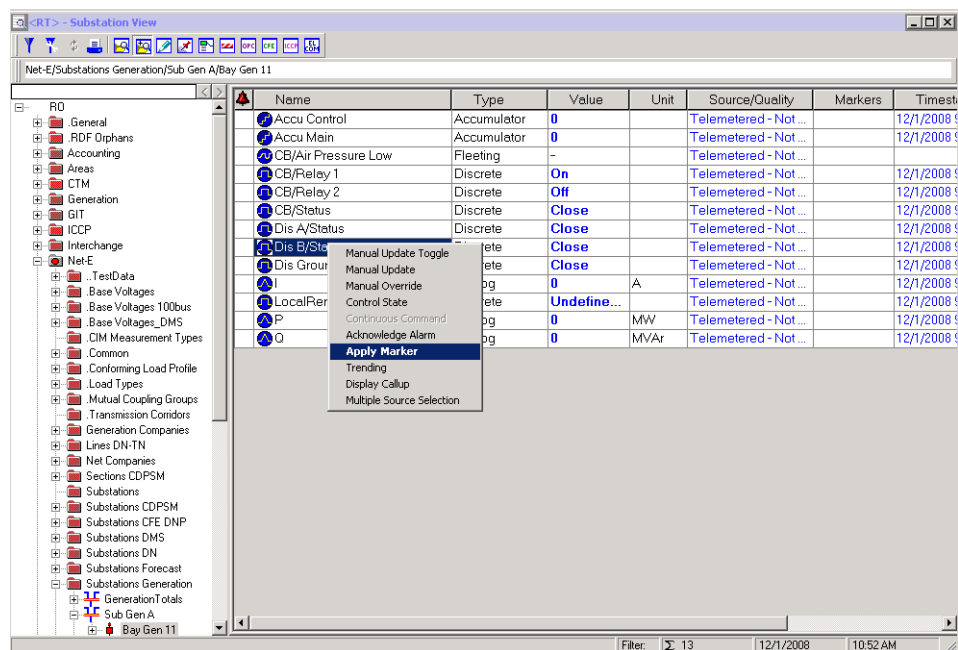


Figure 3-12 Step One of Creating Tags

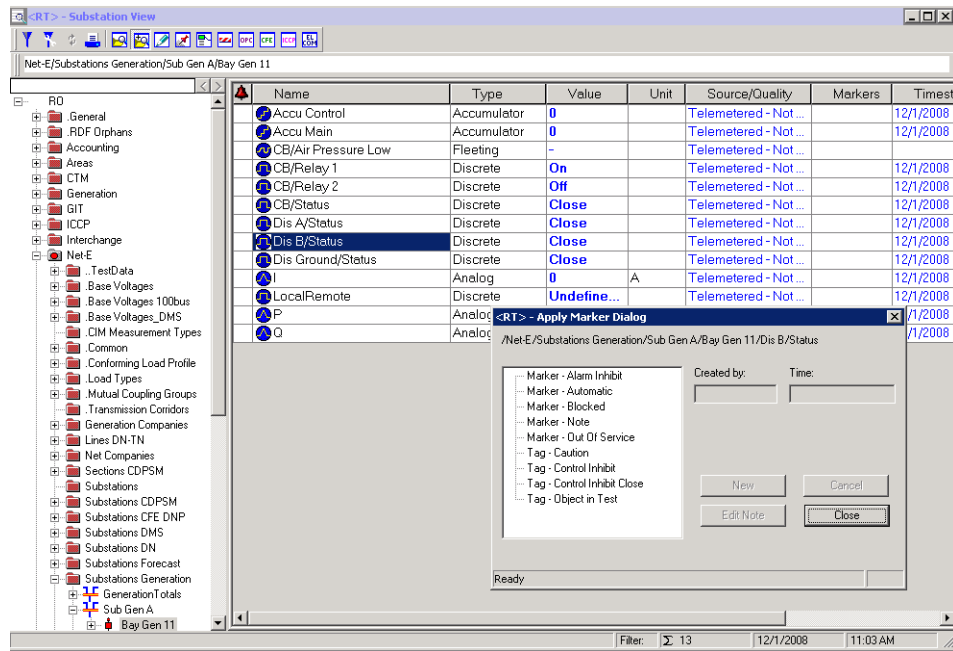


Figure 3-13 Step Two of Creating Tags

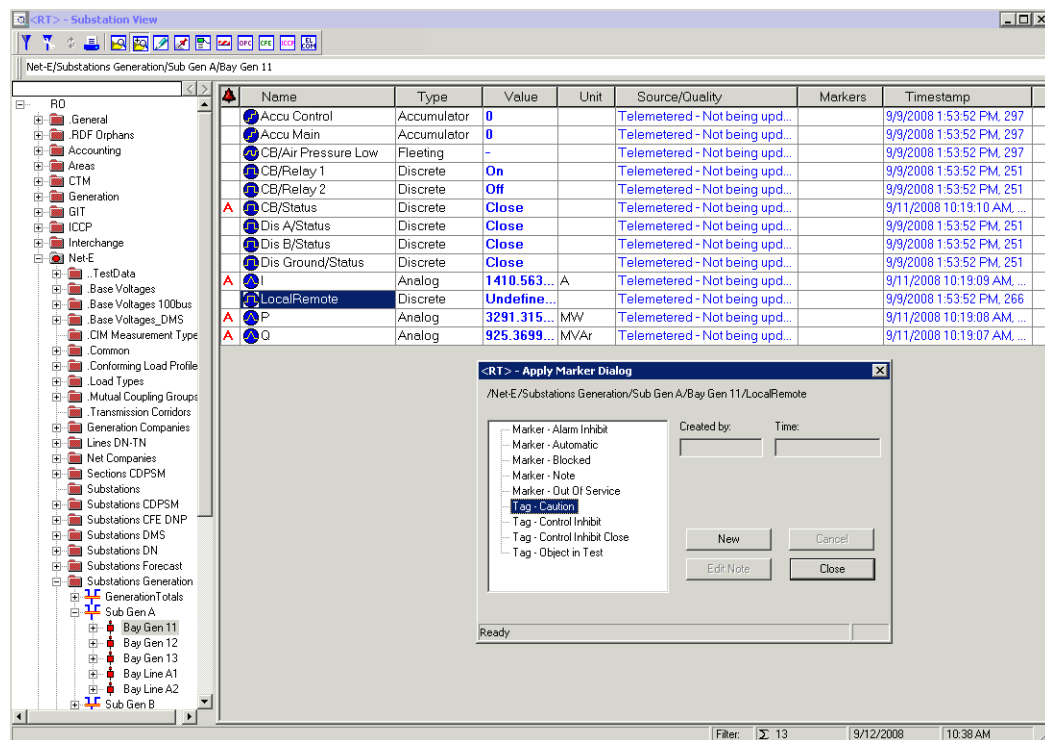


Figure 3-14 Step Three of Creating Tags

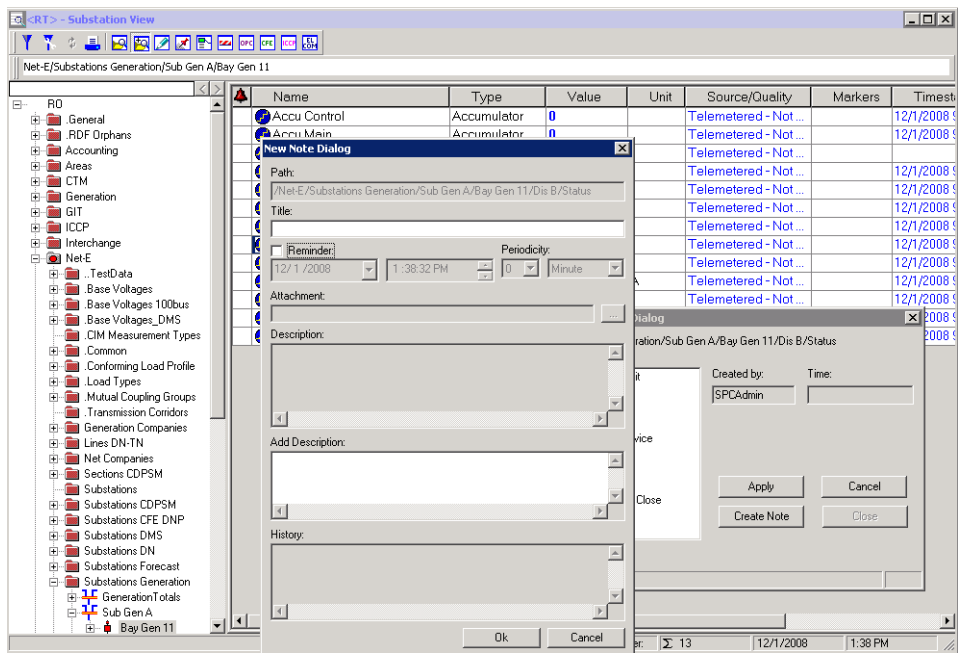


Figure 3-15 Step Four of Creating Tags

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4.1 Setup of the ICCP User Interface

The ICCP runtime user interface (UI) is invoked through the Runtime Explorer user interface of Spectrum PowerCC.

When the ICCP UI is invoked, it is displayed on the Work Pane of the Runtime Explorer. Operators may use the various user interface displays to:

- Enable and disable the ICCP subsystem
- Connect and disconnect ICCP nodes (set the administrative state)
- Control the client transfer groups configured for a remote node.

The ICCP runtime user interface consists of a number of tabbed pages each of which allows the user to access a particular part of the ICCP functionality.

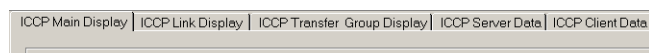


Figure 4-1 ICCP Display Structure

4.2 ICCP Runtime User Interface

4.2.1 Accessing ICCP Functionality

The ICCP user interface is integrated in the Runtime Explorer. The Spectrum PowerCC Runtime Explorer is an application that presents process data in a tabular format.

- Open the Runtime Explorer.
- Click the **ICCP UI** button on the toolbar of the Runtime Explorer.

The **ICCP Main Display** opens.

4.2.2 ICCP Main Display

4.2.2.1 Overview

The **ICCP Main Display** shows the overall administrative and operational status and the administrative and operational status for each ICCP remote.

It provides the operator with the capability to enable/disable ICCP and to connect/disconnect with the listed remote systems.

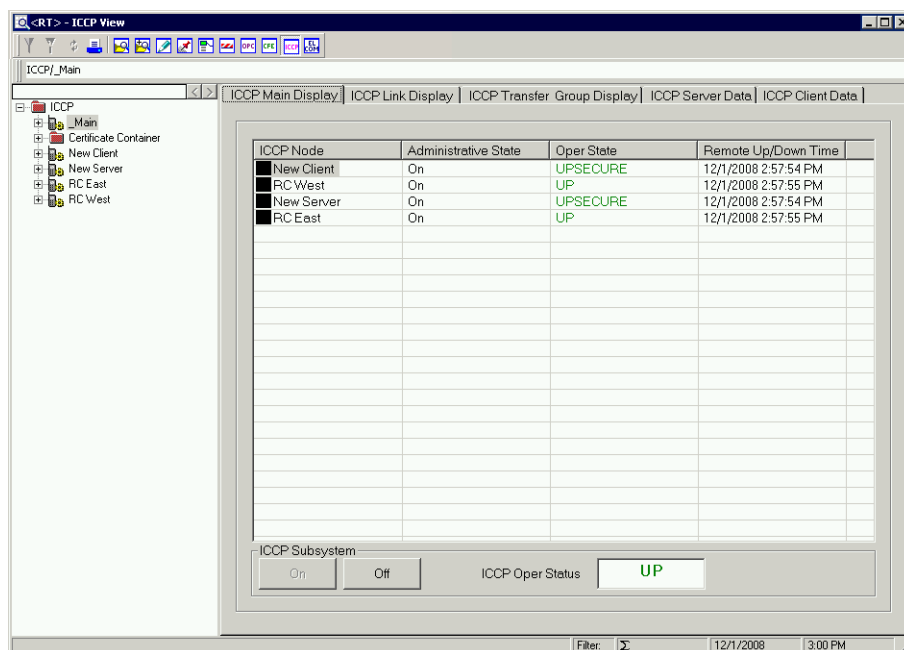


Figure 4-2 ICCP Main Display Window

The **ICCP Main Display** consists of the following components:

- Remote Control Center List
- ICCP Subsystem Area
- Connection Control Pop-up Menu

4.2.2.2 Remote Control Center List

ICCP Node

Name of the Remote control center with the ICCP capability. Right-clicking each remote control center opens the **Connection Control** context menu.

Administrative State

Displays the state desired by the operator. Valid values:

State	Description
On (displayed in green color)	ICCP will attempt to establish a connection with the remote node, either by initiating the connection if the local system is the initiator, or posting a 'listen' if its not the initiator.
Off (displayed in red color)	No attempts are made at establishing a connection, and any established connections are broken.

Operational State

Reflects the actual state of the denoted remote control center. Valid Values:

State	Description
UP (displayed in green color)	An ICCP association is established with a remote control center.
DOWN (displayed in red color)	No ICCP association is established
UPSECURE (displayed in green color)	An ICCP association is established with a remote control center and the connection is encrypted (a TLS connection with encryption was established). If multiple associations are defined for this remote (not the normal configuration), only if all associations are UPSECURE, will the remote operational status show UPSECURE.

Remote Up/Down Time

Shows the time when the operational status of the remote control center has been changed.

4.2.2.3 ICCP Subsystem Area

On

Turns the ICCP Main application on. Clicking the **On** button enables the ICCP subsystem.

Off

Turns the ICCP Main application off. Clicking the **Off** button disables the ICCP subsystem.



Note

The **On** button is disabled when the ICCP subsystem is enabled.

The **Off** button is disabled when the ICCP subsystem is disabled.

ICCP Status

A read-only field that shows the current status of the ICCP subsystem. Valid Values:

State	Description
UP (displayed in green color)	ICCP is running.
DOWN (displayed in red color)	ICCP is not running.

4.2.2.4 Connection Control Context Menu

Right-clicking on the list entries of the Remote Control Center opens the **Connection Control** context menu on the ICCP Main Display.

This pop-up menu contains 3 menu items:

- Connect
- Disconnect
- Remote Fetch

Using these functions, you can change the administrative status of the selected ICCP remote.

Connect

Changes the administrative status of the selected remote control center from **Off** to **On**.

Disconnect

Changes the administrative status of the selected remote control center from **On** to **Off**.

Remote fetch

Allows the operator to fetch new or updated XML objects.

4.2.3 ICCP Link Display

The ICCP Link Display shows the operational status of each ICCP link with the remote system.

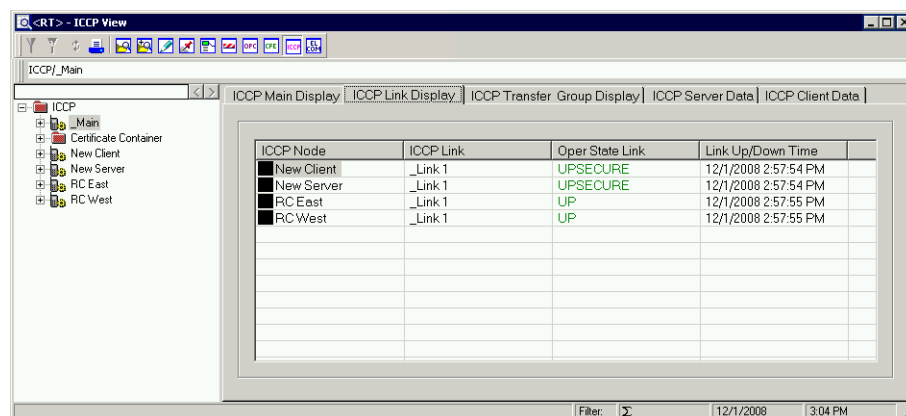


Figure 4-3 ICCP Link Display Window

ICCP Node

Name of the remote control center that has ICCP capability.

ICCP Link

Communication name between any two remote control centers.

Operational State

Reflects the actual state of the denoted ICCP link. When the ICCP link failed, the operational state is DOWN (red color).

Otherwise, it is UP (green color).

4.2.4 ICCP Transfer Group Display

The ICCP Transfer Group display shows the administrative and operational status of **Transfer Groups** for the selected remote control center.

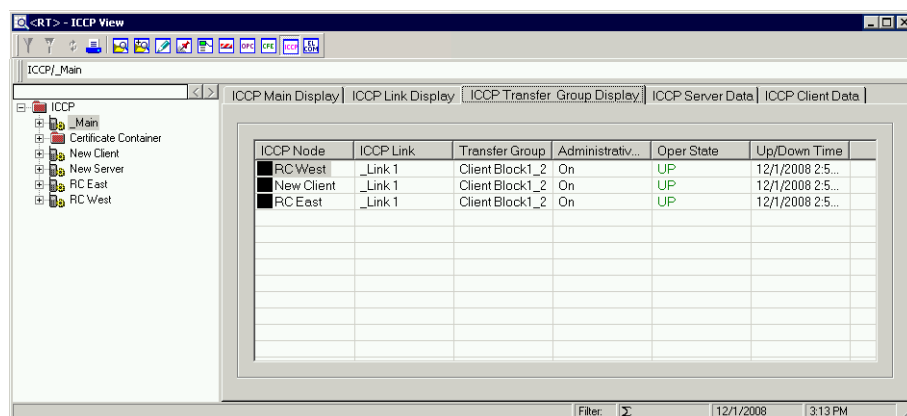


Figure 4-4 ICCP Transfer Group Display Window

ICCP Node

Name of the remote control center that has ICCP capability.

ICCP Link

Communication name between any two remote control centers. Each remote control center may have multiple links.

Transfer Group

ICCP Client data is configured in "containers" called **transfer groups**. A transfer group is defined on the client side and consists of

- the points to retrieve from the server (one or more data objects) and
- the characteristics that control when the data will be transferred from the server.

After the client connects to the server, it writes these characteristics to the server. The server is then supposed to send the data back to the client as defined by these characteristics.

Actually, a transfer group will be broken up into one or more ICCP transfer sets.

Only the locally defined client transfer groups used to retrieve data from the remote control center are displayed.

Administrative State

A state desired by the operator.

Right-clicking on any remote control center name will open a menu containing two items

- Connect
If it is enabled to change the Administrative state from **Off** to **On**.
- Disconnect
If it is enabled to change the Administrative state from **On** to **Off**.

Operational State

Reflects the actual state of that transfer group. Valid Values:

State	Description
Up (displayed in green color)	Any of the data for a transfer group were received.
Down (displayed in red color)	None of the data for a transfer group were received.

4.2.5 ICCP Server Data

The ICCP Server Data Display shows data and values for the server of the selected remote control center.

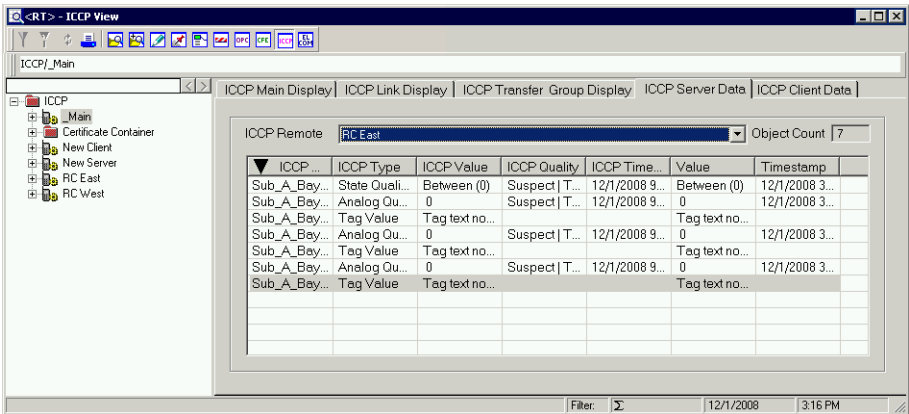


Figure 4-5 ICCP Server Data Display

After selecting the tab, the operator will need to select the remote system to view. Once it is selected, the server data for that remote will appear.

The display is a combination of static and dynamic information. Information such as the ICCP Name and ICCP Type is static configuration information. This information will not change unless there is a database change.

Most of the other columns contain dynamic information. The columns with ICCP in the name, such as ICCP Value, come from the ICCP application. The data for column headers that do not include ICCP, such as Value, come from the DDS SCADA event. In most cases the values should be the same. The exception is the SCADA data does not fit in the ICCP Data type, so this information is lost. If the ICCP type for an ICCP data object does not include a Quality or Timestamp, the display shows 'Not Available' for the ICCP column.

4.2.6 ICCP Client Data Display

The ICCP Client Data Display shows data and values for the client side of the selected remote control center.

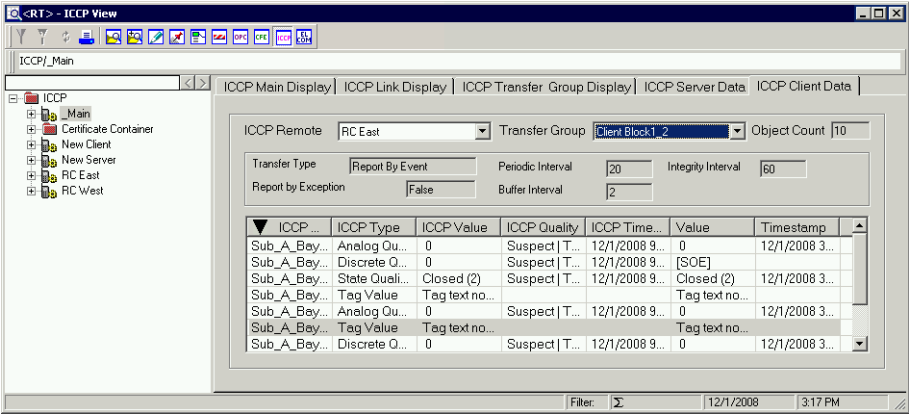


Figure 4-6 ICCP Client Data Display

After selecting the tab, the operator will need to select the remote system to view. Then the transfer group should be selected from the “Transfer Group” list for the selected remote.

Like the Server Data Display, this display includes static and dynamic information. The grayed out fields near the top of the display is static configuration information on the transfer group. Some of the columns information also contains static configuration information, such as the ICCP Name and ICCP Type. This information will not change unless a database change is activated.

Most of the remaining fields contain dynamic information. The fields with ICCP in the column header name, such as ICCP Value, is data from the ICCP application. Data in the columns that do not include ICCP in the column header names, such as Value, comes from the DDS SCADA event.

The data in the ICCP Value and Value columns may be different. The ICCP Value shows the value as it arrived from the remote, before the slope and offset has been applied or type conversion has taken place. The values may also be different if deadband has been configured for the point, and the received ICCP value change does not exceed the deadband, so it has not been fired to SCADA.

5 Engineering

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5.1 Introduction

The Inter-Control Center Communications Protocol (ICCP) is designed to allow data exchange over a Wide Area Network (WAN), a Local Area Network (LAN), etc. between utility control centers. Data exchanged include real-time power system information, such as measured analog values. An individual control center, in its interactions with other control centers, may act as client, server or, both.

ICCP is made up of several components or functions called "Blocks". Block 1 is required by all ICCP implementations. All remaining Blocks are optional. Spectrum PowerCC ICCP currently supports Blocks 1, 2, and 5.

These Blocks are:

- Block 1 – Basic Services

This block function is used to exchange the real-time power system data between Spectrum PowerCC and the remote systems.

- Block 2 – Conditions

This block function is used to exchange the real-time data on exception between Spectrum PowerCC and the remote systems.

- Block 5 – Supervisory Control

The ICCP Subsystem receives control requests (initiated by an operator or application) from the Supervisory Control subsystem. These requests result in device control messages to be sent to a Remote Control Center. Block 5 control messages received from the Remote Control Center are passed to the Supervisory Control subsystem for processing.

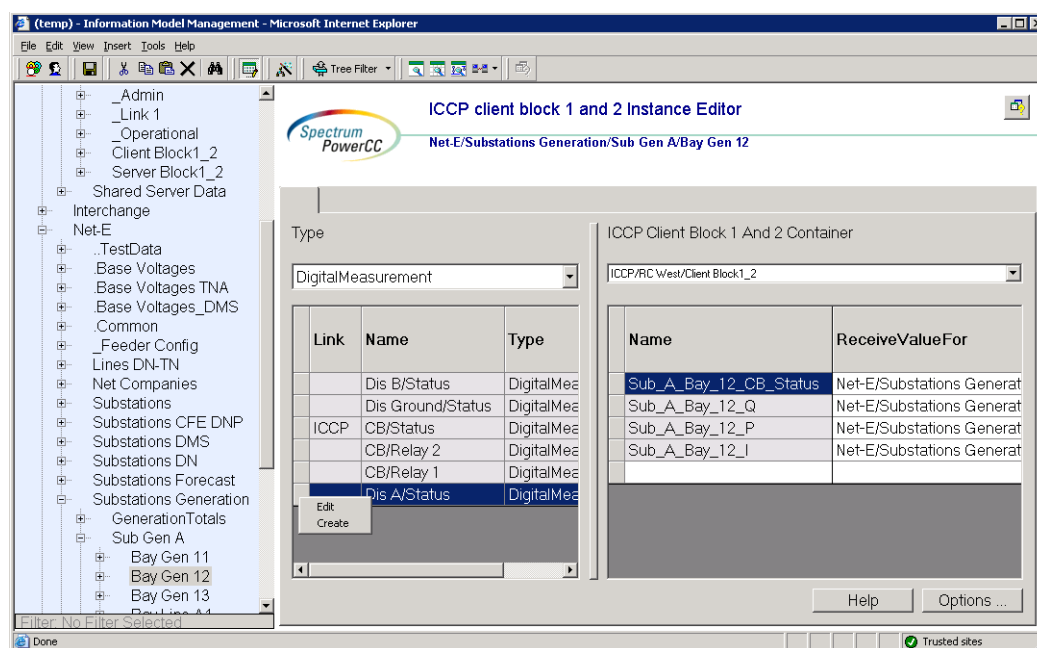


Figure 5-1 ICCP Configuration Window

ICCP Variable Name

Each data object that is passed between the local and remote control centers via ICCP must be identified by a unique name. This identification is called the "Variable Name".

Within the Spectrum PowerCC Engineering System, the Variable Name of the ICCP data object is the instance name of the Client or Server object. A Variable Name (and thus the instance name) must conform to these rules:

- Length of 1 to 32 characters
- Legal characters are: 'A' through 'Z'; 'a' through 'z'; '0' through '9', '\$', and '_'.
- A blank is NOT allowed (cannot be included in name)
- Names shall not begin with a digit.

For a measurement value exchanged between two control centers (between A and B), the ICCP Variable name must match exactly. If the same measurement is to be exchanged with a third control center (between A and C) a different ICCP Variable name can be used.

ICCP Data Entry

Spectrum PowerCC provides a particular kind of Table Editor to simplify the data entry and handling of data related to ICCP Block 1,2, and 5. These kinds of data are ICCP Server Data Objects and ICCP Transfer (Client) Data Objects. The editor provided to configure these data is the Communication Editor for ICCP (refer to [5.3.2 Accessing the Communication Editor for ICCP](#)).

5.2 ICCP Configuration Data Types and Attributes

5.2.1 ICCP Configuration

In Spectrum PowerCC ICCP, the information required to configure ICCP includes the remote/link configuration, the Bilateral Agreements, and the local client transfer conditions configuration.

They are grouped into the following data entities:

- **Remote Center**
This defines a Remote Control Center that communicates with the local Spectrum PowerCC ICCP system. This includes information such as the Bilateral Table Version, Remote Control Center Name, Remote Control Center Type, and Time-out parameters. The shortest periodic interval allowed by the remote is also included here.
- **Link Data**
This defines the addresses required to establish an association between the local Spectrum PowerCC ICCP system and a remote ICCP system.
- **Server Objects**
These are the data objects specified in the Bilateral Agreement as available for the specified Remote Control Centers. Each data object is associated with a measurement within the local Spectrum PowerCC ICCP system. Their ICCP name, type, and access permission by a Remote Control Center are also identified. Since only accessible objects are specified here, the access permission can be read, write, or both. Configuring these objects is equivalent to configuring the ICCP server function.
- **Client Transfer Groups**
These specify the transfer conditions for a group of data objects that are to be received by the local Spectrum PowerCC ICCP system from a Remote Control Center. These transfer conditions specify whether the data is reported periodically or by exception along with the required information for each case.
- **Client Objects**
These are data items that are transferred from a Remote Control Center to the Local Spectrum PowerCC system. Each Client Object is assigned a measurement. This identifier identifies where the Client Object values are stored within the local Spectrum PowerCC system when they are received from a Remote Control Center. Each Client Object is also associated with a transfer group specifying the transfer conditions from the remote system. Configuring these objects is equivalent to configuring the ICCP client function.

Changes in the configuration of ICCP Server Objects will not require a restart of the ICCP program that is supporting communications with the specific Remote. Changes to any other configuration information about the Remote will result in an automatic restart of the ICCP program.

5.2.2 ICCP Data Model

The Spectrum PowerCC Data model is broken up into categories which allow configuration data to be organized based on the subsystem it belongs to. Most of the configuration data relating to ICCP is entered in the sysICCPCategory. Furthermore, a consistent naming convention is implemented in the model for the data types. All of the configuration data types in the data model of ICCP start with 'Ic'.

Configuration Data

ICCP program gets its configuration data from the ICCP ASR.

Table 5-1 ICCP Configuration Data Types

ICCP Type name	Object Description
IcMain	The ICCP Main object is a place holder for attaching measurements representing the overall status of the ICCP application. ICCP uses digital measurements to represent administrative states and operational statuses.
IcRemoteCenter	A Remote Center is an ICCP communication peer. The ICCP peer is typically another control center belonging to a different utility. It may be a client of the local ICCP application, a server to the local ICCP application or both.
IcLink	An ICCP link identifies an association with the Remote Center.
IcLocalAR	It identifies uniquely the local endpoint of exactly one link to a peer. It is necessitated by the administrator of a peer system wishing to connect to the local system.
IcRemoteAR	Each IcRemoteAR identifies the addresses of a peer to which your system will have connections. The information in an IcRemoteAR must match exactly with the information in the IcLocalAR (or equivalent structure) on the peer system.
IcCertificate	Digital certificate used for secure ICCP links. This class is used to specify either a Local Server certificate, a Local AR certificate, a remote AR certificate, or a Certificate Authority (CA) public key certificate.
IcClientBlock1And2Container	Container of client data objects that are requested from a remote ICCP peer as a group. It has attributes that describe transfer conditions such as periodicity, integrity interval, etc.
IcClientBlock1And2Object	This represents a SCADA data point requested from a remote ICCP peer.
IcServerBlock1And2Container	Container of server data objects that are available to the remote ICCP peer.
IcServerBlock1And2Object	This represents a SCADA data point that can be served to a remote ICCP peer.
IcClientBlock5Container	Container of client block 5 control objects.
IcClientBlock5Object	This represents the object that is used for sending control to the remote server.
IcServerBlock5Container	Container of server block 5 control objects.
IcServerBlock5Object	This represents the object that a control from the remote client is acted on.

For additional information on data engineering activities related with ICCP, refer to [Reference Guide ICCP Protocol, \(ICCP\)](#).

5.2.3 ICCP Configuration Data Types and Attributes

The below tables provide a detailed description of the purpose of all ICCP data types as well as their attributes and associations (if any).



Note

The order in which the ICCP data types are listed reflects exactly the order in which they are typically defined in the IMM.

ICCP Main Type

- IcMain

The ICCP Main object is a place holder for attaching measurements representing the overall status of the ICCP application. ICCP uses digital measurements to represent administrative states and operational statuses.

- Associations
AdminState, OperatingStatus

Attribute Label	Possible Values	Description
Alias Name	String	Free text name of the object or instance
Area Of Responsibility Id	- Area A - Area A Gas - Area A Water - Area B - Area B Gas - Area B Water - Area C - Area C Gas - Area C Water - PA - System	Identification of an Area of Responsibility
Exchange Name	String	NERC data exchange name imported from a CIM RDF file
Message Configuration	- Default - E_Message - E_Prio1_Aa - ... etc. - H_Prio3_AaDa - Msg Not Recorded	Unique identification number of the Message Configuration object
Path Name	String (max. 500 characters).	Concatenation of all names from each container

ICCP Remote Center

- IcRemoteCenter
A Remote Center is an ICCP communication peer. The ICCP peer is another control center, typically belonging to a different utility. It may be a client of the local ICCP application, a server to the local ICCP application or both.
- Associations
AdminState, OperatingStatus

Attribute Label	Possible Values	Description
Alias Name	String	Free text name of the object or instance
Application Process Name	String (max. 100 char.)	Name of the application process in System Configuration defined for the remote center As long as the name is not assigned, the remote center will not be connected.
Area Of Responsibility Id	- Area A - Area A Gas - Area A Water - Area B - Area B Gas - Area B Water - Area C - Area C Gas - Area C Water - PA - System	Identification of an Area of Responsibility

Attribute Label	Possible Values	Description
Association Initiator	True, False Default: False	If Yes , then the local (i.e., your) system will initiate associations. If No , the remote system is expected to initiate associations. Since exactly one system should be the initiator, talk to the administrator of the remote peer and decide whether it is to be your system or their system. It is important that exactly one of the communications partners act as the association initiator. More than one will result in (difficult to diagnose) problems. Less than one will result in no creation of an association. NOTE: Normally, the client is the association initiator.
Association Request Retry	Integer number Default: 30	This is the number of seconds between attempts to create an association, i.e., number of seconds to wait after an attempt fails before trying again. See also Association Request Time-out. This is only meaningful if Association Initiator is Yes .
Association Request Timeout	Integer number Default: 30	This is the number of seconds to wait before timing out an attempt to initiate an association. That is, if your system is the association initiator, it will send a message to the peer to attempt to create an association. If a reply is not received within "Association Request Time-out" number of seconds, your system will interpret this as a failure to create the association. It will then wait "Association Request Retry" number of seconds before trying again. This is only meaningful if Association Initiator is Yes .
Bilateral Table Version	String (max. 100 char.)	This is the bilateral table version if the remote peer has implemented ICCP security features. It must exactly match the bilateral table version of the remote peer. Presumably, this matches the version of a Bilateral Agreement (i.e., on paper) between the two control centers.
Client Response Timeout	Integer number Default: 30	When the ICCP client receives either a select or operate request from a supervisory control application and sends the request to the ICCP server then this timer is begun. The acknowledgment of the request from the ICCP server must occur within this time.
Communication Direction	Both server and client Client only Server only	Indicates whether the local system acts as server only, client only, or both server and client
Communication Protocol	TCP/IP, OSI.	Protocol of communication used for all links within the current remote source, OSI or TCP/IP
Default Dead Band Value	Floating point number. Default: 0.	This value will be used by ICCP for all incoming analogs if no dead band values are specified for the individual objects.
Enable Block4	True, False Default: False	This refers to the ICCP protocol writing transfer conditions when the link is associated. When selected block 4 data may be exchanged. Block 4 is currently not supported.
Enable Block5	True, False Default: True	If unchecked the ICCPNT ICCP remote will return "value device not defined" for any request to select or operate a block 5 object. When selected block 5 data may be exchanged.

Attribute Label	Possible Values	Description
Enable Block8	True, False Default: False	Indicates whether or not block 8 transfers are enabled for the remote When selected block 8 data may be exchanged. Block 8 is currently not supported.
Exchange Name	String	NERC data exchange name imported from a CIM RDF file
ICCPVersion	1=1996-08 2=2008-08 Default:1	ICCP (TASE.2) version support by remote control center
Link Heartbeat	True, False Default: True	Indicates whether or not the link heartbeat monitor is enabled for this remote
Link Heartbeat Interval	Integer number Default: 120	Specifies the number of seconds between heartbeat checks, if the heartbeat monitor is enabled. This only determines when the heartbeat is sent. If a heartbeat confirm is not received from the partner by the period specified by the Server Reply Timeout attribute, the connection will be marked down.
Local Domain Name	String (max. 32 char.)	This is the name of the remote system's domain at 'this' system. It is sent by the remote system to access domain scope data on this system. Normally it would be a name which identifies the remote control center. For example, if the local control center is named 'ABC' and the remote control center is named 'XYZ', the local domain name would be 'XYZ'. NOTE: this is only an example configuration. Some systems may use other conventions for naming their domains. The exact names to be used must be negotiated as a part of the bilateral agreement. To avoid opportunities for problems, we recommend that this name be taken from your bilateral agreement(s).
Maximum Data Sets	Integer number Default: 400	Maximum number of data sets allowed per link ICCP divides transfer groups into data sets based on the negotiated PDU size for MMS reports. Small transfer groups may require only one data set, but larger sets must be broken up for delivery.
Maximum PDUSize	Integer number Default: 32000	Maximum PDU size for MMS reports, in bytes. The actual PDU size is negotiated between the communicating peers, and may be less than the maximum. The negotiated PDU size is used to break up large data communications into manageable units for delivery.
Message Configuration	- Default - E_Message - E_Prio1_Aa - ... etc. - H_Prio3_AaDa - Msg Not Recorded	Unique identification number of the Message Configuration object

Attribute Label	Possible Values	Description
Minimum Periodic Interval	Integer number Default: 5	An ICCP peer may request periodic data transfers at an interval specified by the remote. This parameter is the shortest period interval (in seconds) which will be accepted by your system. This parameter is used to prevent an improperly configured remote ICCP peer from overloading your system with data requests. Assume, for example, you have set this value to 30 seconds. Then assume that a client creates a transfer group with a periodicity of 10 seconds. ICCPNT will detect this violation, create a log message, and automatically adjust the periodicity for the transfer group to 30 seconds.
Path Name	String (max. 500 char.)	Concatenation of all names from each container
Remote Domain Name	String (max. 32 char.)	This is the name of this host's MMS domain at the remote system. It is sent to the remote system to access domain scope data on the other system. Normally it would be a name which identifies this control center at the remote system. For example, if the local control center is named 'ABC' and the remote control center is named 'XYZ', the remote domain name would be 'ABC'. NOTE: this is only an example configuration. Some systems may use other conventions for naming their domains (host names for example). The exact names to be used must be negotiated as a part of the bilateral agreement. This name is exchanged via ICCP with the remote system, so depending upon how the remote ICCP implementation functions, it may be required to match a name configured on the remote system.
SBOTimeout	Integer number Default: 120	This timer is begun by the ICCP client after receiving a select request from a supervisory control application and sending the select request to a server. The operate request from the custom application must be received by the ICCP client and sent to the server within this time. It is the responsibility of the supervisory control application to send both the select and operate requests to the ICCP client. This timer is also begun by the ICCP server after it has received the select request from a client and sent the acknowledgment of the select request to the client. The operate request must be received from the client within this time.
Server Reply Timeout	Integer number Default: 120	This is the number of seconds after a message is sent to a server (peer) to wait before timing out the message. If a "Server Reply Time-out" number of seconds pass without a response from the peer being received, either the remote peer or communications to the remote peer have failed and the association is terminated.

Attribute Label	Possible Values	Description
Server Response Timeout	Integer number Default: 10	This time-out value is exclusively used by the ICCP server. The timer is begun after the ICCP client sends a select or operate request and the ICCP server receives the request and sends the request to a supervisory control application. The acknowledgment of the request by the supervisory control application must occur within this time. For smooth operation of ICCP, it is recommended that this time-out value should be less than the ICCP client's Client Response Timeout value.
TagTextCharacterSet	Default: No_CharacterSet_applied	It may be necessary to encode text associated with Tags. If the text is always Latin characters, no character set encoding is needed.

ICCP Link

- IcLink
An ICCP link identifies an association with the Remote Center (i.e. a MMS communication channel used for sending/receiving data to/from an external communication partner).
- Association
OperatingStatus

Attribute Label	Possible Values	Description
Alias Name	Alphanumerical string. Max. length: 100 characters	Free text name of the object or instance
Application Authentication Mode	- Any - No - Yes	Identifies the type of Application Level Authentication to be used on this ICCP link NOTE: Any only applies when the associate icRemote has "Association Initiator" = FALSE.
Area of Responsibility Id	- Area A - Area A Gas - Area A Water - Area B - Area B Gas - Area B Water - Area C - Area C Gas - Area C Water - PA - System	Identification of Area of Responsibility
Critical Data Timeout	Period of time in seconds Default: 20 (this value cannot be changed)	In ICCP, data which declared as critical will receive an application level acknowledgment. This parameter specifies how long the local application should wait for acknowledgment before declaring that the data was not received. The value selected should take into consideration the speed of the link to the remote. It is not implemented in the current release of ICCP. The TASE.2 protocol is unclear about the obligation of the server when the acknowledgment is not received. Siemens has therefore chosen not to implement a feature to handle the situation when an acknowledgment does not arrive.
Exchange Name	String Max. length: 1000 characters	NERC data exchange name imported from a CIM RDF file

Attribute Label	Possible Values	Description
Message Configuration	- Default - E_Message - E_Prio1_Aa - ... etc. - H_Prio3_AaDa - Msg Not Recorded	Assigns the Summary to a message configuration for alarming/message recording purposes
Service Quality	- High quality of service - Low quality of service - All quality of service only one link is allowed - Low and high quality of service - High and other quality of service - Low and other quality of service - Other.	Quality of Service parameter to be associated with the link. Possibilities are: • All • Low • High • Other • Low and High • Other and Low • Other and High If you define the link as High, then only Transfer Groups defined with High quality of service (QOS) will be sent on this link. If you define it as Low and High, then both Low and High QOS transfer groups will be sent. Only one link can be defined for each priority. If All is specified, then only one link is allowed. If you are configuring only block 1 and 2 data then configure single link to a peer and use All for the QOS. For additional information on QOS, see Appendix 4-D.
Transport Layer Security (TLS) Mode	- Secure - Non Secure - Secure Or Non Secure	Identifies the type of Transport Layer Security (TLS) to be used on this ICCP link. (NOTE: SecureOrNonSecure only applies when the associate icRemote has "Association Initiator" = FALSE).

IcLocalAR

- **IcLocalAR**
IcLocalAR entry inherits from AR and contains information specific to local AR definitions. IcLocalAR uniquely identifies the local endpoint of exactly one link to a peer. These parameters must be provided to the administrator of a peer system, which wishes to connect to the local system.
- **Associations:**
Has (IcCertificate)

Attribute Label	Possible Values	Description
AEInvoke ID	Integer number Recommended value see the configuration file example Default: 1	AE Invoke ID For additional information, refer to a text on OSI addressing. This field is optional.
AEQualifier	Integer number Recommended value see the configuration file example Default: 1	AE Qualifier For additional information, refer to a text on OSI addressing. This field is optional.
APIInvoke ID	Integer number Recommended value see the configuration file example Default: 100	AP Invoke ID For additional information, refer to a text on OSI addressing. This field is optional.

Attribute Label	Possible Values	Description
APTitle	Array of up to 16 short decimal integers Recommended value see the configuration file example Default: "1 2 32 2" (without the quotes)	AP title This field is optional.
IPAddress	Alphanumeric string Max. length: 100 characters	A dotted decimal IP Address associated with the remote peer Used for TCP/IP communication protocol only
Presentation Selector	String consisting of up to 32 characters of ASCII encoded hexadecimal values. There must be an even number of digits. Valid: 02, 4444, 123456789A Not valid: 1, 300, 00005	Presentation selector for the peer (Psel) It can be chosen arbitrarily, but you need to tell the administrator of the peer system what value you have chosen. This field is mandatory. Note: The combination of Psel, Ssel and Tsel must be unique. While any value of the 3 may be modified, Siemens varies the Ssel field for its internal testing.
Service Quality	Default: 0	Quality of Service This field is currently not used (it is optional).
Session Selector	String consisting of up to 32 characters of ASCII encoded hexadecimal values. There must be an even number of digits. Valid: 02, 4444, 123456789A Not valid: 1, 300, 00005	Session Selector for the peer (Ssel) It can be chosen arbitrarily, but you have to tell the administrator of the peer system what value you have chosen. This field is mandatory. Note: The combination of Psel, Ssel and Tsel must be unique. While any value of the 3 may be modified, Siemens varies the Ssel field for its internal testing.
Shared	True, False (Default)	Shared the local name with other applications, Yes or No . Default is N .
Subnetwork	Decimal integer. Default: 0.	Indicates the subnetwork number from the subnetwork section. This field is optional.
Transport	Default: "TCP" (without the quotes)	Specifies the Transport Provider Only used when the RemoteSource communication protocol is TCP/IP
Transport Selector	String consisting of up to 32 characters of ASCII encoded hexadecimal values. There must be an even number of digits. Valid: 02, 4444, 123456789A Not valid: 1, 300, 00005	Transport Selector for the peer (Tsel) It can be chosen arbitrarily, but you have to tell the administrator of the peer system what value you have chosen. This field is mandatory. Note: The combination of Psel, Ssel and Tsel must be unique. While any value of the 3 may be modified, Siemens varies the Ssel field for its internal testing.

IcRemoteAR

- IcRemoteAR

A RemoteAR entry inherits from AR and contains information specific to remote AR definitions. The RemoteAR identifies the addresses of all the peers to which your system will have connections. The information in an IcRemoteAR must match exactly with the information in the IcLocalAR (or equivalent structure) on the peer system (i.e. these parameters must match the analogous parameters used in configuring the peer - the administrator of the peer system should provide the parameters used in this section).

- Associations:
Has (IcCertificate)

Attribute Label	Possible Values	Description
AEInvoke ID	Integer number Recommended value see the configuration file example Default: 1	AE Invoke ID For additional information, refer to a text on OSI addressing. This field is optional.
AEQualifier	Integer number Recommended value see the configuration file example Default: 1	AE Qualifier For additional information, refer to a text on OSI addressing. This field is optional.
APInvoke ID	Integer number Recommended value see the configuration file example Default: 100	AP Invoke ID For additional information, refer to a text on OSI addressing. This field is optional.
APTitle	Array of up to 16 short decimal integers Recommended value see the configuration file example Default: "1 2 32 2" (without the quotes)	AP title This field is optional.
IPAddress	Alphanumerical string. Max. length: 100 characters	A dotted decimal IP Address associated with the remote peer Used for TCP/IP communication protocol only
NSAP	String consisting of ASCII encoded hexadecimal values characters. Max. length: 40 characters There must be an even number of digits. Valid: 02, 4444, 1234567890 Not valid: 1, 300, FFFFF0000	This attribute is used for OSI communication protocol only. The Network Service Access Point (NSAP) is one of two types of hierarchical addresses (the other type is the network entity title) used to implement Open Systems Interconnection (OSI) Network layer addressing. The NSAP is the logical point between the network and Transport layers where network services are delivered to the transport layer; the location of this point is identified to the OSI network service provider by the NSAP address.
Presentation Selector	String consisting of up to 32 characters of ASCII encoded hexadecimal values. There must be an even number of digits. Valid: 02, 4444, 123456789A Not valid: 1, 300, 00005	Presentation Selector for the peer (Psel). It can be chosen arbitrarily, but you need to tell the administrator of the peer system what value you have chosen. This field is mandatory. Note: The combination of Psel, Ssel and Tsel must be unique. While any value of the 3 may be modified, Siemens varies the Ssel field for its internal testing.
Service Quality	Default: 0	Quality of Service This field is currently not used (it is optional).
Session Selector	String consisting of up to 32 characters of ASCII encoded hexadecimal values. There must be an even number of digits. Valid: 02, 4444, 123456789A Not valid: 1, 300, 00005	Session Selector for the peer (Ssel). It can be chosen arbitrarily, but you have to tell the administrator of the peer system what value you have chosen. This field is mandatory. Note: The combination of Psel, Ssel and Tsel must be unique. While any value of the 3 may be modified, Siemens varies the Ssel field for its internal testing.
Static Route	Default: "False" (without the quotes) This is the only allowed value within the Siemens product.	Indicates that a remote entry's ES-IS routing record (MAC Address) must be configured. Use of static routing would require code changes. Used for OSI communication protocol only. *** Not visible.
Subnetwork	Decimal integer Default: 0	Indicates the subnetwork number from the subnetwork section This field is optional.

Attribute Label	Possible Values	Description
Subnetwork Point Of Attachment	True, False (Default).	This field is required only if StaticRoute is "Y". It contains the SubNetwork Point of Attachment (MAC Address). Used for OSI communication protocol only. *** Not visible.
TLS Port	Integer number [1025..20000] Default: 3782	Identifies the TCP port used to access the TLS Layer on the remote system
Transport	Default: "TCP" (without the quotes)	Specifies the Transport Provider. Only used when the RemoteSource communication protocol is TCP/IP.
Transport Selector	String consisting of up to 32 characters of ASCII encoded hexadecimal values. There must be an even number of digits. Valid: 02, 4444, 123456789A Not valid: 1, 300, 00005	Transport Selector for the peer (Tsel) It can be chosen arbitrarily, but you have to tell the administrator of the peer system what value you have chosen. This field is mandatory. Note: The combination of Psel, Ssel and Tsel must be unique. While any value of the 3 may be modified, Siemens varies the Ssel field for its internal testing.

ICCP Certificate Data Type

- IcCertificate

Digital certificate used for secure ICCP links. This data type is used to specify either a Local Server certificate, a Local AR certificate, a remote AR certificate, or a Certificate Authority (CA) public key certificate.

Attribute Label	Possible Values	Description
Certificate Type	- Local System Certificate - Certificate Authority Certificate - Local AR Certificate - Remote AR Certificate	Type of certificate For additional information, refer to 5.5.3 Security Configuration .
Issuer	Issuer string from the certificate	This value must match the issuer information of the associated certificate in the local system certificate store.
Serial Number	Serial number string from the certificate	This value must match the serial number of the associated certificate in the local system certificate store.
Subject	Subject string from the certificate	This value must match the Subject information of the associated certificate in the local system certificate store.

IcClientBlock1And2Container

- IcClientBlock1And2Container

Container of client data objects for Block 1 and 2 that are requested from a remote ICCP peer as a group. It has attributes that describe transfer conditions such as periodicity, integrity interval, etc.

- Associations

AdminState, OperatingStatus

Attribute Label	Possible Values	Description
Alias Name	String	Free text name of the object or instance
All Changes Reported	True, False (Default)	This attribute has meaning if 'Report by Exception' is True and Buffer Interval is non-zero. When these conditions are met, if this attribute is True, the server should report all changes to an object within the stated buffer period. If False, the server only needs to report the last change (current value). If the remote control center does not support ICCP version 2000-08 (or newer versions), this attribute is ignored.

Attribute Label	Possible Values	Description
Area Of Responsibility Id	Keyword to select using an option menu: - Area A - Area A Gas - Area A Water - ... Area C Water PI.	Identification of an Area of Responsibility
Buffer Interval	Integer number Default: 2	This specifies the amount of time in seconds to wait after the event (i.e., a change in the value of a data object in the transfer group) before the server initiates the transfer. This is used to delay transfer for a short time in case some other data values change also; commonly a change in one data object in a transfer group will trigger a change in one or more others. This minimizes multiple transfers which may result from the same event.
Critical Data	True, False (Default)	This specifies whether the data in this transfer group are to be considered critical. Receipt of critical data will be acknowledged (application layer acknowledgment) by the receiving ICCP application. (This value cannot be changed. It is not implemented in the current release of ICCP. The TASE.2 protocol is unclear about the obligation of the server when the acknowledgment is not received. Siemens has therefore chosen not to implement a feature to handle the situation when an acknowledgment does not arrive.)
Exchange Name	String	NERC data exchange name imported from a CIM RDF file.
Integrity Interval	Integer number Default: 1800	(This attribute is only meaningful if the Transfer Group is By Event or Periodic/Report By Exception.) It specifies the amount of time in seconds after which the transfer group will be transferred even if no data change occurs.
Message Configuration	- Default - E_Message - E_Prio1_Aa - ... etc. - H_Prio3_AaDa - Msg Not Recorded	Assigns the Summary to a message configuration for alarming/message recording purposes

Attribute Label	Possible Values	Description
Periodic Interval	Integer number Default: 60	This attribute specifies the interval (in seconds) at which this transfer group will be transferred. This interval is probably specified in a Bilateral Agreement. Care must be exercised in selecting this value, as too small a value (i.e., frequent transfers) may result in excessive loading of communication links or of the server. The exact behavior of Periodic Interval is modified somewhat by the setting of Periodic Offset as follows: - If Periodic Offset is set to -1, Periodic Interval specifies a pattern of transmission times which begin at Start Time. In this version of ICCP, Start Time will be the time at which a transfer group is created (i.e., when you select the Finish button on the window used to define a transfer group. - If Periodic Offset is set to 0, Periodic Interval specifies a pattern of transmission times which are synchronized to the "top of the hour". (The default for Periodic Offset is 0.) - If Periodic Offset is set to a positive integer, Periodic Interval specifies a pattern of transmission times which are synchronized to the top of the hour, but offset by Periodic Offset number of seconds. Periodic Interval must be set to a value by which 3600 may be evenly divided (i.e., no remainder). Values like 5, 10, 30, 60, 120, 600 are good. Values by which 3600 is not evenly divisible (e.g., numbers like 49, 603, 22, etc.) may produce inconsistent results.
Periodic Offset	Integer number Default: 0	The Periodic Interval establishes a pattern of transmission times (see description above) at which the server will transfer data to the client. If you wish to shift this pattern with respect to the top of the hour, you may use the Period Offset to accomplish this. Periodic Offset can move the pattern forward in time by the offset amount. It can be used to ensure that the server does not attempt to transfer several transfer groups at once, thereby possibly overloading communications links or computer resources. If several transfer groups are configured for the same server, each can be offset by a different amount. This value must always be less than the periodic interval. A value of -1 indicates that the Periodic Offset will not be used, and the periodic intervals will be based on when the association is established with the remote.
Report By Exception	True, False (Default)	This option controls which data objects in the transfer group will be transferred. If the option is selected, only those data objects which have changed will be transferred. If it is not selected, then the entire transfer group, including both those data objects which have changed and those which have not, will be sent.

Attribute Label	Possible Values	Description
Service Quality	- High quality of service - Low quality of service - All quality of service only one link is allowed - Low and high quality of service - High and other quality of service - Low and other quality of service - Other	Quality of Service parameter to be associated with the link. Possibilities are: • All • Low • High • Other • Low and High • Other and Low • Other and High If you define the link as High, then only Transfer Groups defined with high quality of service (QOS) will be sent on this link. If you define it as Low and High, then both Low and High QOS transfer groups will be sent. Only one link can be defined for each priority. If All is specified, then only one link is allowed. If you are configuring only block 1 and 2 data then configure single link to a peer and use All for the QOS. For additional information on QOS, see Appendix 4-D.
Transfer Type	Select either Periodic or Report by Event	Periodic data are normally transferred at some periodic interval Report by Event data are transferred when a change occurs to any item of data in the transfer group.

IcClientBlock1And2Object

- IcClientBlock1And2Object
Represents a SCADA data point requested from a remote ICCP peer
- Associations
ReceiveValueFor

Attribute Label	Possible Values	Description
Deadband	Floating point number	If the absolute value of the received value subtracted from the previously propagated value is greater than the deadband, the received value will be propagated. Otherwise it will be suppressed. The deadband will not be applied to received accumulators or digitals. If the deadband is zero, the Default Dead Band value for the remote will be used. Applies for Block1 and 2 data. Scope: [ICCP].
Iccp Datatype	- Analog Value - Analog plus Quality - Analog plus Quality plus Times-tamp - Analog plus Quality plus Times-tamp plus Change of Value - Analog plus Quality plus Times-tamp Extended - Discrete Value - Discrete plus Quality - Discrete plus Quality plus Times-tamp - Discrete plus Quality plus Times-tamp plus Change of Value - Discrete plus Quality plus Times-tamp Extended - State Value - State plus Quality - State plus Quality plus Times-tamp - State plus Quality plus Times-tamp plus Change of Value - State plus Quality plus Times-tamp Extended	Specifies the ICCP data type transferred.
Offset	Floating point number Default: 0	Analog values are converted by a formula $Y = \text{Slope} * X + \text{Offset}$. Scope: [ICCP].
Scope	System, Domain Default [recommended]: System	Indicates the ICCP scope of the variable: System or Domain. The default is the system. The 2 control centers must agree on the Scope of the variable.
Slope	Floating point number Default: 1	Analog values are converted by a formula $Y = \text{Slope} * X + \text{Offset}$.
SOE Data	- ClientObjectSOEData - No-process as telemetered data only - Yes-process only as SOE data - Both-Process as telemetered and SOE data - Default Value: No - process as telemetered data only.	This attribute is used to indicate how incoming ICCP data (from remote system) is propagated into SCADA. This attribute is ignored if the ICCP data type does NOT include milliseconds or the remote control center does not support ICCP version 2000-08 (or newer versions).

IcClientBlock1And2Tag

- IcClientBlock1And2Tag

Used to send tags via Block1&2. New tag will have two 'Attributive' associations that indicate which Markers/Tags to send and an association to the Measurement whose tags should be exchanged.

Attribute Label	Possible Values	Description
Scope	Default: 0 0=System 1=Domain	Indicates the ICCP scope of the variable: system or domain. The default is system. The 2 systems must agree on the Scope of the variable.

IcServerBlock1And2Container

- IcServerBlock1And2Container
Container of server data objects that are available to the remote ICCP peer.

IcServerBlock1And2Object

- IcServerBlock1And2Object
Represents a SCADA data point that can be served to a remote ICCP peer.

Attribute Label	Possible Values	Description
Iccp Datatype	- Analog Value - Analog plus Quality - Analog plus Quality plus Times-tamp - Analog plus Quality plus Times-tamp plus Change of Value - Discrete Value - Discrete plus Quality - Discrete plus Quality plus Times-tamp - Discrete plus Quality plus Times-tamp plus Change of Value - Digital Value - Digital plus Quality - Digital plus Quality plus Times-tamp - Digital plus Quality plus Times-tamp plus Change of Value	Specifies the type of data transferred
Permission	Read, Write, Read/Write	This specifies the permissions associated with the data object.
Scope	- System - Domain - Default [recommended]: System	Indicates the ICCP scope of the variable If you specify Domain, the data will be available only to this peer and cannot be made available to any other peer. If you specify System, the data will be accessible by all peers. We recommend using System unless a particular ICCP peer insists on Domain.
SOE Data	Default: 1 1 - No: send telemetered values of object. 2 - Yes: send SOE values of object.	This attribute is used to indicate what value ICCP should send (server) to the remote system. This attribute is ignored if the ICCP data type does NOT include milliseconds of the remote control center does not support ICCP version 2008-08 (or newer versions).

New IcServerBlock1And2Tag

- Used to send tags via Block1&2. New tag will have two 'Attributive' associations that indicate which Markers/Tags to send and an association to the Measurement whose tags should be exchanged.
- Associations Send for ICCP Close-Only-Inhibit, Marker of Type Send for ICCP Open-Close-Inhibit, Marker of Type Sends Value From Measurement

Attribute Label	Possible Values	Description
Scope	System, Domain Default (recommended): System	Indicates the ICCP scope of the variable: system or domain. The default is system. The 2 control centers must agree on the Scope of the variable.
Permission	Read, Write, Read/Write Default: Read	This specifies the permissions associated with the data object. The permission choices are "Read", "Write", and "Read/Write".

IcClientBlock5Container

- IcClientBlock5Container

Container of client block 5 control objects registered as a unit at the remote ICCP peer. It receives control and sends the control command to the remote ICCP peer.

Attribute Label	Possible Values	Description
Service Quality	- High quality of service - Low quality of service - All quality of service only one link is allowed - Low and high quality of service - High and other quality of service - Low and other quality of service - Other	Quality of Service parameter to be associated with the link. Possibilities are: • All • Low • High • Other • Low and High • Other and Low • Other and High If you define the link as High, then only Transfer Groups defined with high quality of service (QOS) will be sent on this link. If you define it as Low and High, then both Low and High QOS transfer groups will be sent. Only one link can be defined for each priority. If All is specified, then only one link is allowed. If you are configuring only block 1 and 2 data then configure single link to a peer and use All for the QOS. For additional information on QOS, see Appendix 4-D.

IcClientBlock5Object

- IcClientBlock5Object

Represents the object used for sending control to the remote server (i.e. the capability of ICCP to receive a SCADA supervisory control point from a remote ICCP peer).

- Associations

SendControlFrom

Send for ICCP Close-Only-Inhibit, Marker of Type

Send for ICCP Open-Close-Inhibit, Marker of Type

Attribute Label	Possible Values	Description
Iccp Datatype	- Command - Set Point Real - Set Point Discrete	Specifies the type of data transferred
Offset	Floating point number Default: 0	Analog values are converted by a formula $Y = \text{Slope} * X + \text{Offset}$. Scope: [ICCP].

Attribute Label	Possible Values	Description
Scope	System, Domain Default [recommended]: System	Indicates the ICCP scope of the variable. If you specify Domain, the data will be available only to this peer and cannot be made available to any other peer. If you specify System, the data will be accessible by all peers. We recommend using System unless a particular ICCP peer insists on Domain.
Slope	Floating point number Default: 1.0	Analog values are converted by a formula $Y = \text{Slope} * X + \text{Offset}$. Scope: [ICCP].
Tag Info	UseOnlyForControl Value: 0 Use_Only_For_Control, Use_For_Control_And_Tag Default: Use_Only_For_Control	Indicates if this control object is used only for control requests or if it should also be used for sending Tags

IcServerBlock5Container

- IcServerBlock5Container
Container of server block 5 control objects.

IcServerBlock5Object

- IcServerBlock5Object
Represents the object that a control from the remote client is acted on (i.e. the capability of ICCP to supply a SCADA data processing or supervisory control point to a remote ICCP peer).
- Associations
ApplyControlForMeasurement
Process for ICCP Close-Only-Inhibit, Marker of Type
Process for ICCP Open-Close-Inhibit, Marker of Type

Attribute Label	Possible Values	Description
Iccp Datatype	Command, Set Point Real, Set Point Discrete	Specifies the type of data transferred
Permission	Read, Write, Read/Write	Specifies the permissions associated with the data object
Scope	System, Domain Default [recommended]: System	Indicates the ICCP scope of the variable. If you specify Domain, the data will be available only to this peer and cannot be made available to any other peer. If you specify System, the data will be accessible by all peers. We recommend using System unless a particular ICCP peer insists on Domain.
Tag Info	Use_Only_For_Control, Use_For_Control_And_Tag Default: Use_Only_For_Control	Indicates if this control object is used only for control requests or if it should also be used for sending Tags

5.3 Communication Editor for ICCP

5.3.1 User Interface

The Communication Editor for ICCP allows the user to configure the measurements that will be exchanged with remote ICCP nodes. The Communication Editor for ICCP is an engineering application that is invoked on the Engineering Explorer.

The ICCP Communication Editor is a table editor whose basic layout is the same regardless which kind of ICCP data will be viewed or edited, but the meaning of the displayed components may vary depending on the selected instance and on the selected ICCP data mode.

For additional information on the capabilities of a table editor, refer to [Reference Guide Information Model Management \(IMM\)](#).

Header

The header of the Communication Editor for ICCP shows the name of the instance that was selected from the instance tree on the Work Pane.

Body

The body of the Communication Editor for ICCP mainly consists of two pull-down lists and two table grids.

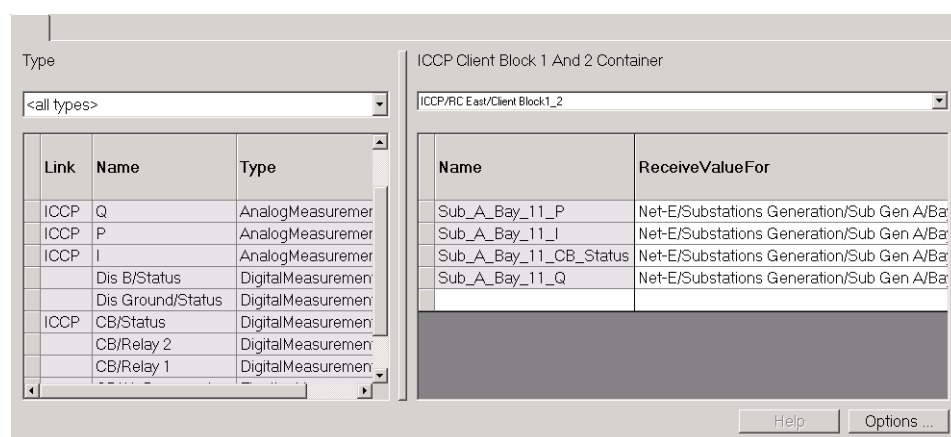


Figure 5-2 Body of the Communication Editor for ICCP

The Pull-Down Lists and Table Grids components may have different meanings depending on the selected ICCP instance and on the selected ICCP mode. [Table 5-2](#) shows the meanings of these components depending on the selected ICCP mode.

Table 5-2 Body of the Communication Editor for ICCP

	Left Part of the Body	Right Part of the Body		
	Pull-Down List Type	Table Grid	Pull-Down List	Table Grid
Client Mode	Contains the types available for filtering	Shows the measurements of the instance selected from the overview tree. ¹	Contains all instances of the types Remote Source and Transfer Group.	Shows all linked instances of the selected transfer group.
Server Mode	Contains the types available for filtering	Shows the measurements of the instance selected from the overview tree. ²	Contains all instances of the type Remote Source.	Shows all server instances of the selected remote source.

1. Depending on the option selected from the pull-down list type, the content of the table grid may show all measurements or only a specific type of measurement.
2. Depending on the option selected from the pull-down list type, the content of the table grid may show all measurements or only a specific type of measurement.

Type List

The Type pull-down list lists the types on which the instances shown in the Table Grid beneath the list are based.

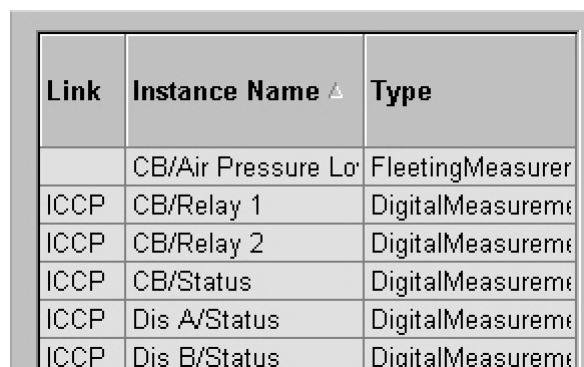
The pull-down list Type may be used to filter the Table Grid. Selecting a certain type from the list reduces the scope of the instances displayed on the Table Grid to those that are based on the selected type.

Table Grid

The Table Grid on the left body part shows attributes of the measurements of the selected instance in a tabular form.

The Table Grid consists of columns and rows. The columns represent the attributes. The rows represent the instances. Each row displays attribute values of one single instance. By default, the Table Grid shows the attributes that are contained in all measurements. Normally, the shown attributes are "Link", "Instance Name" and "Type" (see description below).

The column headings are static text fields that denote the purposes of the columns. The heading of a column may be clicked to sort the table grid by the column values. A small arrow indicates the sorting order. Successive clicks reverse the sorting order. For example, the content of the Table Grid shown in [Figure 5-3](#) below is sorted by ascending Instance names.



Link	Instance Name ▲	Type
	CB/Air Pressure Lo	FleetingMeasurer
ICCP	CB/Relay 1	DigitalMeasureme
ICCP	CB/Relay 2	DigitalMeasureme
ICCP	CB/Status	DigitalMeasureme
ICCP	Dis A/Status	DigitalMeasureme
ICCP	Dis B/Status	DigitalMeasureme

Figure 5-3 Column Headings of a Table Grid

The number of shown measurements is determined by the settings specified in the Advanced block of the Options Dialog Box. For additional information on these settings, refer to [5.3.3.2 Options Dialog Box](#).

Column Heading	Description
Link	Indicates whether a communication link exists for the respective measurement. In addition, the displayed text indicates the communication application for which the link was established. Valid values: CFE: Link to a CFE data point ICCP: Link to an ICCP data point
Instance Name	Shows the instance path of the respective measurement.
Type	Shows the name of the type on which the measurement is based. The type name may be double-clicked to filter the Table Grid. This reduces the content of the table grid to instances based on the clicked type. This may also be achieved by using the -down list Type. For details, refer to its description above.

Pull-Down List

The pull-down list located in the right body part lists the appropriate container instances (client data or server data) of the configured ICCP peers.

- Client mode (ICCP Blocks 1 and 2)
A container represents a set of linked SCADA data points that are requested from a remote ICCP peer as a group. It has attributes that describe transfer conditions such as periodicity, integrity interval, etc.
- Client mode (ICCP Block 5)
A container represents a set of linked control objects that are requested from a remote ICCP peer as a group. Client control objects are instances used to send control commands to the remote server.
- Server mode (ICCP Blocks 1 and 2)
A container represents a set of server data objects (SCADA data points) that can be served to a remote ICCP peer as a group.
- Server mode (ICCP Block 5)
A container represents a set of server control objects that are requested from a remote ICCP peer as a group. Server control objects are instances to which a control commands from the remote server will be applied.

The Table Grid on the right body part shows instances belonging to the selected container instance in a tabular form.

Column	Description
Instance Name	Shows the path name of the ICCP instance.
ReceiveValueFor	Shows the name of the (linked) local data point receiving the value from the remote system.
SendsValueFromMeasurement	Shows the name of the (linked) local data point that will be sent (transmitted) to the remote system.



Note

The column ReceiveValueFor is available only in Client mode. The column SendsValueFromMeasurement is available in Server mode.

Command Area

The **Options** button (located in the lower left corner of the body) opens the **Options Dialog Box**. This dialog box allows the user to customize the content of the table grids. For additional information on the **Options Dialog Box**, refer to [5.3.3.2 Options Dialog Box](#)

Context Menu for the Left Part of the Body

This pop-up menu is opened by clicking the right mouse button on a table cell.

Create

Establishes a link between one of the instances available on the Table Grid on the Left Part of the Body and the currently selected ICCP container instance on the Right Part of the Body.

Instead of selecting an instance from the Table Grid on the left body part and selecting the **Create** menu item, you may also drag the desired instance from the Table Grid on the left body part and drop it onto the Table Grid on the right body part.

When a link is established, an ICCP instance is created under the currently selected ICCP container instance. Both the name of that instance and the ICCP Variable Name are derived from the instance name of the selected data point. The new ICCP instance is automatically inserted into the Table Grid on the right body part.



Note

Create is disabled when a link has already been established for the respective instance.

Edit

Enables you to edit one of the instances available on the Table Grid on the Left Part of the Body. Using this button, you can change the instance name and/or the type of the selected ICCP instance.

Context Menu for the Right Part of the Body

This pop-Up menu is opened by clicking the right mouse button on a table cell.

Unlink

Unlinks a data point. The reference (i.e. the link) to the data point is deleted. The ICCP instance itself is not deleted. The last known value of the unlinked data point continues to be transmitted.

Rename

Changes the instance name of the ICCP instance, and thus, the ICCP Variable Name.

Delete

Removes the selected ICCP instance.

Both the ICCP instance and its reference to a data point are deleted. The respective ICCP instance is removed from the Table Grid on the right body part.

5.3.2 Accessing the Communication Editor for ICCP

The Communication Editor for ICCP is invoked in two steps:

- Select any instance on the Overview Pane of the Engineering Explorer.
- Click one of the following menu items on the toolbar of the Engineering Explorer to choose the desired ICCP mode (client or server):
 - Communication Editors > IcClientBlock 1 and 2
Opens the Communication Editor for ICCP to view or configure ICCP Block 1 and 2 client data objects for the selected instance.
 - Communication Editors > IcClientBlock 5
Opens the Communication Editor for ICCP to view or configure ICCP Block 5 client data objects for the selected instance.
 - Communication Editors > IcServerBlock 1 and 2
Opens the Communication Editor for ICCP to view or configure ICCP Block 1 and 2 server data objects for the selected instance.
 - Communication Editors > IcServerBlock 5
Opens the Communication Editor for ICCP to view or configure ICCP Block 5 server data objects for the selected instance.

In each case, the Communication Editor for ICCP opens in the Work Pane of the Engineering Explorer (refer to the example shown in [Figure 5-4](#) below).

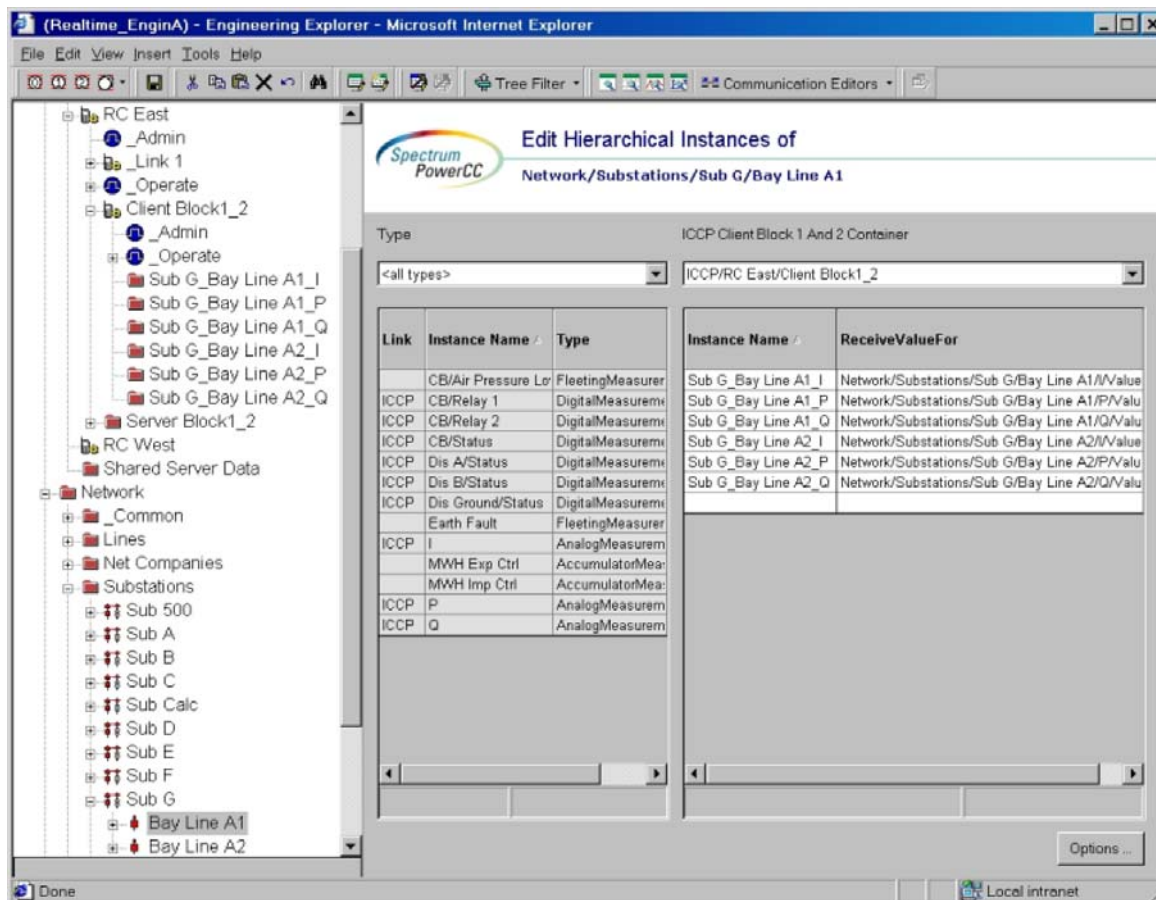


Figure 5-4 ICCP Communication Editor (displayed in Engineering Explorer)

5.3.3 Basic Editor Components

5.3.3.1 User Interface

The Communication Editor for ICCP allows the user to configure the measurements that will be exchanged with remote ICCP nodes. The Communication Editor for ICCP is an engineering application that is invoked on the Engineering Explorer.

The ICCP Communication Editor is a table editor whose basic layout is the same regardless which kind of ICCP data will be viewed or edited, but the meaning of the displayed components may vary depending on the selected instance and on the selected ICCP data mode.

For additional information on the capabilities of a table editor, refer to [Reference Guide Information Model Management \(IMM\)](#).

Header

The header of the Communication Editor for ICCP shows the name of the instance that was selected from the instance tree on the Work Pane.

Body

The body of the Communication Editor for ICCP mainly consists of two pull-down lists and two table grids.

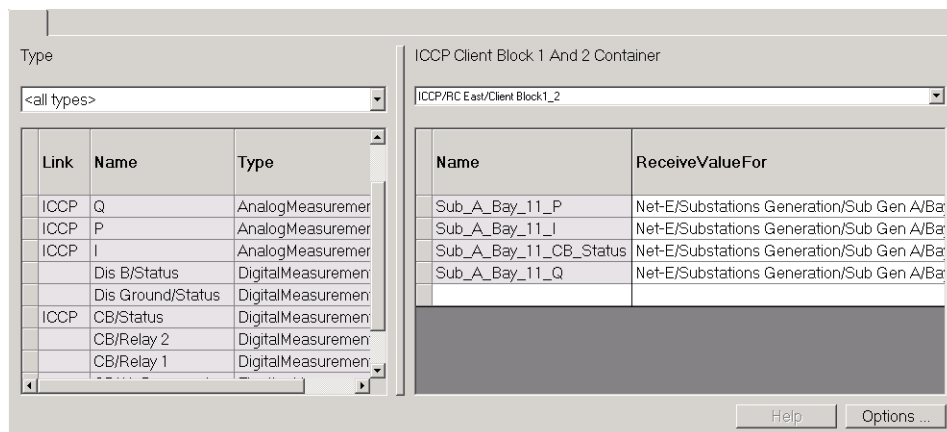


Figure 5-5 Body of the Communication Editor for ICCP

The Pull-Down Lists and Table Grids components may have different meanings depending on the selected ICCP instance and on the selected ICCP mode. [Table 5-3](#) shows the meanings of these components depending on the selected ICCP mode.

Table 5-3 Body of the Communication Editor for ICCP

	Left Part of the Body	Right Part of the Body		
	Pull-Down List Type	Table Grid	Pull-Down List	Table Grid
Client Mode	Contains the types available for filtering	Shows the measurements of the instance selected from the overview tree. ¹	Contains all instances of the types Remote Source and Transfer Group.	Shows all linked instances of the selected transfer group.
Server Mode	Contains the types available for filtering	Shows the measurements of the instance selected from the overview tree. ²	Contains all instances of the type Remote Source.	Shows all server instances of the selected remote source.

1. Depending on the option selected from the pull-down list type, the content of the table grid may show all measurements or only a specific type of measurement.
2. Depending on the option selected from the pull-down list type, the content of the table grid may show all measurements or only a specific type of measurement.

Type List

The Type pull-down list lists the types on which the instances shown in the Table Grid beneath the list are based.

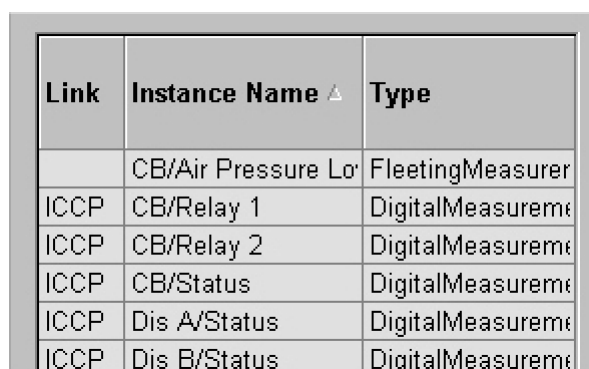
The pull-down list Type may be used to filter the Table Grid. Selecting a certain type from the list reduces the scope of the instances displayed on the Table Grid to those that are based on the selected type.

Table Grid

The Table Grid on the left body part shows attributes of the measurements of the selected instance in a tabular form.

The Table Grid consists of columns and rows. The columns represent the attributes. The rows represent the instances. Each row displays attribute values of one single instance. By default, the Table Grid shows the attributes that are contained in all measurements. Normally, the shown attributes are "Link", "Instance Name" and "Type" (see description below).

The column headings are static text fields that denote the purposes of the columns. The heading of a column may be clicked to sort the table grid by the column values. A small arrow indicates the sorting order. Successive clicks reverse the sorting order. For example, the content of the Table Grid shown in [Figure 5-6](#) below is sorted by ascending Instance names.



Link	Instance Name ▲	Type
	CB/Air Pressure Lo	FleetingMeasurer
ICCP	CB/Relay 1	DigitalMeasureme
ICCP	CB/Relay 2	DigitalMeasureme
ICCP	CB/Status	DigitalMeasureme
ICCP	Dis A/Status	DigitalMeasureme
ICCP	Dis B/Status	DigitalMeasureme

Figure 5-6 Column Headings of a Table Grid

The number of shown measurements is determined by the settings specified in the Advanced block of the Options Dialog Box. For additional information on these settings, refer to [5.3.3.2 Options Dialog Box](#).

Column Heading	Description
Link	Indicates whether a communication link exists for the respective measurement. In addition, the displayed text indicates the communication application for which the link was established. Valid values: CFE: Link to a CFE data point ICCP: Link to an ICCP data point
Instance Name	Shows the instance path of the respective measurement.
Type	Shows the name of the type on which the measurement is based. The type name may be double-clicked to filter the Table Grid. This reduces the content of the table grid to instances based on the clicked type. This may also be achieved by using the -down list Type. For details, refer to its description above.

Pull-Down List

The pull-down list located in the right body part lists the appropriate container instances (client data or server data) of the configured ICCP peers.

- Client mode (ICCP Blocks 1 and 2)
A container represents a set of linked SCADA data points that are requested from a remote ICCP peer as a group. It has attributes that describe transfer conditions such as periodicity, integrity interval, etc.
- Client mode (ICCP Block 5)
A container represents a set of linked control objects that are requested from a remote ICCP peer as a group. Client control objects are instances used to send control commands to the remote server.
- Server mode (ICCP Blocks 1 and 2)
A container represents a set of server data objects (SCADA data points) that can be served to a remote ICCP peer as a group.
- Server mode (ICCP Block 5)
A container represents a set of server control objects that are requested from a remote ICCP peer as a group. Server control objects are instances to which a control commands from the remote server will be applied.

The Table Grid on the right body part shows instances belonging to the selected container instance in a tabular form.

Column	Description
Instance Name	Shows the path name of the ICCP instance.
ReceiveValueFor	Shows the name of the (linked) local data point receiving the value from the remote system.
SendsValueFromMeasurement	Shows the name of the (linked) local data point that will be sent (transmitted) to the remote system.



Note

The column ReceiveValueFor is available only in Client mode. The column SendsValueFromMeasurement is available in Server mode.

Command Area

The **Options** button (located in the lower left corner of the body) opens the **Options Dialog Box**. This dialog box allows the user to customize the content of the table grids. For additional information on the **Options Dialog Box**, refer to [5.3.3.2 Options Dialog Box](#)

Context Menu for the Left Part of the Body

This pop-up menu is opened by clicking the right mouse button on a table cell.

Create

Establishes a link between one of the instances available on the Table Grid on the Left Part of the Body and the currently selected ICCP container instance on the Right Part of the Body.

Instead of selecting an instance from the Table Grid on the left body part and selecting the **Create** menu item, you may also drag the desired instance from the Table Grid on the left body part and drop it onto the Table Grid on the right body part.

When a link is established, an ICCP instance is created under the currently selected ICCP container instance. Both the name of that instance and the ICCP Variable Name are derived from the instance name of the selected data point. The new ICCP instance is automatically inserted into the Table Grid on the right body part.



Note

Create is disabled when a link has already been established for the respective instance.

Edit

Enables you to edit one of the instances available on the Table Grid on the Left Part of the Body. Using this button, you can change the instance name and/or the type of the selected ICCP instance.

Context Menu for the Right Part of the Body

This pop-Up menu is opened by clicking the right mouse button on a table cell.

Unlink

Unlinks a data point. The reference (i.e. the link) to the data point is deleted. The ICCP instance itself is not deleted. The last known value of the unlinked data point continues to be transmitted.

Rename

Changes the instance name of the ICCP instance, and thus, the ICCP Variable Name.

Delete

Removes the selected ICCP instance.

Both the ICCP instance and its reference to a data point are deleted. The respective ICCP instance is removed from the Table Grid on the right body part.

5.3.3.2 Options Dialog Box

The **Options** dialog box provided by the Communication Editor for ICCP allows you to customize the content of the table grids. This dialog box has the same functions as the **Options** dialog box provided by table editors. However, **within the Options** dialog box of the Communication Editor for ICCP, some of the functions may not be enabled.

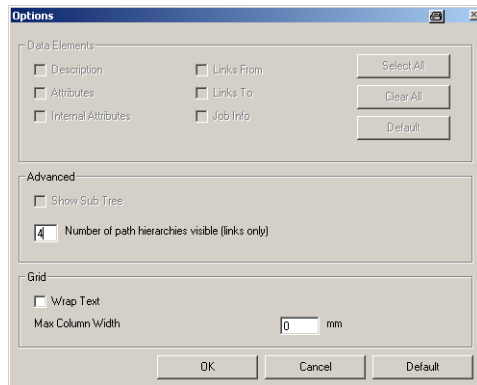


Figure 5-7 Options Dialog Box of the Communication Editor for ICCP

The following table describes the use of the functions provided by the Options dialog box.

Domain	Function(s)	Description
Data Elements	Description, (Internal) Attributes, Method, Links From/To, Attributive Links	The functions are disabled for the ICCP Communication Editor.
	Select All	Selects all check boxes of the Data Elements group
	Clear All	Deselects all check boxes of the Data Elements group
	Default	Applies default settings to the check boxes of the Data Elements block
Advanced	Show Sub-tree	This function is disabled for the ICCP Communication Editor.
	Number of Path Hierarchies Visible	Specifies the number of hierarchy levels shown in a link path on the table grid. E.g. when set to 3, the lowest three hierarchy levels of the path name are displayed. Valid values: integer value (practically unlimited). If no input or if set to 0, then all hierarchy levels are displayed.
Grid	Wrap text	Specifies whether text shown in a cell of the table grid may wrap. When enabled, the character string shown in a cell of the table grid is wrapped if it exceeds the maximum column width (specified in the Max Column Width field).
	Max Column Width	Specifies the maximum width of a table cell [mm]. Valid values: real number values (practically unlimited). If no input or set to 0, then the width of the selected table column is determined by its widest cell (i.e. the cell containing the longest text).
Push-buttons for the entire Options dialog box (at its bottom)	OK	Applies the configured option settings. The appearance of the table grid and the number of instances shown on the table grids change accordingly. The Options dialog box is closed.

Domain	Function(s)	Description
	Cancel	Discards the option changes. The Options dialog box is closed. No action takes place.
	Default	Resets all option settings and applies default settings to the options of the Options dialog box.

5.4 ART Configuration

5.4.1 ART Configuration

The primary binaries of the ICCP subsystem are ICCPMain.exe and ICCPR.exe. ICCPMain is considered the 'parent' process and its main responsibility is to start up (and restart) the ICCPR processes. ICCPR takes care of communications with a specific remote. For each configured remote in the Real-time context there needs to be one ICCPR process.

ICCPMon.exe is another binary in the ICCP subsystem that must run on a non-ICCP server (usually the RTS). Its sole purpose is to monitor the ICCPR processes and make sure they are producing the operational status for the remote. If ICCPMon.exe detects that the ICCPR process is no longer producing the operational status, it changes the operational state of the remote from UP to DOWN. The primary scenario where this could happen is if the ICCP Server(s) experience a hard failure or isolated from the rest of the network. ICCPMon will detect this and ensure that the operational status of the remotes are set to DOWN.

ICCPMain, ICCPMon, and ICCPR are all 'ART' aware applications (ART = Application Run Time) and thus, must be configured under the appropriate "Software Packages" in the System Management Information (SMI) context.

An example configuration should already be present if the standard 'Demo' data model has been loaded on the system. If not, perform the following steps:

- Add a software package called 'ICCP' (suggested name) to "Software Packages";
- Link a Hardware Component (such as ICCP) to this newly created Software Package.
- Add a software group called 'ICCP Applications - Group HotStandby' (suggested name) to the 'ICCP' software package;
- Add the 'ICCPMain' Application to 'ICCP Applications - Group HotStandby'. Use \$SR5_INSPATH\Apps\ICCP\ICCPMain.exe as path attribute. Set the 'Startup Type' attribute to 'Automatic';
- Create a link from the 'Depends On Active' to the DDS Server on the RTS.
- Add an 'ICCPR_XXXX' Application (recommend XXXX be the remote name from the Realtime context) to 'ICCP Applications - Group HotStandby'. For each instance of IcRemoteCenter in the Realtime context, there must be one 'ICCPR_' application added. If there are 5 IcRemoteCenter instances, there will need to be 5 "ICCPR_" Application entries.
- IMPORTANT NOTE: The 'ICCPR_XXXX' name given to the Application instance MUST be entered in the 'ApplicationProcessName' attribute of the corresponding IcRemoteCenter in the Realtime context.
- Use \$SR5_INSPATH\Apps\ICCP\ICCPR.exe as path attribute. Set the 'Startup Type' attribute to 'Interactive' (ICCPMain will start up the ICCPR processes).
- Go to the Links tab and create a 'Depends on Active' link to the DDS Server on the RTS package and to the Marker Manager Monitor in SCADA package.

It is assumed that other software packages containing applications such as "DDS" and "SourceValueSelection" are already defined.

An example of the ICCPMon configuration should already be present if the standard 'Demo' data model has been loaded on the system. ICCPMon should NOT be configured on the 'ICCP' Software package. Normally it is configured to run on the RTS Server.

5.5 Stack Configuration

5.5.1 Overview

[6.1 Mandatory Configuration Parameters](#) provides you with the information to be exchanged with a remote system in order to establish an ICCP association. While all of this information is needed to communicate with a remote, not all of it is actually needed within the ICCP **application** layer at the **top of the protocol stack**. Some of it, mainly the **addressing** information, is only needed by the **lower layers** of the protocol stack. These layers get this information out of a text file called **osill2.cfg**.

5.5.2 Maintaining osill2.cfg File

With Spectrum PowerCC, there are two ways to maintain the osill2.cfg file:

- By manual editing: if you have multiple servers (e.g. in case of redundancy), you will need to manually edit this file on all servers.
- By entering all information into the IMM and running the **OSIFileCreator** program (see further down).

SIEMENS recommends the use of the OSIFileCreator tool for maintaining the osill2.cfg file. If you are interested in manually updating the osill2.cfg file or just interested in what goes in it, refer to [6.3.1 Overview](#) containing OSI Config file info.

OSI File Creator

The OSIFileCreator tool reads the configuration information out of IMM and then writes it into the osill2.cfg file on each of the ICCP servers. Normally, it is run on the PSOS server.

Starting the OSI File Creator

The program is located in \SIEMENS\PowerCC\Apps\ICCP (you may want to create a shortcut to it on your desktop).

- Double-click on its name. A dialog box will appear. It contains a menu for selecting a job name and a CreateFile push-button to be used for creating of a new osill2 file.

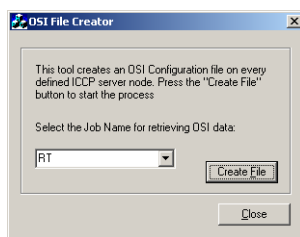


Figure 5-8 OSI File Creator Dialog Box

Creating New osill2 Files

- Click on the **CreateFile** push-button.

A message pop up will appear informing you that if you continue, all osill2.cfg files will be overwritten.

- If you are not sure, select **No**.

As a result no file will be created.

- If you wish to create a new file, select **Yes**, then the existing files will be overwritten.
If the creation operation is successful, another message appears saying "ICCP was populated successfully" which means that all of the files were populated successfully. Any problems will also be reported in a pop up message box.

**Note**

When you run the OSIFileCreator, it overwrites the existing osill2.cfg files. Thus any manual changes to the osill2.cfg file will be lost. You can make temporary changes to the osill2.cfg file for testing purposes, but if you plan on running the OSIFileCreator later, make sure you enter the changes into IMM.

Ways of Running OSF File Creator

The OSIFileCreator can be run at anytime, i.e. you can run it either:

- in **RT** (Real-time) or
- by reading the information out of a job.

Running OSF File Creator in RT

If the menu shown in the above OSI File Creator dialog box displays **RT**, the OSIFileCreator reads the current information out of RT and writes it into the osill2.cfg files. To get the ICCP applications and stack to pick up any changes in the file, the ICCP applications need to be restarted. This can be done by bringing up the ICCP user interface and turning off the ICCP subsystem and then turning it back on (giving it time to fully stop before starting again). Stopping the ICCP subsystem will stop communications with all remotes.

Running OSF File Creator for a Job

When doing this, the OSIFileCreator reads the information out of a job instead of what is currently in RT. During this, the OSIFileCreator reads the current image of everything contained within the job (i.e. it does not just read the changes) and writes it to the configuration file. Advantage: when the job is eventually activated, only the ICCP process associated with the change will restart, causing it to read the new configuration. Thus, the other processes not modified by the change will not be impacted.

Using Information Changed within a Job

In order to have the OSIFileCreator use information changed in a job, the following steps must be followed:

- **Create** job and make changes.
- **Prepare** the job.
- Once the job reaches the 'Prepared' state, run OSIFileCreator.
- Instead of using **RT**, you must enter the internal job name which can be found by looking at properties of job (example of a full job name to enter: **RT.Job47**).
- Let OSIFileCreator finish before proceeding.
- **Activate** the job.

The ICCPR process that was impacted by this change should go away and be restarted shortly.

Undoing Job Changes

To undo these changes:

- **Prepare Undo** of the job.
- Once the job reaches the 'Undo Prepared' state, run OSIFileCreator. Again, instead of using the default RT, you must enter the internal job name (the same as that used to prepare the job, e.g. **RT.Job47**). Let OSIFileCreator finish before proceeding.

- **Undo** the job.

The ICCP process impacted by this change will go away and be restarted shortly.

5.5.3 Security Configuration

Siemens recommends that the user first configure and establish ICCP association in a Non-Secure mode. Once that has been successfully accomplished, then attempt to make it secure.

Here is the check list for establishing secure associations:

- Establish association in Non-Secure mode
- Get Certificates (How to do this is outside the scope of this document).
- Insert certificates into Microsoft Certificate store.
- Configure certificate information in IMM.
- Configure ICCP to use certificates and run in secure mode.
- Update the osll2.cfg file and secman.xml using the OSIFileCreator. See section 5.5 (make link) for details.
- Bring up the ICCP UI and enable the ICCP subsystem and ICCP remote. See section 4 (make link) for additional information on ICCP UI.

Configuring Certificates for Secure Connections in IMM

One of the main steps in establishing a secure connection is the use of a certificate. The actual certificates that ICCP uses for establishing secure connections must be imported into the Microsoft Certificate store (for additional information, refer to [5.6.1 Certificate Management Using Security Stores](#)). Information that is used to identify the certificate in the Microsoft Certificate store must be entered into IMM.

IcCertificate Data Type

This data type has been created specifically for ICCP security purposes. When you create in IMM an instance of this type, it enables you to enter information that identifies a certificate in the Microsoft certificate store. This identity information is used by the lower layers of the protocol stack to find the certificate in the certificate store.

The ICCP Certificates are contained in a IcCertificate container under the "ICCP" Branch in the IMM RealTime context.

The following list presents the 4 **attributes** representing the type of certificate provided for ICCP security:

- Certificate Type
- Serial Number
- Issuer
- Subject

Certificate Type Attribute

This attribute allows you to select one of the following values each of which corresponds to a specific type of certificate:

- Local System Certificate
- Remote AR Certificate.
- Certificate Authority Certificate
- Local AR Certificate

The use of the above certificates is explained in the following.

- **Local System Certificate:**

Only one Local System Certificate must be configured for TLS (Transport Layer Security) authentication. This certificate is sent to the remote ICCP system as the client or server TLS certificate (depending on

association initiator attribute). Only one Local System Certificate is allowed and is used by all Spectrum PowerCC ICCP servers. The Local System Certificate is configured in IMM and must also be installed in the Windows certificate store.

- **Remote AR Certificate:**

This certificate is received from a remote ICCP system during TLS authentication. A unique Remote AR Certificate must be configured for all ICCP links configured for secure mode. The Remote AR Certificates are configured in IMM and should be installed in the Windows certificate store.

NOTE: In the past, when the local system was not the initiator, a Remote AR was not needed, this is still true for Non Secure associations. For Secure associations, Remote ARs will need to be defined in order to define the Remote certificate.

- **Certificate Authority Certificate:**

A Certificate Authority (CA) Certificate must be configured in IMM for all certificates **received from remote ICCP**. The CA Certificate is used to verify that the received certificate was issued and signed by a valid CA. A separate CA Certificate must be configured for each unique "Issuer" of the received certificates. The Certificate Authority (CA) Certificates are configured in IMM and must also be installed in the Windows certificate store.

- **Local AR Certificate**

For ACSE authentication a Local AR and a Remote AR Certificate must be configured in IMM. The same certificate can be used for ACSE authentication on more than one link. The Local AR Certificate is sent to the remote system as the ACSE level certificate. Any ACSE certificate received from the remote system must also be configured as a Remote AR Certificate. Currently if both TLS and ACSE authentications are configured, the same Remote AR Certificate is used for both. Local AR and Remote AR Certificates are configured in IMM and must also be installed in the Windows certificate store.

Certificate Identification Attributes

In IMM, under ICCP, you will have to create a 'CertificateContainer' that will hold all of the certificate identification information. For each certificate, you will have to create an instance of an 'ICCP Certificate'. Thus for each ICCP Certificate you insert, you must specify the way ICCP has to use it - this is done by selecting one of the above values for the certificate type attribute. The remaining attributes of the IcCertificate data type are used to identify the certificate:

- Serial Number
- Issuer
- Subject

In the following we describe how to enter (paste) these 3 certificate identification attributes into IMM.

You can get the certificate identification information by using the MMC Certificate Plug In (for additional information, refer to [5.6.1 Certificate Management Using Security Stores](#) which discusses in detail, how to import and access certificates in the Microsoft Certificate store).

The following sections discuss the way how to fetch the information to be pasted into IMM for the 3 certificate identification attributes.

- In the MMC Certificate Plug In user interface, double click on the certificate you want to get the information on.
- Then click on the **Details** tab.

The Details tab display appears within the Certificate window.

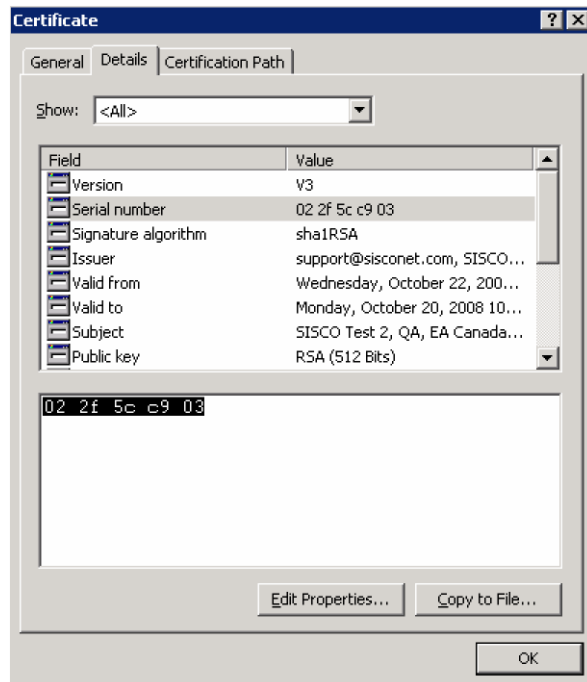


Figure 5-9 Certificate Details Tab Display

The list shown in the above figure ([Figure 5-9](#)), you can find among the others the 3 fields needed for each ICCP Certificate specifications: Serial Number, Issuer, and Subject (in our example presented in the above figure the Serial number field is selected).

At the bottom portion of the window, you can find the value of the field currently selected in the list (in our example "02 2f 5c c9 03").

- Select the value and then
- Copy it to your clipboard.

The OSIFileCreator just reads this out of IMM and writes it into the secManCfg.xml configuration file. Therefore you will have to edit what has been placed into your clipboard:

- Before pasting such a string into the IMM, paste it into a text editor first (e.g. Notepad).
- Modify the string as shown in the examples further down.
- Once the modified certificate information has been entered into the ICCP Certificate, you can link the Local and Remote ARs to the appropriate certificates.

The following examples present the information cut and pasted from the Details tab and the format needed when being entered into IMM.

Serial Number

From Details tab:

02 2f 5c c9 03

Entry into IMM:

022f5cc903 (instead of spaces every 2 characters, the space is every 4 characters)

Issuer

From Details tab:

E = support@sisconet.com

CN = SISCO Test CA (NO LIABILITY ACCEPTED)

O = SISCO, Inc.

L = Sterling Heights

S = Michigan

C = US

Entry into IMM:

(notice quotes and reverse order of parameters):

C="US" S="Michigan" L="Sterling Heights" O="SISCO, Inc." CN="SISCO Test CA (NO LIABILITY ACCEPT-ED)" E="support@sisconet.com"

Subject

From Details tab:

CN = SISCO Test 2

OU = QA

O = EA Canada

L = Quebec City

S = Quebec

C = CA

Entry into IMM:

(notice quotes and reverse order of parameters):

C="CA" S="Quebec" L="Quebec City" O="EA Canada" OU="QA" CN="SISCO Test 2"



Note

Much of the ICCP Secure connection functionality is implemented in the protocol stack. Just as the stack addressing information resides in the osill2.cfg file, the security information is contained within a file called **secManCfg.xml**.

Even if you have not purchased the ICCP Security function, you should be aware that the OSIFileCreator also creates the secManCfg.xml file. This file can be read by any normal xml or text file editor. Its primary content is the certificate identification (not the certificate) and the possible **association** of the certificate to a specific **remote** system.

- In the same way as with the osill2.cfg file, the secManCfg.xml file changes can be maintained either by manual editing, or by entering the necessary information into IMM and running the OSIFileCreator.
- Unlike the osill2.cfg file, an additional option is provided by the MMS vendor, called the **Security Configuration Utility**.

For a additional information on the the security relevant attributes IcLink, IcCertificate and IcRemoteAR, refer to [5.2.3 ICCP Configuration Data Types and Attributes](#).

Configuring ICCP

In addition to configuring the certificates described in the previous section, some of the instances that were used to establish the non-secure associations will need to be modified.

They are as follows:

- IcLink

There are two attributes on this instance related to Secure Associations, they are 'Application Authentication Mode' and 'Transport Layer Security (TLS) Mode'.

- IcLocalAr

This IMM type has a Link for linking an instance of the IcLocalAr to a Local Ar Certificate instance.

- IcRemoteAr

This IMM type has a Link for linking an instance of the IcRemoteAr to a Remote Ar Certificate instance.

For non-secure associations, it was not necessary to define a IcRemoteAr instance if the system was not the initiator. In order to associate Certificates properly to a link, it is necessary to define IcRemoteAr instances for secure associations, even if the local system is not the initiator.

5.6 Certificate Management for Security

5.6.1 Certificate Management Using Security Stores

The Spectrum PowerCC ICCP Security makes extensive use of Microsoft Windows' "certificate store." That is, it utilizes the certificates and keys that are stored, managed, and protected by the operating system. It is assumed that the user of the Spectrum PowerCC ICCP Security has knowledge of certificate management for their particular version of Microsoft Windows.

The Spectrum PowerCC ICCP Security makes use of the following store folders when working with certificates:

Table 5-4 Description of Store Folders

Store Folder	Description
Personal	Used for loading local SSL and MACE certificates
Trusted Root Certification Authorities	Used for loading CA certificates and CRLs
Intermediate Certification Authorities	Used for loading CA certificates and CRLs
SISCO Remote Certificates	Used by the SecCfg when selecting remote certificates. Also used for loading CRLs

NOTE:

The SISCO Remote Certificates folder is created by the SecCfg the first time the user right clicks on the **Remote Certificates** tree item.

5.6.2 SISCO Test Certificates

For demonstration and testing purposes, SISCO has provided a set of test certificates along with a sample Root (Trusted Certificate Authority) certificate; these are located in the TestCertificates subdirectory. Once these certificates are installed, they can be used to experiment with the various Spectrum PowerCC ICCP Security features.

Table 5-5 Description of SISCO Test Certificates

File Name	Common Name	Description
TestCA.cer	SISCO Test CA (NO LIABILITY ACCEPTED)	Sample root trusted certificate authority certificate
TestCRL.crl		Sample CRL issued by SISCO Test CA that revokes the certificate SISCO Revoked .
ST1.pfx – ST10.pfx	SISCO Test 1 – SISCO Test 10	10 valid certificates issued by SISCO Test CA . These have varying key sizes and in some cases, varying signature algorithms. Note that filenames that contain the words "(No MACE)" have private key sizes that are too large for a MACE encoded MMS PDU. However, they are still suitable for use with SSL.
SE.pfx	SISCO Expired	Expired but otherwise valid certificate issued by SISCO Test CA .
SN.pfx	SISCO Knave	Valid certificate issued by Nefarious Test CA , which is not to be trusted!
SR.pfx	SISCO Revoked	Valid certificate issued by SISCO Test CA but revoked by the CRL.

5.6.3 Certificate Store Access

The Spectrum PowerCC ICCP Security operates using the 'Local Computer' certificate store. This particular certificate store gives all users access to the contained certificates. Furthermore, applications which run as Windows Services in protected user contexts will also have access to the contained certificates without additional configuration.

To install each of the test certificates in the Windows 'Local Computer' certificate store:

- first start the Microsoft Management Console (MMC) and add the Certificates snap-in.
- When prompted, choose the 'Computer account' as the store to be managed.
- You may want to save this current MMC as it will be used often for certificate management.
- Follow the instructions below for details how to install each of the test certificates.

5.6.4 Installing the SISCO Test Certificate Authority

- Using the MMC configured in the section, Certificate Store Access, select the Trusted Root Certification Authorities\Certificates folder.
- Then right-click to bring up the Certificate Import Wizard:

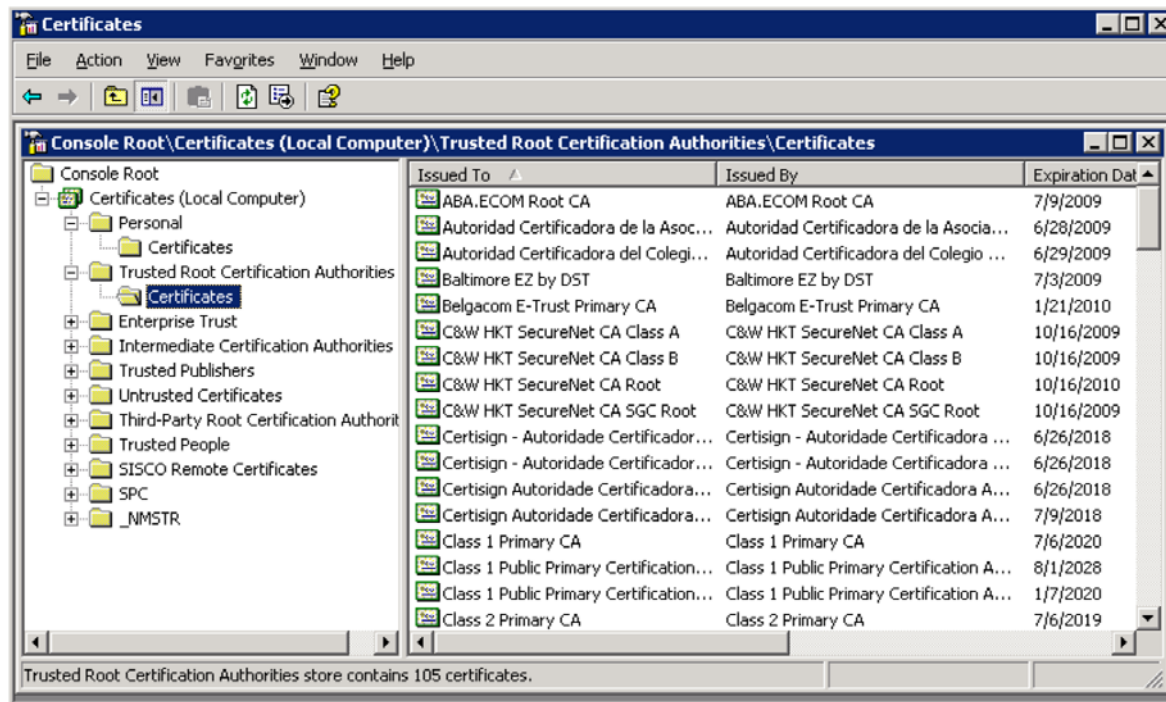


Figure 5-10 Installing the SISCO Test Certificate Authority

- Browse for the file "TestCA.cer" and
- Select the default options for the remaining prompts to install this certificate.
- Once the import has been successful, you can browse the Trusted Root Certificates for the newly-installed SISCO test CA certificate.

5.6.5 Installing the SISCO Test Certificates

- Select the Personal folder and right-click to bring up the Certificate Import Wizard.
- Select the desired certificate to install by double-clicking on the file.
- Once the Certificate Import Wizard is browsing for files, you must select Personal Information Exchange from the drop-down combo box per the screen shot as shown below:

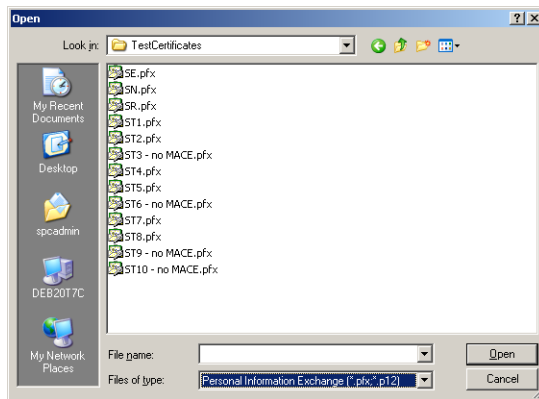


Figure 5-11 Installing the SISCO Test Certificates

Once the import wizard continues, it will ask for a password among other options. The password for all of the test certificates is "test".

- Then, check the **"Mark the private key as exportable"** check box

Default options may be accepted for any remaining prompts.

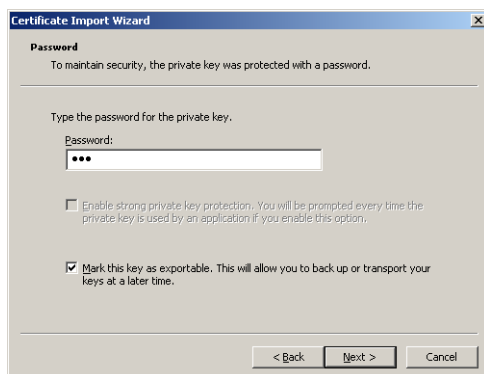


Figure 5-12 Certificate Import Wizard

■

6 Additional Information

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6.1 Mandatory Configuration Parameters

The table below lists important ICCP configuration parameters that need to be provided in order to establish an ICCP association between the local PowerCC system and the remote control center. Each company should provide this information about their system to the other company. This is the information the other company will need to connect to the local system. For example, if TCP/IP is used, the IP address of the local ICCP PC should be provided.

In the following table, the term "Company A" is the company providing the information. "Company B" refers to the company that will be receiving the information.

The 'Parameter' column indicates where the information being provided to the remote company is located in the local (Company A's) IMM (or where it needs to be entered). The 'Usage by Company B' column indicates where the information received from the remote company (Company A) should be entered within the local IMM.

Parameter	Usage by Company B
Server name The name Company A uses to refer to its own system. This field is not electronically transmitted during any ICCP transaction, it is only used to facilitate verbal communication between Company A and Company B.	Company B can use this as the instance name of an IcRemoteCenter.
Company A's domain name The domain name which Company B will use to access data on Company A's remote control center. (This is what Company A entered as the LocalDomainName attribute for its IcRemoteCenter instance).	This is the RemoteDomainName attribute for the IcRemoteCenter instance.
Requested Company B's domain name The domain name on Company B's remote control center which Company A requests to be created. This is only a suggestion as this name is designated by Company B.	If Company B accepts the suggested Domain Name, it would enter it in the LocalDomainName attribute for its IcRemoteCenter instance.
Bilateral Table ID This is the table name that the two companies must agree to (e.g. "1.1"). (This is the BilateralTableVersion attribute for the IcRemoteCenter).	This is the BilateralTableVersion attribute for the IcRemoteCenter instance.
Supported ICCP services A list of the conformance blocks supported by Company A. (This is set with the EnableBlock5 attribute of the IcRemoteCenter).	Blocks 1 and 2 are always enabled. Block 5 can optionally be enabled by setting EnableBlock5 attribute to true for the IcRemoteCenter instance.
ICCP Version The version of ICCP running on the server (e.g. PowerCC ICCP can support TASE.2 Version 1996-08 or Version 2000_08).	If other system can support it, recommend using 2000-08. If 1996-08 is specified, some attributes will not apply/work.
Minimum periodic interval Time in seconds at which Company A's data is being updated. Company B should request data at a periodic rate equal to or greater than this value. (This is the MinimumPeriodicInterval attribute for the IcRemoteCenter instance.)	At Company B, for periodic client transfer groups (IcClientBlock1And2Container), the PeriodicInterval attribute should be set to a value equal to or greater than this value.
IP Network Address If TCP/IP is used, this is the IP address(es) of Company A's server(s). (This is the IPAddress attribute in the PersonalComputerNode instance of the ICCP Server in the SysConfig context).	This value is used as the IPAddress attribute of the appropriate instance of the IcRemoteAR type.
AP Title Optional object identifier representing the Application Process Title given this application. (This is the APTitle attribute in the IcLocalAR instance).	This value is used as the APTitle attribute of the appropriate instance of the IcRemoteAR type.

Parameter	Usage by Company B
AP Invoke ID A long integer used to identify an invocation instance of the application process. Typically not specified. (This is the APInvokeID attribute in the IcLocalAR instance).	This value is used as the APInvokeID attribute of the appropriate instance of the IcRemoteAR type.
AE Qualifier A long integer (32 bit signed) is used to qualify the application entity. (This is the AEQualifier attribute in the IcLocalAR instance).	This value is used as the AEQualifier attribute of the appropriate instance of the IcRemoteAR type.
AE Invoke ID Used to identify an invocation instance of the application entity. Typically not specified. (This is the AEInvokeID attribute in the IcLocalAR instance).	This value is used as the AEInvokeID attribute of the appropriate instance of the IcRemoteAR type.
Presentation Selector (PSEL) 2 or 4 byte number used to select the correct instance of the presentation layer. Typically in the form of '00 09' or '00 00 00 09'. (This is the PresentationSelector attribute in the IcLocalAR instance).	This value is used as the PresentationSelector attribute of the appropriate instance of the IcRemoteAR type.
Session Selector (SSEL) 2 or 4 byte number used to select the correct instance of the session layer. Typically in the form of '00 09' or '00 00 00 09'. (This is the SessionSelector attribute in the IcLocalAR instance).	This value is used as the SessionSelector attribute of the appropriate instance of the IcRemoteAR type.
Transport Selector (TSEL) 2 or 4 byte number used to select the correct instance of the transport layer. Typically in the form of '00 09' or '00 00 00 09'. (This is the TransportSelector attribute in the IcLocalAR instance).	This value is used as the TransportSelector attribute of the appropriate instance of the IcRemoteAR type.
Complete NSAP If OSI is used, this number represents the OSI network address(es) for company A's Server(s). The NSAP can be up to 20 bytes long. (Use of OSI addressing is seldom used anymore).	This value is used as the NSAP attribute of the appropriate instance of the IcRemoteAR type.
Association Type Indicates if Company A will provide just Server or Client functionality on this association, or both Server and Client. SIEMENS product allows Company A to be both a client and server on a single association while others restrict it to be one or the other. (This is the CommunicationDirection attribute for the IcRemoteCenter instance).	If the other company has specified it wants the association to provide both Server and Client functionality, then the CommunicationDirection attribute for the IcRemoteCenter instance should be set to 'Both server and client'. If the other company specified its end will be a Server Only, then this end is a Client, this end must select 'Client Only'.
Maximum MMS PDU Size Size, in bytes, of the maximum MMS protocol data unit, typically 8 k bytes or more. (This is the MaximumPDUSize attribute for the IcRemoteCenter instance.)	This could be used as the MaximumPDUSize attribute for the IcRemoteCenter instance or a different value could be specified. (MMS will negotiate the Max PDU size if necessary when the association is established.)
Association Initiator Under the strictest interpretation of the ICCP standard, the client end of the association would be the initiator of the association. Because SIEMENS product allows each end to be both a client and server, both ends must specify who the initiator of the association. (This is the AssociationInitiator attribute of the IcRemoteCenter instance).	This is the AssociationInitiator attribute for the IcRemoteCenter instance. If the other control center indicates it is going to be the initiator (True), then this value must be set to False. Or Vice-versa. (Only one side can be the initiator and both side have to agree who that is.)

6.2 Optional Configuration Parameters

The table below lists optional PowerCC specific configuration parameters for monitoring ICCP links.

Some background: TCP/IP has some long timeouts for detecting a connection break (possibly in hours). There are some TCP/IP settings that can be changed to reduce these timeouts, but doing this impacts all TCP/IP connections (not just ICCP). An application using TCP/IP can implement its own methods for detecting a broken connection. This is what PowerCC's ICCP implementation does with the 'Server Reply Timeout' attribute. When PowerCC sends a ICCP message that it expects a reply for, it starts a reply timer. If it does not get a response by the time 'Server Reply Timeout' has expired, it assumes the connection is broken.

It is possible that an ICCP remote node may be configured so that messages that the remote needs to respond to are few and far between (many, many minutes). By configuring ""heartbeat"", you are ensuring that a message is being sent that requires a response. (Heartbeat is not a standardized ICCP term, in PowerCC's ICCP implementation it is a request of the remote server's identification.) The Heartbeat Interval indicates how often this message is sent. The worst case for detecting a broken connection is the 'Heartbeat Interval' plus the 'Server Reply Timeout'.

Note: Depending on the parameters chosen it may happen that something else (such as TCP/IP) is detecting the loss of the connection before this, so you may see the connection go down earlier than the 'Heartbeat Interval' plus 'Server Reply Timeout'.

Parameter	Remark
ServerReplyTimeout The Partner Server has to respond to a data request within the time specified through this parameter. Recommended value is 30 seconds.	- ServerReplyTimeout - LinkHeartbeat - LinkHeartbeatInterval are attributes of the ICCP instance type IcRemoteCenter. These parameters serve to monitor the ICCP link to the remote control center. In particular, if the transmission of user data is rather rare, the Link Heartbeat can help to detect broken ICCP links sooner. Note: the worst case for detecting a broken ICCP link is the Heartbeat Interval plus the Server Reply Timeout.
LinkHeartbeat This parameter is used for activating / deactivating the periodic link monitoring routine. This feature can be used for links operating with only sporadic user data transmission.	
LinkHeartbeatInterval This parameter is used for setting the time interval for transmitting the heartbeat message to the remote center. Recommended value is 60 seconds.	

6.3 OSI Configuration

6.3.1 Overview

ICCP is based on MMS, which used an OSI Stack in communication with remote systems. An option was added to ICCP to allow it to use a TCP/IP stack. Since TCP/IP is easier to use, there are very few customers still using the OSI Stack for ICCP. The osill2.cfg file is used to configure the OSI and TCP/IP stacks. This appendix provides a little more information on the OSI and TCP/IP configuration and where it goes in the osill2.cfg file.

OSI addresses are long, complex, not always easy to understand and to enter without errors, and could well be the subject of a separate document. This Appendix discusses a few topics relevant to ICCP installation and configuration. This Appendix is not intended to be a complete and all inclusive description of OSI addressing. Before starting to configure OSI addresses, the following is strongly recommended:

- Read and comprehend the remainder of this section before beginning the job of configuring OSI addresses.
- Contact the network administrator to learn about any system which the own company may have in place for assigning OSI addresses, names, etc.
- If invention of systems for assigning names and addresses is planned, this should be made before configuring OSI addresses. Otherwise, the need to change addresses may arise later.
- Don't configure a mass of addresses all at once. Get one link to work and once it is working, copy and paste the instances to get a head start on configuring additional addresses.

6.3.2 OSI Address Overview

6.3.2.1 OSI Addresses

The MMS product used by Spectrum PowerCC can obtain its configuration from two different sources, one is an ASCII text configuration file and the other is a tool provided by the MMS provider. SIEMENS supports the ASCII text configuration file.

The ASCII text configuration file can be maintained by one of 2 ways:

- One way is to modify it directly with a traditional text editor.
- The other means is to configure the necessary parameters in the Spectrum PowerCC database and run the OSIFileCreator.

The 2nd method is recommended for maintaining the configuration file. The OSIFileCreator will create a new osill2.cfg file and write it to the PCs supporting ICCP.

The following s describe the various components of this configuration file. If using the OSIFileCreator to maintain the configuration file, you do not need to worry about the layout of this file, but the meaning of the parameters can be very helpful when determining what to enter into the Spectrum PowerCC IMM.

If it was installed with default file locations, the configuration file that will be used by the system will be located in: Program Files\SISCO\osill2

<drive letter>:\Program Files\SISCO\osill2</drive>

To edit the file, use a text editor. It is advisable to save a copy of the old file each time it is changed, and to place a copy in a different directory as this configuration file may be replaced with a default version when re-installing Spectrum PowerCC or if the OSIFileCreator is run.

Begin and **End** statements in the example divide this file into several identifiable s. For example, the subnet-work begins with the statement "**Begin_Subnet**" and ends with the line "**End_Subnet**". The various s of the file are described in the following s. For the following parameters, unless otherwise indicated or if you have a good understanding of the parameters, the values specified as the default should be used.

6.3.2.2 Subnetwork Configuration Section

The subnetwork configuration is used to configure all subnetworks in the system. Simply put, each subnetwork represents a single network (e.g., Ethernet) adapter. If there is one Ethernet adapter in the Spectrum PowerCC server used for ICCP traffic, there is one subnetwork. If two Ethernet adapters are used for ICCP traffic, there have to be two subnetworks.

The subnetwork begins with:

Begin_Subnet

and ends with:

End_Subnet

Each subnetwork has its own "Begin_Subnet" (and "End_Subnet") keywords.

The subnetwork contains the following parameters:

Parameter	Meaning
Type	Indicates the subnet type (0 = Ethernet, 1 = Token Ring, 2 = FDDI, 3 = Token Bus). The default is 0 (Ethernet). The default is recommended.
NSAP	This is an octet string representing the local NSAP (Network Service Access Point) of this subnetwork. This parameter is required if using the OSI stack for communicating with any of the remote systems. This number should be obtained from the network administrator. (Use of OSI addressing is seldom used anymore).
IP_Address	This is the IP address (in dotted decimal notation) for the LAN interface being configured. This parameter is required if using TCP/IP for communicating with any of the remote systems.
ES-CT	This is the End System Configuration Timer (in seconds). This determines the interval between ES "Hello" PDUs issued by ES-IS routing. The range is from 1 to 300 seconds. The default is 60. It is recommended to use the default.
ES-HT	This is the End System Holding Timer (in seconds). This determines how long information will be kept in ES-IS routing tables before being flushed. The range is 1 to 300 seconds. Default is 60. It is recommended to use the default.
Error_Bit_On	This determines whether the CLNP entity turns the error bit of the PDU header on or off. Default is N. It is recommended to use the default.
Internet_Type	This determines whether the internet is segmenting (1), non-segmenting (2), or NULL (0). Default is 2. It is recommended to use the default.
Checksum	This determines whether network checksumming is used. Default is N. It is recommended to use the default.
Driver	This is the name of the driver. Unless an own driver is provided (bad idea), this will be "osillc1".

For most parts of the subnetwork the default values should be used. If using the OSI stack, the NSAP must be set to the proper value for the subnetwork. If using TCP/IP, the IP_Address must be set to the proper value for the subnetwork.

A sample subnetwork might look like the following:

Begin_Subnet

Type= 0

NSAP= 49 00 0F 4F 46 46 00 01 00 01 00 60 97 1E 64 74 01

IP_Address = 161.254.254.67

ES_CT= 60

ES_HT= 60

Error_Bit_On = N

Internet_Type = 2

Checksum= N

Driver= osilc1

End_Subnet

If the system has multiple Ethernet adapters, there will be multiple subnetwork s.

6.3.2.3 Transport Configuration Section

The transport is used to configure transport parameters. The configuration file must have one and only one transport .

The transport begins with:

Begin_Transport

and ends with:

End_Transport

The transport contains the following parameters

Parameter	Meaning
Max_TPDU_Size	This is the maximum size of a transport protocol data unit (TPDU). The value must be a power of two. The range is 128 to 8192. Default is 1024. It is recommended to use the default.
Credits	This is the number of credits (i.e., outstanding unacknowledged TPDUs) used by a transport connection. Range is 1 to 15. Default is 4. Use the default.
Max_Retrans	This is the number of attempts to retransmit a TPDU before issuing a disconnect request. Range is 1 to 19. The default of 6 is a good compromise between the need to recognize network problems fairly quickly, but not over-react.
Retrans_Time	This is the maximum time the local transport will wait for acknowledgment before retransmitting a PDU (in 0.1 seconds). The range is 10 to 1000 (i.e., 1 second to 100 seconds). Default is 100. The default value of 100 (10 seconds) is recommended. For slow networks, this may need to be increased.
Ack_Time	This is the maximum time that can elapse between the receipt of a TPDU by the transport layer (from the network layer) and the transmission of the corresponding acknowledgment (in 0.1 seconds). Range is 10 to 1000 (i.e., 1 second to 100 seconds). The default value of 10 (1 second) is recommended. For slow networks, this may need to be increased.
Window_Time	This is the maximum time that transport will wait before transmitting up to date window information (in 0.1 seconds). Range is 10 to 1000. (i.e., 1 second to 100 seconds). Default is 100.
Extended_Format	This parameter is negotiated during the transport class 4 connection negotiation. There are two possible TP4 formats: Extended and Normal. Possible values are Y, y, N, or n. Default is N (use Normal format).
Checksum	This parameter causes a transport checksum to be placed in the variable part of the transport frame. Possible values are Y, y, N, or n. Default is N (no checkQOS means "quality of service"; this parameter allows QOS to be set. The range is currently undefined. Default is 0. sum).

Parameter	Meaning
Default_QOS	QOS means "quality of service"; this parameter allows QOS to be set. The range is currently undefined. Default is 0.
Receive_Window	Do not modify this value from the supplied default (5) unless advised by Siemens.
TPDU_Q_Size	Do not modify this value from the supplied default (10) unless advised by Siemens.
Alloc_Quantum	Do not modify this value from the supplied default (512) unless advised by Siemens.

In most cases, there will be no need to modify anything in the transport supplied with the Spectrum PowerCC installation materials.

6.3.2.4 Session Configuration Section

Normally, there is no need to modify the session configuration supplied with the Spectrum PowerCC installation materials, unless advised by Siemens.

6.3.2.5 General Configuration Section

The general is used to specify certain general configuration parameters. The configuration file must have one and only one general. The general configuration begins with:

Begin_General

and ends with:

End_General

The transport contains the following parameters

Parameter	Meaning
Max_connections	This is the total number of connections which will be supported. Default is 256, probably enough for almost any configuration.
Log_Level	This is a two-byte hexadecimal number which acts as a mask to specify log levels. The default is 00 (no logging). This will normally not be changed unless advised by Siemens as an aid to problem resolution.

6.3.2.6 Local DIB Entry Section

This is used to configure all local names. In this some configuration parameters must be edited. Since each local configuration uniquely identifies the local endpoint of exactly one link to a peer, there will probably be multiple local configurations. The information used in the local configuration has to be provided to the administrator of a peer system, which wishes to connect to the system.

A local configuration begins with:

Begin_Local

and ends with:

End_Local

The local contains the following parameters:

Parameter	Meaning
AR_Name	Each logical connection has two end points, one local and one remote. These end points are identified by fully qualified OSI addresses. OSI addresses are long strings of hexadecimal digits – not particularly amenable to being typed correctly. So, the AR_Name is the name by which the local connection end point will be known. This name is merely shorthand for the complete OSI addressing information, which is specified by the other parameters in the local DIB configuration (so these numbers only have to be typed once). This name does not appear on the network and will not be known to the remote ICCP peer system. There is no requirement for it to match any name on the remote peer, although certainly names will be chosen which do match. This name is used in performing the task of configuring a Spectrum PowerCC data link. Each link must have a unique local AR_Name. This field is required. NOTE: This name is case - sensitive.
AP_Title	This is the local AP title. This field is optional. It is recommended to use the value in the example configuration file.
AP_InvokeID	This is the local AP invoke ID. This field is optional. It is recommended to use the value in the example configuration file.
AE_Qualifier	This is the local AE qualifier. This field is optional. It is recommended to use the value in the example configuration file.
AE_InvokeID	This is the AE invoke ID. This field is optional. It is recommended to use the value in the example configuration file.
Psel	This is the local presentation selector. It is required. Psel can be chosen arbitrarily (see ¹), but the administrator of the peer system has to be informed what value has been chosen.
Ssel	This is the local session selector. This field is required. In internal testing, Siemens varies this field – that is, all other fields in all remote s will be identical except for Ssel and AR_Name. Ssel can be chosen arbitrarily (see ²), but the administrator of the peer system has to be informed what value has been chosen.
Tsel	This is the local transport selector. Tsel can be chosen arbitrarily (see ³), but the administrator of the peer system has to be informed what value has been chosen.
Subnet	This is a decimal integer, which indicates the subnetwork number from the subnetwork . This field is optional. Default is 0.

1. In the MMS context, the layer selectors (Psel, Ssel, and Tsel) have no significant meaning. Therefore, they may be assigned arbitrarily. Since all the parameters in this are assembled into a fully qualified OSI address, and since each OSI address must be unique, Siemens recommends to make either the Ssel, Psel, or Tsel value different for each link. (At Siemens, for testing, it has been decided to use Ssel for this function.)
2. In the MMS context, the layer selectors (Psel, Ssel, and Tsel) have no significant meaning. Therefore, they may be assigned arbitrarily. Since all the parameters in this are assembled into a fully qualified OSI address, and since each OSI address must be unique, Siemens recommends to make either the Ssel, Psel, or Tsel value different for each link. (At Siemens, for testing, it has been decided to use Ssel for this function.)
3. In the MMS context, the layer selectors (Psel, Ssel, and Tsel) have no significant meaning. Therefore, they may be assigned arbitrarily. Since all the parameters in this are assembled into a fully qualified OSI address, and since each OSI address must be unique, Siemens recommends to make either the Ssel, Psel, or Tsel value different for each link. (At Siemens, for testing, it has been decided to use Ssel for this function.)

All parameters in this (except AR_name) are assembled into an OSI address. When a remote ICCP peer wishes to establish a connection, it uses this address, including layer selectors, which match those chosen here.

Each logical link between a local system and a remote ICCP peer is identified by its two-endpoint addresses (AR names). Thus, if three logical links between the own system and one remote peer shall be established (i.e., one link for High priority data, one for Low, and one for Other), three local AR names are required. All of these will map to addresses, which (if the above recommendation is followed) vary only in terms of Ssel (or Psel or

Tsel). If three logical links to each of two remote ICCP peers shall be established, six local AR names are required, also varying only in terms of Ssel.

6.3.2.7 Remote DIB Entry Section

This is the in which the addresses of all the peers to which the own system will have connections are configured. As with the local , all components specified in a remote configuration (again, with exception of the AR name) are assembled into a fully qualified OSI address of an application instance on a remote ICCP peer. Therefore, the parameters must match the analogous parameters used in configuring the peer – the administrator of the peer system should provide the various parameters used in this .

Since each remote configuration uniquely identifies the remote endpoint of exactly one logical link to a peer, there will probably be multiple local configuration s.

A remote configuration begins with:

Begin_Remote

and ends with:

End_Remote

The remote contains the following parameters:

Parameter	Meaning
AR_Name	As it has been said in discussing local AR Names, each logical connection has two end points, one local and one remote. These end points are identified by fully qualified OSI addresses. OSI addresses are long strings of hexadecimal digits – not particularly amenable to being typed correctly. So, the AR_Name is the name by which the remote connection end point will be known. This name is merely a shorthand for the complete OSI addressing information which is specified by the other parameters in this remote DIB configuration (so these numbers only have to be typed once). This name does not appear on the network and will never be seen by the remote ICCP peer system. There is no requirement for it to match any name on the remote peer, although certainly names will be chosen which do match. This name is used in performing the task of configuring a Spectrum PowerCC data link. Each link must have a unique remote AR_Name. This field is required. NOTE: This name is case - sensitive.
IP_Address	This is the IP address of the remote communication end point for this Remote AR.
Transport	This indicates the transport layer, either OSI or TCP. Almost all projects use TCP.
AP_Title	This is the remote node's AP title. This field is optional. It is recommended to use the value in the example configuration file.
AP_InvokeID	This is the remote node's AP invoke ID. This field is optional. It is recommended to use the value in the example configuration file.
AE_Qualifier	This is the remote node's AE qualifier. This field is optional. It is recommended to use the value in the example configuration file.
AE_InvokeID	This is the remote node's AE invoke ID. This field is optional. It is recommended to use the value in the example configuration file.
Psel	This is the remote node's presentation selector. This field is required. The administrator of the remote node should tell the Psel.
Ssel	This is the remote node's session selector. This field is required. The administrator of the remote node should tell the Ssel.
Tsel	This is the remote node's transport selector. The administrator of the remote node should tell the Tsel.

Parameter	Meaning
Subnet	This is a decimal integer, which indicates the subnetwork number from the subnetwork . This field is optional. Default is 0.
QOS	This field is optional and currently not used. Default is 0.
Static_Route	This flag indicates that a remote entry's ES-IS routing record (MAC Address) must be configured. Siemens recommends that static routing not be used. Possible values are Y and N. Default is N (no static routing).
SNPA	This field is required only if Static_Route is Y. It contains the SubNetwork Point of Attachment (i.e., a MAC address).

6.3.2.8 If You Are Using TCP/IP Networking

If you are using TCP/IP networking:

- In the subnetwork configuration (s), configure an IP address, where IP_Address is the dotted decimal IP address which matches the IP address of the local network adapter, for example:
IP_Address = 128.1.1.9
- Configure a local name and add the parameter Transport:
Transport = TCP
- Configure a remote name and the parameters Transport and IP_Address, where IP_Address is the dotted decimal IP address of the remote node. Remove the NSAP parameter from the remote . Transport = TCP
IP_Address = 129.1.1.123

6.3.2.9 Putting It All Together

In the past, many ICCP configurations used OSI networking for their ICCP connectivity. Because TCP/IP addressing is easier to work with, there has been a move to use TCP/IP for ICCP connectivity. Spectrum PowerCC is able to support either OSI or TCP/IP.

AR name (OSI address) planning can become a bit complex. Before starting, the person who will be performing the configuration of ICCP logical links, transfer groups, etc., should be contacted and the concepts should be agreed, as well as questions such as "How many logical links will we have to XYZ utility's ICCP peer?". As a starting point, the following should be considered:

- When defining a logical link using the Spectrum PowerCC User Interface, a Quality of Service is selected for that link. Possibilities for QOS are: ALL, LOW, HIGH, OTHER, Low & High, Other & Low, and Other & High. If a link is defined with QOS of HIGH, then only transfer groups defined with HIGH quality of service (QOS) will be sent on this link. If it is defined as LH, then both LOW and HIGH QOS transfer groups will be sent. Only one link can be defined for each priority. If ALL is specified, then only one link is allowed.
- When defining a transfer group, a QOS is attached to the transfer group too. As discussed above, the transfer group will travel over the link with a matching QOS.
- When installing Spectrum PowerCC, one of the tasks is to configure OSI networking. In configuring OSI addresses, one of the parameters is QOS (refer to Appendix Appendix 3-B: Quality Of Service (QOS)). In order for different ICCP logical links with different priorities to really work, they must map to OSI links with corresponding QOS parameters.
- A logical link is defined by its end points. The end points are the (fully qualified) addresses of the applications on each end of the link. Since AR names are used as shorthand for addresses, a pair of AR names defines a logical link: exactly one local AR name and exactly one remote AR name.

Armed with these facts, the procedure to configure one logical link between the own system and an ICCP peer and use this link for all traffic, could be as follows:

- Define one local AR name, that is one "local configuration" in the OSI configuration file. This will map to one local address.

- Define one remote AR name, that is one "remote configuration" in the OSI configuration file. This will map to one remote address.
- When defining a link to the remote system (or, when the person responsible for configuring ICCP defines that link), use this AR name pair to define the link. It will be assigned a QOS of ALL.
- All transfer groups will traverse this link.

This is a relatively common configuration.

There are approximately as many systems for organizing names and addresses as there are people configuring systems. It is recommended to develop a naming system, which is understandable to the respective users and meets their needs. The following example assumes that there are two machines named PCCTST8 and PCCTST9. In this case, PCCTST8 is going to be configured to be the association initiator.

A local on PCCTST9 might look like:

Begin_Local

AR_Name= LocalAR_for_CC_PCCTST8

AP_Title = 1 2 32 2

AP_Invokeld = 100

AE_Qualifier= 1

Psel = 00 00 00 01

Ssel = 00 01

Tsel= 00 01

Subnet= 0

Shared = N

QOS = 0

Transport = TCP

End_Local

A local on PCCTST8 for this same link might look like:

Begin_Local

AR_Name =LocalAR_for_CC_PCCTST9

AP_Title= 1 2 32 3

AP_InvokeID= 100

AE_Qualifier = 1

Psel= 00 00 00 05

Ssel= 00 05

Tsel= 00 05

Subnet= 0

Shared= N

QOS= 0

Transport = TCP

End_Local

A remote on PCCTST8 would be:

Begin_Remote
AR_Name= RemoteAR_for_PCCTST9
IP_Address = 161.254.254.67
AP_Title= 1 2 32 2
AP_Invokeld= 100
AE_Qualifier= 1
AE_Invokeld= 1
Psel= 00 00 00 01
Ssel= 00 01
Tsel= 00 01
NSAP= not used
Subnet= 0
Transport = TCP
End_Remote

6.4 Quality Of Service (QOS)

6.4.1 Overview

As mentioned previously, especially major vendors of internet working equipment (e.g., routers) do not universally implement the OSI Quality of Service (QOS) feature.) This discusses this topic at some length, and provides some suggestions as to how a "pseudo-QOS" might be achieved.

What is QOS? In simple terms, it is part of a priority scheme. It was designed to provide networking devices (e.g. routers) which perform queuing with a mechanism to prioritize traffic. The intent is that, if the device is forced to queue packets for routing over a slow or congested path, it would be able to make decisions based upon the Quality of Service (or priority) of individual packets. Higher priority packets would receive preferential treatment. The absence of such a mechanism is often criticized as a failing in many network protocols.

6.4.2 Using QOS in Spectrum PowerCC

In installing and configuring the Spectrum PowerCC system, the subject of QOS comes up in a few key places:

- During installation and initial configuration, QOS values have to be entered into the OSILL2.CFG file.
- When configuring links (described in the Spectrum PowerCC Configuration), a QOS has to be selected for each link.
- When configuring transfer groups (also described in the Spectrum PowerCC User Configuration), a QOS has to be selected for each transfer group.

Spectrum PowerCC uses QOS as follows:

- When configuring a logical link, it is given a value for QOS (low, high, other, high and low, other and low, high and other, or all).
- When configuring a transfer group, a value for QOS is specified too. The value given to a transfer group for QOS determines which logical link it will be transferred on. A link with the appropriate QOS must have been defined before.
- When adding a data object to a transfer group, the QOS associated with that transfer group ultimately determines the "priority" at which that data object will be handled.



Note

Our intent is not to comment on OSI networking, nor any vendor's products. The fact is that OSI networking has not achieved the widespread implementation once envisioned. Furthermore, the full OSI 7-layer stack is quite complex. In the world of networking, "complex" is often synonymous with "expensive", both in time and money. As a result, many vendors of networking equipment, driven by economic and competitive forces, and faced with limited demand, have not fully implemented the OSI protocol suite. Specifically, some have chosen to not implement QOS. This can have significant implications for users of ICCP.

6.4.3 Considering QOS Implementation

Whether someone should care about QOS depends upon the individual needs. If someone:

- exchanges data via ICCP with a peer over a slow or congested data link, and / or
- wishes to assign higher priority to some data,

then he probably should care.

Therefore this topic shall be discussed in a bit more detail. Be aware that it is not sufficient for the ICCP subsystem to implement QOS, nor is it sufficient for the ICCP subsystem's MMS service provider to implement

QOS. In order for it to really work, QOS must be implemented throughout the network. Consider the simple configuration below:

The following figure shows the sample configuration.

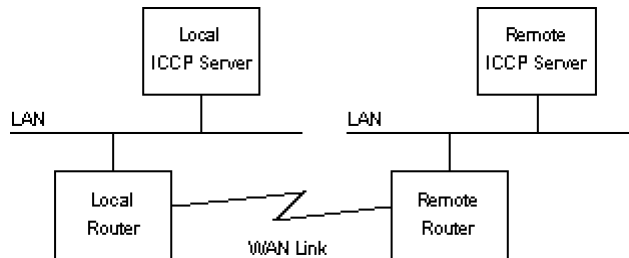


Figure 6-1 Sample Configuration

In this sample configuration, we have two ICCP servers, one on a local LAN and the other on a remote LAN. Each LAN has a router, and the routers are connected by a WAN link of some sort. Presumably, the "local" parts belong to the own utility, and the "remote" parts belong to another utility with which data shall be exchanged. For simplicity, modems, SCADA systems, etc. have not been considered.

Some important observations that can be made about this configuration include:

- The LANs are probably 10 Mb/Sec or 100 Mb/Sec Ethernet.
- The WAN link is probably much slower. At most, it might be a T1 link (1.544 Mb/Sec). It is much more likely to be something like a 64 Kb/Sec, 56 Kb/Sec, or even as low as 9.6 Kb/Sec.

If priorities shall be assigned to data traffic in such a configuration using OSI's QOS, it is important that all the devices in the configuration understand and honor QOS. Actually, since the LAN bandwidth is high, it isn't so much a problem if, for example, the MMS provider software or the ISO networking in the ICCP server platform doesn't fully implement QOS. The ability to set a priority from the ICCP subsystem's User Interface is of little use if the remainder of the system does not honor that priority setting. In the above example, where priority setting really needs to be honored is in the router(s). It is the router that must perform queuing when the WAN link is congested, so it is the router that must make intelligent queuing decisions.) The fact is that many router vendors have not implemented OSI's QOS for all WAN protocols.

So, what to do? There are actually a number of possibilities:



Note

It really isn't as bad as it might seem. Even if the router does not understand QOS, as the link becomes increasingly congested, throttling will occur (i.e., the transmission window size will be decreased). This throttling will eventually reach the MMS service provider and the ICCP subsystem. If they understand QOS, they will begin giving preferential treatment to high priority data. However, these high priority packets may be queued behind low priority packets in devices which do not understand QOS.

- Ignore the whole problem, hope it never happens, etc. This is a fairly common approach. It is certainly simple to implement. However, it rarely produces acceptable results.
- Buy networking equipment that implements QOS. This will work quite well. However, it could be expensive (not to mention hard to find), especially if links and equipment which are already in place shall be used. Furthermore, in order to effect a complete solution, not only the correct equipment has to be bought, but also the communications peers must be convinced to do so as well.
- Provide WAN links that are fast enough that the problem never occurs. This too will work quite well. However, it is also expensive. In some localities, it may not be possible; fast data links may simply not be available.

- Carefully calculate traffic and configure the system(s) so that the available bandwidth is not exceeded. This can mitigate the problem, but it is difficult:
 - In order to perform these calculations, not only the traffic has to be included which results directly from the data objects that shall be exchanged, but also the overhead. For ICCP, overhead cannot be ignored, as it is significant; in fact, if the bits on the wire were counted, under most conditions one will probably find more overhead bits than data bits. In order to perform these calculations accurately, one must have a detailed knowledge of the protocols and encoding methodologies involved. Furthermore, these calculations can be quite tedious.
 - Whenever the configuration shall be changed, calculations have to be checked. This will ultimately lead to negotiations along the lines of: "If I add this data object, which data object must I delete?"
 - Even if the calculations are perfect, it is nearly impossible to account for the unpredictability that is inherent in an electrical network. Some of the data objects that are exchanged ultimately derive from events that occur in the electrical network. ICCP's ability to trigger data exchange on occurrence of events allows to exchange such data only when changes occur. If the events are themselves unpredictable, it follows that the data resulting from those events are similarly unpredictable. Thus, traffic can be "bursty", and occurrences of the bursts are unpredictable.
- Depending upon features of the ICCP subsystem, attempt to load level the traffic. PowerCC has two such features. First, when defining a transfer group which is to be transferred periodically, it is possible to offset the transfer from the "top of the hour". Through use of this feature, it is possible to offset each of the transfer groups by a different amount so they are not all transferred simultaneously. Second, for "by event" transfer groups, a buffer interval can be specified. This actually delays data transfer so that additional data changes that arise from the same event can be reported in the same transfer group. Without the delay, these additional data changes could result in additional "events", which result in transfer of additional "by event" transfer groups. Both of these measures will help significantly. However, depending upon the circumstances, they may not provide a completely acceptable solution.

6.5 Tag Creation

6.5.1 Tag Modification

The ICCP standard allows a user to send a 'Tag' and 'Tag Value' a remote control center.

Block 1&2 is used to report when a tag is set on the system that directly controls the device.

Block 5 is used when the control center that does NOT directly control the devices wants to set a tag on the Control Center that does control the device.

Block 1&2

- The user on Control Center 'A' places a Tag on a point that it provides to Control Center 'Z' (Control Center 'A' "Owns" point).
- The Tag information is reported to Control Center 'Z' using Block 1/2 reporting. On Control Center 'Z', a flag should be set to indicate the tag is set by the remote.



Note

The ICCP Standard defines how to support, for a measurement, a single tag of a specific type. PowerCC allows the user to set multiple tags of the same time on a measurement. For example: A PowerCC user can set multiple Control Inhibit tags (instances) on a single measurement.

If a Tag of a specific type exists on a Control Center 'A' then it is reflected via Block 1&2 as a single tag on Control Center 'Z'

Block 5

- A user on a Control Center 'Z' places a Tag on a point that is provided by Control Center 'A' (Control Center 'A' "Owns" point).
- A Block 5 Tag control request is used to set the Tag on Control Center 'A'. A flag should be set on Control Center 'A' to indicate the tag is set by the remote.



Note

The ICCP standard only defines how to support a single tag of a specific type for each measurement. The same approach that was taken for Block 1&2 is taken for Block 5.

When Block 5 is used to set a tag, Block 1/2 must also be set up to give feedback to the Control Center initiating the Block 5 request. The feedback serves two purposes:

- It confirms to the user on Control Center 'Z' that a tag was successfully set on Control Center 'A'.
- If a user at Control Center 'A' removes all tags on the point, including the tag set by Control Center 'Z', the removal of the tag gets reported to Control Center 'Z'.



7 Troubleshooting

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7.1 ICCP Troubleshooting

This section provides some different things to try if you experience problems getting ICCP to work properly. Because it is impossible to cover all of the different problems that you can run into, this section provides some additional background information on ICCP, in hopes that it will provide the user with the means to resolve issues not covered. This section does assume you are familiar with ICCP and how to configure ICCP in IMM.

There are two main aspects of ICCP that the customer may need to troubleshoot, the first is the ICCP association establishment with the remote control center. The second is the exchange of the data with that remote control center after the association is established.

7.2 Association Establishment

7.2.1 Association Establishment

First, some background on ICCP associations. An ICCP association must be established between the two entities that need to exchange ICCP data. When the ICCP standard first came out, this association was an OSI association that included, as part of its addressing definition, a Network Service Access Point (NSAP). This addressing scheme was not as easy to understand as the more familiar IP addressing scheme. So fairly quickly OSI usage was replaced with IP. PowerCC ICCP still supports OSI, but Siemens focus, including this troubleshooting section, is IP.

7.2.2 Association Configuration and SISCO Programs

7.2.2.1 Association Configuration and SISCO Programs

Siemens ICCP products use a 3rd party product called MMS-EASE from SISCO to implement the lower layers of the ICCP protocol (ICCP sits on top of MMS). This 3rd party product has its own configuration, most of the configuration relating to addressing for the associations. The 3rd party product also provides some separate programs that are part of the communication stack.

7.2.2.2 The OSIFileCreator

The OSIFileCreator takes information out of IMM and writes it to the configuration files needed for the 3rd party product (see [5.3.2 Accessing the Communication Editor for ICCP](#)). The reason the OSIFileCreator exists is because the PowerCC approach is to have the user enter all configuration information into a single place, IMM. This simplifies it for the customer because they do not need to figure out how to set up different configuration files.

The OSIFileCreator creates two configuration files for the SISCO 3rd party products. It creates the osil2.cfg file that contains most of the addressing information for connecting to the remote host (see osil2.cfg). If secure ICCP associations are also configured, it creates the second file, secManCfg.xml (see secManCfg.xml).

If things are not set up correctly, the OSIFileCreator can not complete its task, so it will put out an error message box. Attempts were made to make the error messages descriptive enough so additional information is not needed. But some errors need some additional background information, which is provided here:

Error: No PC object found for the 'ICCP' software package of ICCP type.

The OSIFileCreator needs information about the PC the ICCP software is running on. It gets this information by searching the SMI for Software Packages with the type of 'ICCP', and then following the 'Runs On Server' link to the hardware server. This error message is generated when the 'Runs On Server' link is not set. To correct this problem, link the ICCP software package to the appropriate hardware server.

Error: No IP Address defined for the '\$HOSTNAME' PC Object in DOI

The configuration file needs to contain the local IP address it is supposed to use. Currently, the OSIFileCreator gets this out of the SMI configuration. It is the 'IP Address' attribute of the "Personal Computer Node" (under Hardware Components) where ICCP runs. Please make sure the 'IP Address' attribute is set for the servers ICCP is configured to run on.

Error: Cannot open the '\\\$HOSTNAME\SiscoCfg\$losil2.temp' file for writing. Make sure the disk is not full or write protected

The OSIFileCreator should be run on the PSOS. After it creates the configuration file, it needs to write it to a shared directory on the ICCP Servers (where the 3rd party product runs). This error is generated when it is

unable to write the configuration file to this shared directory. The share on the ICCP server is called SiscoCfg\$. This share should have been set automatically by the PowerCC installation script.

Possible causes:

- OSI Stack installed after PowerCC. When the PowerCC installation script runs, it checks to see if the OSI Stack has been installed, if it has been installed, then it sets the share on this OSI Stack directory. If it does not detect the OSI Stack, it does not set up the share. The preferred solution is to uninstall PowerCC, install the OSI Stack, and then re-install PowerCC. If this is not an option, contact SIEMENS PowerCC system support.
- The permissions on this shared directory do not allow OSIFileCreator to write to this directory. Check the permissions and change as necessary. If default installation settings are used, the share is set on the directory: C:\Program Files\SISCO\osill2.

If default settings were not used, check this registry setting to see where the share should be set:
HKEY_LOCAL_MACHINE\SOFTWARE\SISCO\OSILL2\CurrentVersion\Parameters\CfgPath

There may be some issues with what has been configured in IMM that does not prevent the OsiFileCreator from creating a new osill2.cfg file (it may prevent the secManCfg.xml file from being created). If this occurs, the OSIFileCreator will pop up a message box indicating 'ICCP' was populated but errors were detected, see NT Application Event log for details'. Use the NT Event Viewer to check the application log for more details.

osill2.cfg

This is the file created by the OSIFileCreator (and used by the 3rd party product). More information on the contents of this file can be found in Section 6.3 [6.3.1 Overview](#) OSI Configuration. It is a text file that can be edited.

SIEMENS recommends the customer use the OSIFileCreator for maintaining this file. The customer could make a temporary change to this file, but realize that any time the OSIFileCreator is executed; it will overwrite any changes manually made to the file. Changes that should be permanent should be made in IMM.

If a change is made to the osill2.cfg file, the easiest way to have ICCP pick up the changes is to stop and restart the ICCP subsystem.

Log Levels for osill2d

The Log Level for the osill2d program is defined by the Log_Level parameter in the osill2.cfg file. The OsiFileCreator builds the osill2.cfg file. It sets the Log_Level based on the 'GeneralLogLevel' attribute that currently resides in the Application Definition section of SMI. The default value for this attribute in IMM is currently 0. The only value SIEMENS recommends this being changed to would be 80, so errors are generated in the log file.

If the project is having problems with the osill2d program, and wants to turn on a higher log level, this would be a situation where manually editing the osill2.cfg file would be acceptable. It is assumed such a change is temporary. After capturing the osill2d trace, make sure you turn off the log level.

The Log Level parameter is a bit mask. Here is information provide by SISCO regarding the definition of this parameter:

- **Error 80 (recommended to be set always)**
- **Debug 40**
- **ACSE 20**
- **Presentation 10**
- **Session 08**
- **Transport 04**
- **Low Interface 02**
- **High Interface 01**

In cases when detailed information is required to resolve Stack communication problems, the mask C0 (80 | 40) would be recommended to obtain a full trace.

The Log Level parameter is a 2 digit hex field. Do NOT precede it with 0x. If you want all levels set, you would just set it to: ff.

See **osill2d** section below to see where the log file resides.

osill2d

This is the 3rd party product that handles the lower layer protocol.

Startup:

There are two ways osill2d can be started up. By default, it is started by the ICCPR processes when they start up. Then when the ICCPR programs stop, the osill2d program stops. This is the preferred method for starting osill2d.

The other method for starting osill2d is by making it automatically start up as a service (it is already set up as a service, but the 'Startup Type' is set to Manual). If it is to run as a service, care must be taken in making sure it runs as a user that the ICCPR processes can communicate with.

When ICCPR starts up osill2d, the osill2d program normally runs hidden (will not be seen on local login). Sometimes it is helpful to see the osill2d process so you can verify its status. To make it so it is NOT hidden, the following registry setting can be changed to 0:

HKEY_LOCAL_MACHINE\SOFTWARE\SISCO\OSILL2\CurrentVersion\Parameters\RunHidden.

After making this change in the registry, you need to stop the ICCP subsystem and restart it. After the restart, a "SISCO OSI STACK" window should appear in the login where PowerCC was started from.

If you no longer want it to be visible, set the RunHidden value to 1 and stop and restart the ICCP subsystem.

osill2d log file – osill2d.log

The osill2d program generates a log file that may contain information that may be useful for resolving problems. If the default installation is used for installing the SISCO stack, the location of the log file should be c:\Program Files\SISCO\osill2.

If the log file is not appearing in this location, there is a registry entry that indicates where the osill2d.log file should be written to. It is found in:

HKEY_LOCAL_MACHINE\SOFTWARE\SISCO\OSILL2\CurrentVersion\Logging\LogFile

What osill2d writes to the log file is determined by the setting of the Log Level parameter in the osill2.cfg file. See previous section for more information on setting this attribute.

Failure of ICCPR to communicate with osill2d

In our development labs where we frequently use a debugger with the ICCPR programs, we have experienced on a few rare occasions, a situation where the ICCPR programs are no longer able to communicate with the osill2d program. Some symptoms of this are that NONE of the ICCPR process go active (green). And in the NT Application Event log, there are log messages from ICCP that include this 'Description' text:

"SDK Log: 3080 ACSE SUIC Error: Lower Layers not Operational PROCESS=ICCPR SOURCE=ICCP CLASS= METHOD=New Server USER=SPC5DEV\SPCAAdmin"

The solution is to stop the ICCP Subsystem, and then to use the Windows Task Manager to do an 'End Process' on the osill2d program (also see description of 'snap' below). Restarting the ICCP subsystem should cause a new instance of the osill2d program to start (IMPORTANT, make sure all ICCPR processes have terminated before restarting ICCP subsystems. If ICCPR process is not going way, do an 'End Process' on it.)

We are not aware of this happening in normal conditions, but feel it is something that should be included in this trouble shooting section.

secManCfg.xml

This file is similar to the osill2.cfg file. It is created by the OSIFileCreator (and used by the 3rd party product). But the content of this configuration file is information for secure associations. It is an .xml file that can be edited.

SIEMENS recommends the customer use the OSIFileCreator for maintaining this file. The customer could make a temporary change to this file, but realize that any time the OSIFileCreator is executed; it will overwrite any changes manually made to the file. Changes that should be permanent should be made in IMM.

If a change is made to the secManCfg.xml file, the easiest way to have ICCP pick up the changes is to stop and restart the ICCP subsystem.

SecCfg

SecCfg is a tool provided by SISCO to configure the secManCfg.xml file. But SIEMENS approach is to have the user enter everything in IMM, and have the OsiFileCreator create the secManCfg.xml file. So SecCfg does not need to be used.

While it does not need to be used to configure secManCfg.xml, it can be used as a tool to validate the secManCfg.xml file. It can also be used to make temporary changes to the secManCfg.xml file, such as modifying the log setting for the snap program.

Startup:

There are two ways you can start SecCfg:

- When you installed the SISCO Security Toolkit, it should have added the tool to the Start Menu: Start->Programs->SISCO->Security Toolkit for MMS-EASE->Security Configuration Utility Or
- If the installation used the defaults, the binary should have been installed in: C:\Program Files\SISCO\Security\bin. The executable name is SecCfg.exe. Just double clicking on it will start it.

Validating secManCfg.xml file:

When you start SecCfg, if it encounters any validation problems with secManCfg.xml, it will immediately pop up a "Validate Configuration File" window. This window will point out possible errors or warnings in the configuration file. In some cases, it will actually attempt to fix the problem it detected.

It is possible that SecCfg will not pop up the "Validate Configuration File", even though there are still problems with the configuration. After starting the SecCfg tool, it may be necessary to 'browse' the different sections of the secManCfg.xml file and see if there are any problems.

The cause of all of these errors and warnings need to be figured out, and properly corrected in IMM.

Contact SIEMENS if you need additional help in figuring out the cause of warning or error messages from the SecCfg tool.

Log Levels for snap

The SecCfg tool can also be used for setting the Log levels in the secManCfg.xml file that the snap program uses.

Do this by starting the SecCfg program. Once it is started, on the left side of the SecCfg window, you should see 'SNAP' with 'Log Masks' under it. Select 'Log Masks', and on the right side of the window you will see the different masks that can be set. If investigating a problem and in doubt on what to set, then set everything. Hit OK and then exit SecCfg.

The easiest way to ensure that these new settings are picked up is to stop and then restart the ICCP Subsystem.

See section below on **snap log file** to see where the log file is located.

Be aware that these changes to the secManCfg.xml file will be wiped out if the OsiFileCreator is run.

snap

This is another program that is part of the 3rd party product that handles the lower layer protocol. It is specifically used if secure ICCP associations are configured and used. In fact, it is only installed if the SISCO Security toolkit package is installed and only runs if secure associations are configured and enabled.

Startup

The snap program is started by the osill2d program. When the osill2d program is started, and if security is configured and enabled, the osill2d program starts snap. When the osill2d program is stopped, the snap program should go away.

If there are problems with communications between ICCPR and osill2d, and it becomes necessary to end the osill2d process, it is important that the snap process is also ended.

snap log file – snap.log

The snap program generates a log file that may contain information that may be useful for resolving problems. If the default installation is used for installing the SISCO stack, the location of the log file should be C:\Program Files\SISCO\Security\bin.

The log settings are controlled in the secManCfg.xml file. See the description of the SecCfg tool in the previous section.

7.2.3 Establishing Non-Secure ICCP associations

This section assumes you have configured the ICCP associations in IMM, and believe everything is configured correctly, but the ICCP association is not coming up. Here are a few things to try or look at:

7.2.4 Telnet

When working on establishing an ICCP association to a remote control center, one of the problems is making sure you are getting through all of the firewalls to the end server. And one of the simplest ways of checking this is by using the standard telnet command. Just open a command prompt on the ICCP PC you are establishing the association from, and then enter:

```
telnet "IP" 102
```

"IP" is the IP address you are attempting to connect to. For non-secure associations, ICCP uses well known TCP/IP port 102.

If telnet can not connect to this well known port, then ICCP will not be able to. telnet will put out a message like this if it can not connect:

```
Connecting To xxx.xxx.xxx.xxx...Could not open connection to the host, on port 102: Connect failed
```

Pay attention to how long it takes for this message to appear. If the message appears rather quickly, it could indicate it found the server, but the server refused the attempted connection to port 102. If it takes a while for the response, it may indicate that it was not able to get to the indicated IP address. Just the timing of the response is not a good indication of what is going on; there could be other explanations on why the response comes back immediately or why it does not. (If you do a telnet from an AIX server, you will actually get different messages that indicate if the remote refused the connection or if the attempt to the remote timed out).

If telnet is able to get a connection, on Windows, the behavior is that the command prompt window is cleared out. (On AIX, telnet will print out a message indicating 'Connected to xxx.xxx.xxx.xxx').

7.2.5 Check NT Application Event Log for Error messages from ICCP

If ICCP is able to create an association with the remote control center, it will put out an Informative Event log message stating something like this:

SDK Log: 1017 Association established: Negotiated PDUSize=32000

After the association is established, ICCP needs to exchange 3 pieces of information before it declares the association as up. If there is a problem with any one of these 3 pieces of information, ICCP puts out a message into the NT Application Event log indicating what the problem is.

Here are the pieces of information and the type of thing that is wrong:

- **Bilateral Table Version:** Both sides must specify the same value for the Bilateral Table Version. If there is a mismatch, ICCP will put out two Error messages that look something like this:
A). ICCP SDK ErrorCallback: 'Bilateral Table Version' (ID) mismatch, association concluded. Review setting in database and confirm with remote control center. See previous log message for additional details.
B). SDK Log: 3106 u_mp_read_conf, BLT ID mismatch, received (version) expected (version1)
The second message (B) indicates what was received (in this case 'version' and what was expected 'version1'. The 'what was expected' value is the value of the Bilateral Table Version attribute on the IcRemoteCenter instance in IMM. Either the remote control center needs to change or this attribute in IMM needs to change so they both match.
- **ICCP Version:** Both sides need to agree on what version of ICCP they will be supporting on this association. If there is a mismatch, ICCP will put out two Error messages that look something like this:
A). ICCP SDK ErrorCallback: ICCP Major version number does not match between local control center and remote control center, association concluded. Review with remote control center. See previous log message for additional details.
B). SDK Log: 3105 u_mp_read_conf, remote ICCP Major Version (1996) is not (2000)
The second message (B) indicates what was received (in this case '1996' and what was expected '2000'. The ICCP Version attribute on the IcRemoteCenter instance in IMM is how this is defined. Either the remote control center needs to change or this attribute in IMM needs to change so they both match.
- **Local and Remote domain name:** As per the protocol definition, the ICCP client needs to read the Bilateral Table Version variable out of a specific domain on the remote server. If the local system does not correctly specify the remote systems domain when it attempts to read this variable, you will see Warning messages like this:
A) SDK Log: 2008 u_mp_read_conf, Bilateral_Table_ID read failure (4), verify domain name is correct.
On the server side, there will be a Warning message like this: A) SDK Log: 2052 Domain (RC) error for Object Bilateral_Table_ID from client RC East
The domain names are attributes on the IcRemoteCenter instance in IMM. The Local Domain Name attribute is our server's domain name that the remote client must specify. The Remote Domain Name is the name our client will use to try and read out of the remote server's domain.
These Domain names are not Windows domains. They are 'ICCP' domains. The Local Domain Name must match the Remote Domain Name, and vice versa.

7.2.6 Telnet works, but Application Event Log does not contain an 'Association Established' log

Here are a couple of things to look into:

Association Initiator

As stated elsewhere, SIEMENS ICCP implementation allows one association to support both client and server functionality. But only one side can initiate the association and the other listens. So on the IcRemoteCenter, there is an 'Association Initiator' attribute, which is either true or false.

If you are not seeing any ICCP messages in the NT Application Event log for this remote on either system, then it could mean that both sides have 'Association Initiator' set to false.

If both systems are seeing a Connection Terminated Warning message in the event log for this remote, then it could mean that both sides have 'Association Initiator' set to true.

The solution is to make sure one side has it set to true, and the other side has it set to false.

Stack Address incorrect

If telnet indicates you are getting to the ICCP port on the box, but you are still not seeing any messages indicating Association Established, it probably means the stack address (PSEL, SSEL, and TSEL) are not correct.

In general terms, the IP address gets the association to the server, the well known port (102) gets the association to the osill2, and the PSEL, SSEL and TSEL gets the association to the ICCPR process supporting the communication with the remote.

On the listening side (non initiator), each Local AR entry on an ICCP server MUST have a unique combination of PSEL (presentation selector), SSEL (session selector), and TSEL (transport selector). These three addresses combined define a unique end point for the association, so the osill2d program can associate it to a specific process.

In the Remote AR entry on the remote system that is initiating the association, it must specify a PSEL, SSEL, and TSEL that exactly matches the PSEL, SSEL, and TSEL of the Local AR on the system that is listening.

7.2.7 Establishing Secure ICCP Associations

7.2.7.1 First get association working in Non-Secure mode

It has been our experience that it is more difficult to get a secure association established than a Non-Secure association. So we recommend the user first work through the issues that may arise from Non-Secure communications (see previous section). Once the Non-Secure issues have been worked out, and an association is established, then attempt to make it a secure association.

7.2.7.2 Getting the association secured

Here are some of the problems we have seen when trying to get secure associations to work. There is also some information on things to try.

Osill2d does not work

Once we start using secure associations, the osill2d program will need to start a program called 'snap' (see section above on **snapp**). If osill2d is able to successfully start and communicate with snap, the SISCO OSI Stack Window (if RunHidden is false in registry) will show a 'Status' of "Operational (secured mode)".

If osill2d is not coming up or staying up, or does not appear to be working, check if the osill2d log file provides any information. Check for an error like:

2008-09-25 16:21:25.157: ERROR: Start of SNAP failed

If there are problems trying to get the osill2d program to start up snap, it may be useful to manually start snap. Here are the steps to follow:

- If you have not already done so, stop the ICCP Subsystem.
- Bring up a command prompt, and cd to the directory the snap executable is located (the default installation installs it in c:\Program Files\SISCO\Security\bin)
- In the command window, enter 'snap'.

If snap is able to run properly, it will print out some versioning information, followed by 'Starting...' and 'Running...'. Then it will just sit there. This means snap is working properly. (This is not the proper way to start snap, so enter Ctrl-C to stop it). Please examine the osill2d log file closer to see if there are any additional messages indicating why it is having problems starting. You can also attempt to run osill2d from a command prompt.

If you do not see 'Running...', or you are given the entry prompt again, it probably means the problems are with snap. If snap does not provide any information in the command window, check the snap log file (see section on **snap**). It should provide more details on the problem.

Some of the problems with snap are as follows:

- It is unable to find the information it expects in the secManCfg.xml file. The OsiFileCreator should be creating this file based on information entered into IMM, so it could indicate the necessary information was not entered into IMM, (it could also mean there is a problem with the OsiFileCreator). Another tool for validating the contents of the secManCfg.xml file is SecCfg.
- Snap can not find the certificates that have been specified in the secManCfg.xml file, so either they have not been imported into the Microsoft certificate store, or they were imported incorrectly.

Review [5.6.1 Certificate Management Using Security Stores](#) to make sure the certificates were properly configured.

Make sure when setting up the Certificate snap in that 'Computer account' was specified. And when importing the certificate, make sure the 'Mark the private key as exportable' was checked.

Telnet Test for Secure Associations

Just like the test for non-secure associations, telnet can be used to verify the path to the TCP/IP port used for secure ICCP associations. For non-secure associations, ICCP uses port 102. For secure associations, the default TCP/IP port used is 3782. See Section on **telnet** on how to use telnet to verify the association is getting to this port.

7.3 Data Configuration Issues

7.3.1 Analog Value not changing (Deadband Configuration)

Sometimes analog values on the server are changing, but they are not showing up on the client system. If everything indicates that the serving system is sending the analog value, then it is possible that the receiving ICCP process (the client) is not reporting the change to SCADA because the value change did not exceed the Deadband.

There is an attribute on the `IcClientBlock1And2Object` called `Deadband`. This value is applied to incoming Analog Measurements (it is not applied to Accumulator Measurements). If no `Deadband` value, or a `Deadband` of 0 (zero) is specified on the `IcClientBlock1And2Object`, then ICCP uses the 'Default Dead Band Value' attribute defined on the `IcRemoteCenter` instance as the `Deadband`. Users sometimes put a `Deadband` of 0 on the `IcClientBlock1And2Object`, and then expect any change in the analog value to be reported, but a 0 just means the Default Dead Band will be used.

This `Deadband` is NOT a percentage of the value. It is compared, as is, against the incoming analog value. This is the logic applied to the incoming value:

If $(\text{fabs}(\text{lastValueICCPProcessed} - \text{receivedvalue}) \leq \text{Deadband})$, then process (save as `lastValueICCPProcessed`) and report value

Setting the `Deadband` to 0 (on both the `IcClientBlock1And2Object` and `IcRemoteCenter`) still means that the value needs to change before ICCP will report the value.

ICCP is not aware of any change to the value on the local system. If some tool is used to change the value in SCADA on the local system, ICCP is not aware of this change. In this situation, even with a `Deadband` of 0, ICCP will not re-fire the value just because the remote system has resent the same value. Only if the received value changes by the `Deadband` will ICCP report the value.

Depending on the situation, you may want ICCP to always fire the value it receives, even if it has not changed. To get this behavior, set the `Deadband` to -1 on the `IcClientBlock1And2Object`. This means that every time ICCP receives a value, even if it hasn't changed, ICCP will process and report the value. Siemens recommends this only be done on a point by point basis, when really deemed necessary. Only on very special cases should a should-1 be used as the Default `Deadband` on the `IcRemoteCenter` instance.

7.3.2 Remote system Only Supports Pure Server and Pure Client Associations

Siemens solution supports on a single association, both the ICCP client and server functionality. There are some ICCP implementations that do not support this. If a remote system does have this limitation, it is necessary for two 'remotes' to be configured on the local system. One 'remote' implements the pure 'Client' functionality of ICCP, and the other supports the pure 'Server' functionality.

Here are some of the configuration changes for this type of support:

IcRemoteCenter

There is a 'Communication Direction' attribute on the `IcRemoteCenter`. This attribute indicates how the local ICCP process is going to behave. Normally this is set to 'Both server and client'. For the 'remote' configuration that is to implement the pure 'Client' functionality, this attribute should be set to 'Client Only'. If implementing the Server functionality, this attribute should be set to 'Server Only'.

Server Objects

If this 'remote' is a 'Server Only', then the user should only configure ICCP server objects on this 'remote'. ICCP Transfer groups or client objects should NOT be configured on this remote.

Client Objects

If this 'remote' is a 'Client Only', then the user should only configure ICCP client objects and Transfer groups under it. Server objects should NOT be configured on this remote.

7.3.3 AllChangesReported

In the 2000-08 revision of the ICCP standard, a new attribute was added to the DSTransmissionPars model called 'All Changes Reported'. It is an attribute that the client sets on the server. This attribute was added so multiple changes to a single object can be sent, instead of only the current value being sent.

Because the client writes this to the server, the configuration of this occurs on the client side of the association. Here are the setting needed in IMM on the Client configuration to request that the server report all changes for points in a specific transfer group:

IcRemoteCenter

ICCP Version- This capability is introduced in the 2000_08 version, so the ICCP version must be set to 2000_08.

IcClientBlock1And2Container:

All Changes Reported- must be set to true.

Buffer Interval- must be greater than zero.

Report By Exception (RBE)- must be set to true

Transfer Type- must be set to "Report by event".

For the most part, the server function is just reacting to what the client sets up. The only configuration setting that must be set is the ICCP Version on IcRemoteCenter, it must be set to 2000_08. Once the server receives the enable for a Transfer Set that has All Changes Reported set to true (along with the other settings described above), it will start reporting all changes it receives from SCADA after that moment, until either the Transfer Set is disabled on the association with the remote is terminated.

7.3.4 SOE Support

Siemens is attempting to use the AllChangesReported capability defined in ICCP, along with the new data types with millisecond resolution, to provide a way to treat the data exchanged as SOE data. The standard does not provide any SOE capability like those defined in some traditional RTU protocols, such as 'SOE Buffer' full indication. Siemens is using the capabilities provided as a means to send data across ICCP that will be treated as SOE data. The rules that apply for All Changes Reported apply here (see previous section on AllChangesReported).

Internal to PowerCC, data is sent as a Telemetered event or an SOE event. If the data should be treated as both, then it needs to be sent as a Telemetered event and as an SOE event. The SOE event type is only supported for Analogs, Accumulators, and Digitals (all discretes except Tap Position) measurements.

If ICCP is the client receiving an 'AllChangedReported' value from the remote, it is possible for ICCP to fire this into SCADA as either a Telemetered event, an SOE event, or both (fire both events). How the data is to be fired is set on the IcClientBlock1And2Object, with the SOE Data attribute.

If ICCP is sending the data to the remote (server), it will only be configured to send either the Telemetered event or the SOE event for a specific ICCP object. This is done by setting the SOE Data attribute on the IcServerBlock1And2Object.

If you want to treat this as SOE data, it is required that the ICCP data type that is used includes the millisecond time stamp (timestamp extended).

7.3.5 Remote Fetch

Remote Fetch button grayed out

If the user brings up the ICCP UI, and then right clicks on the remote, if the Remote Fetch button is grayed out, it probably means the user role or the console being used does not have the 'Engineering Dataset Administration' Access Right. Please check with your administrator to be given this access right.

Impact of Remote Fetch

The actual retrieval of the names and types from the remote system takes place when the association is established. When you select the 'Remote Fetch' option in the ICCP UI, it is only setting a flag to ICCP to perform the 'fetch' process the next time the association is established.

The retrieving of the names and types can take some time and WILL prevent the exchange of the data already configured in IMM. How much it impacts depends on many objects and the types of those objects.

The first exchange is the names and is fairly quick. The ICCP standard allows the names to be exchanged as lists. How many names can be exchanged in a single request/response exchange depends on the size of the PDU. If the PDU size is small, and there are a lot of long object names, there will need to be more request/responses to get all of the names across, which means it will take a little more time.

The fetch of the types is what usually takes the most time. The mechanism to get the detailed type description only allows for a single object per request/response. This handshaking, when multiplied by hundreds or thousands of objects, can take a relatively large amount of time.

But there is an alternative to getting the detailed type description. By performing a read of all the objects, ICCP can use the type information included in the read response to determine the type of most of the objects. (Many objects can be put in a single read request, how many depends on the PDU size.) But some types can not be distinguished, specifically the "State plus Quality plus Timestamp" and the "Discrete plus Quality". For objects that have this type, the method for getting the detailed type must be used.

Siemens solution first attempts the read of all objects to see if it can determine the type this way. For those objects it can not distinguish, it uses the detailed method, which requires more handshaking with the remote, thus takes more time. So how long the remote fetch takes also depends on the data types.

Remote Fetch Unable to distinguish some ICCP data types

The detailed method can be used to distinguish between the 'State plus Quality plus Timestamp' and the 'Discrete plus Quality', but it can not distinguish between 'State Value' and 'State plus Quality'. The remote fetch process assumes that objects that have a type that can be either as 'State plus Quality'.

With the 2000-08 revision, Tag types can now be exchanged as Block 1&2 data. If there is an ICCP Tag object defined on the server, there is probably a ICCP server object defined for the corresponding value. Both the Tag object and the object for the value would be linked to the same measurement. Our Fetch process is not able to determine that the two ICCP objects are associated with the same measurement, so it will create two independent measurements on the client system to dump the received values. One is for the Tag object and the other is for the Data. On the client system, when looking at the tag measurement in the Runtime Explorer, the data value for the tag measurement will never get updated, but the tag still comes across. If a user is able to determine what Tag object goes with what data object, they could modify the appropriate association between the ICCP client object and measurement so both point to the same measurement.

7.3.6 Shared Server Objects

In IMM, the user can define ICCP Shared Server objects. In IMM, these shared server objects can be linked to specific remotes with the expectation that they would only be available (served) to those remotes. This link is currently not honored. Any shared server object that is defined is available to all remotes.

7.4 ICCPR Trace Capabilities

7.4.1 Overview

ICCPR generates trace/log information at several different layers. Where the trace goes is dependent on the layer. The ICCPR code that is aware of the PowerCC interfaces has its trace information it generates. The ICCP SDK, which the ICCPR code uses, has its trace information. The ICCP SDK also has a separate trace for MMS messages exchanged.

As with any trace, the more that is being generated, the more of an impact it will have on performance.

7.4.2 PowerCC Trace of ICCPR

The layer of code in ICCPR that interfaces with the PowerCC applications (ASR, DDS, etc), uses the standard PowerCC trace mechanism. The trace is controlled via the PowerCC System Diagnostics UI. Using the UI, the trace level can be set and the user can indicate if they want it logged to a private file instead of a single file with all of the other PowerCC trace.

Turning on trace for ICCPR causes it to trace for all ICCPR processes. If 'Log to Private File' is selected, there is a separate trace file generated for each ICCPR processes. Part of the file name includes the Process ID (PID) of the ICCPR process the trace belongs to.

The 'Trace Info1' will result in the more important, single occurrence, type messages. 'Trace Info 4' is the other extreme, with very repetitive messages ('Enter function, Exit function, etc.). The 'Trace Info2' and 'Trace Info3' are in between as far as what type of trace is produced.

7.4.3 ICCP SDK Trace

The bulk of the ICCP protocol is implemented within the ICCP SDK. It was developed and used in other Siemens ICCP products. It supports 10 debug levels. The Trace Info levels set for ICCPR using the Diagnostics UI get mapped to the ICCP SDK debug levels. Here is the mapping:

PowerCC trace level	MsgLog Config Setting					
TRACE INFO4	√		√			
TRACE INFO3		√	√	√		
TRACE INFO2	NA		NA	√	√	
TRACE INFO1	NA	NA	NA	NA	NA	√
SDK Debug_Level	10	8	7	5	3	1

In order to get ICCP SDK debug level 7, the user sets Trace Info 4 and Trace Info 3 for ICCPR. NA means Not Applicable, thus with Info 4 and Info 3 set, the setting of Info 2 and Info1 does not matter.

SDK Debug level 10 is going to be the most messages, Debug level 1 is the fewest. Setting Trace Info 4, 3, 2 and 1 provides the best combination of ICCPR PowerCC and ICCP SDK trace without producing too much.

The SDK trace comes out to the same log file the ICCPR PowerCC trace goes to. In the trace file, if the Source is ICCP, it means it comes from the ICCP code. If the Class is SDK, then it came from the ICCP SDK. If Class does not specify SDK, then it came from the ICCP code that interfaces with PowerCC.

The SDK also puts the same SDK trace output into its own log file. This log file is located in: %SR5_TEMP%\IC-CPR\Log\Remote host name\log1.txt SR5_TEMP is usually C:\SIEMENS\PowerCC\Temp. The digit after 'log' is the number of the log file. The ICCP SDK always keeps around the log1.txt file from when the process starts. Once log1.txt is full, it starts writing to log2.txt. Then it starts writing to log3.txt. Once it starts writing to log4.txt, it will delete log2.txt. So you will only see a log1.txt, and then a log(n-1).txt and a log(n).txt.

7.4.4 MMS Trace

The ICCP SDK also has a separate trace message for the messages exchanged with the MMS-EASE program. This trace can be set to 5 levels. Again, the ICCPR debug level is mapped to the MMS trace level. Here is the mapping of the different Info levels to the MMS trace level:

PowerCC trace level	MsgLog Config Settings				
TRACE INFO4		√			
TRACE INFO3	√	NA			
TRACE INFO2		NA	√	√	
TRACE INFO1	NA	NA	NA	NA	√
MMS Trace Level	5	4	3	2	1

Setting Trace Info 4 for ICCPR provides a detailed trace of the object. It looks like this:

16:10:30 R READ CONF 1 Local AR:LocalAR_West

Read variables - variable count:4

Siemens read kind: Initial dataset read

Variable names not present in PDU

1. Type:StateQTime Sub_A_Bay_12_CB_Status 2 Quality:SUSPECT/TELEM/NORM_VALTIM

Time:9/30/2008 16:10:3

2. Type:RealQTime Sub_A_Bay_12_P 500.200012 Quality:SUSPECT/TELEM/ABNORM_VALTIM

Time:9/30/2008 16:10:3

The MMS trace file goes to the same directory as the SDK trace. But it goes to a different file. It goes to trace_mms1.txt

7.5 ICCP_DS Display: Program to View Negotiated Datasets

The first time ICCP on the local system and remote control center connect, they negotiate several things, one of the things is the contents of the datasets. If there are a lot of points configured, or points missing on the server, the negotiation can take a relatively large amount of time. The results of these negotiations are stored in files in directories under %SPC_TEMP%\ICCPR\ . Every time the associations are re-established, the two sides see if the other side has the datasets that were previously negotiated (ICCPR looks at the contents of these directories). If they do, they just use those dataset definitions. This speeds up the negotiation process considerably. If the files are deleted, the entire dataset negotiation process needs to be repeated.

The results of these negotiations are just text files that can be viewed by the user. Under the ICCPR there are actually two different directories that hold these definitions, ICCPR\ActiveClientDataSets and ICCPR\ActiveServerDataSetDef. Sometimes it is helpful to look in these files to see how objects in a transfer group are broken up into different datasets.

Instead of people viewing these files and potentially corrupting them, a simple UI was created that can be used to view these files in a non-editing mode. The program is located in \$SR5_INSPATH\Apps\ICCP. It is called ICCP_DSDisplay.exe.

If you double click on ICCP_DSDisplay.exe, it will bring up a UI that is pretty self-explanatory. This UI is just a tool for viewing the datasets that have been negotiated, and nothing more. It does not display the values of the objects that have been exchanged.



